SUSTAINABLE RIVERS PROGRAM

JENNINGS RANDOLPH LAKE & NORTH BRANCH POTOMAC RIVER ENVIRONMENTAL FLOWS

STATE OF THE SCIENCE REPORT

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Baltimore District

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Acronyms

AMD: Acid Mine Drainage

AVF: Artificially Varied Flow

BWRA: Barnum Whitewater Recreation Area

CWMS: Corps Water Management System

CFS: Cubic Feet per Second

CO-OP: Cooperative Water Supply Operations on the Potomac

DO: Dissolved Oxygen

FA-E: Flow Alteration-Ecology

HA: Hydrologic Alteration

ICPRB: Interstate Commission on the Potomac River Basin

IHA: Indicators for Hydrologic Alteration

JASON: July, August, September, October, November

JRL: Jennings Randolph Lake

MBSS: Maryland Biological Stream Survey

MCPR: Mineral County Parks and Recreation

MD DNR: Maryland Department of Natural Resources

MDE: Maryland Department of the Environment

NBPR: North Branch Potomac River

NBPAC: North Branch Potomac River Advisory Committee

NTU: Nephelometric Turbidity Unit

PCD: Project Construction Datum

QC: Quality Control

RRM: Reservoir Regulation Manual

USACE: United States Army Corps of Engineers

SpCond: Specific Conductance

TDS: Total Dissolved Solids

TNC: The Nature Conservancy

TSS: Total Suspended Solids

USFWS: United States Fish and Wildlife Services

USGS: United States Geological Survey

WV DNR: West Virginia Division of Natural Resources

WWTP: Wastewater Treatment Plant

Chapter 1. Introduction

1.1. Background

In 2021, the North Branch Potomac River was added to the Sustainable Rivers Program. "The Sustainable Rivers Program (SRP) is a 22-year partnership between the U.S. Army Corps of Engineers (USACE) and The Nature Conservancy (TNC). The SRP aims to improve ecosystem health by adjusting [USACE-managed] reservoir operations to benefit natural communities. Modifying operations to further consider environmental needs while still accomplishing authorized project purposes is an effective way to restore, protect and sustain riverine and floodplain habitats" (Winget, 2020). This report presents the underlying literature and data review informing an understanding of the natural flow, flow requirements of the native species, and communities of the North Branch Potomac River. For the purposes of this report, natural flow is defined as: the flows the river system would have without any impoundments or human action (development, dams, levees, etc.). Natural flow values are either observed, calculated or a combination of both.

Jennings Randolph Lake, also commonly referred to as JRL or Randolph, serves as both the main reservoir and dam on the North Branch. It straddles the Maryland and West Virginia state lines, placing the reservoir in both Garrett County, Maryland and Mineral County, West Virginia. The dam is located 58 miles upstream of the confluence of the North Branch and South Branch Potomac Rivers. JRL is operated by the USACE, Baltimore District (NAB), with the authorized project purposes of flood risk management, domestic and industrial water supply, water quality control, and recreation. Regulation of Randolph was historically based, in part, on poor water quality conditions in the watershed from acid mine drainage (AMD) and industrial uses. Water quality and other watershed conditions have improved since JRL was constructed, which suggests that re-evaluation of how the reservoir's water quality storage is regulated is warranted. For the purposes of this report, North Branch Potomac River will also be referenced as the North Branch or NBPR.

The goal of this effort is to evaluate the existing release philosophies for Randolph, identify current downstream environments and environmental needs, and identify any areas of potential re-evaluation which could result in revisions to the JRL Reservoir Regulation Manual (RRM). Historically, there have been water quality issues in the watershed, and regulation plans such as Artificially Varied Flow (AVF) releases were implemented to address environmental problems, which will be discussed in further detail later in this report. AVF releases were developed as short-term high-flow releases of at least 1,000 cubic feet per second (cfs) made in the late summer months to flush organic solids that settle on the river bottom. AVF releases were implemented as an environmental strategy in the 1980s whereas environmental flows (e-flows) are a contemporary protocol based on detailed input to hydrologic models of needs for species and may help identify areas of refinement of reservoir regulation. Multiple stakeholders have

differing opinions on the most effective way to simultaneously utilize the JRL releases for downstream fisheries and recreational activities.

The next steps following this report are: (1) Provide the State of the Science Report to workshop participants, (2) conduct a face-to-face stakeholder workshop to evaluate and identify environmental opportunities to sustain or improve the ecosystem health for the North Branch Potomac River below Randolph, and (3) complete a workshop summary report to summarize the recommendations from the workshop.

Results and recommendations identified via the SRP process are expected to have a positive and long-term impact to ensure the best use of the water quality storage in JRL. This will benefit environmental factors which could include improved streambed health and flows for water temperature management for temperature sensitive fish species downstream. The metrics evaluated, data interpreted, and recommended operational changes will all inform the RRM update, especially for the water quality authorized purpose.

1.2. Study Area

The study area is the segment of the North Branch Potomac River basin from downstream of Jennings Randolph Dam to the United States Geological Survey (USGS) gage at Luke, MD. The North Branch Potomac River runs through steep hilly terrain, and the watershed is mostly undeveloped and agricultural land. There is a history of mining and logging in the region. Towns along the river (from upstream to downstream) include Kitzmiller, MD; Barnum, WV; Luke, MD; McCoole, MD; Pinto, MD; and Cumberland, MD. Cumberland is the largest population center with about 18,700 people. There was a papermill located in Luke that shut down in 2019, which was one of the industrial point sources in the watershed.

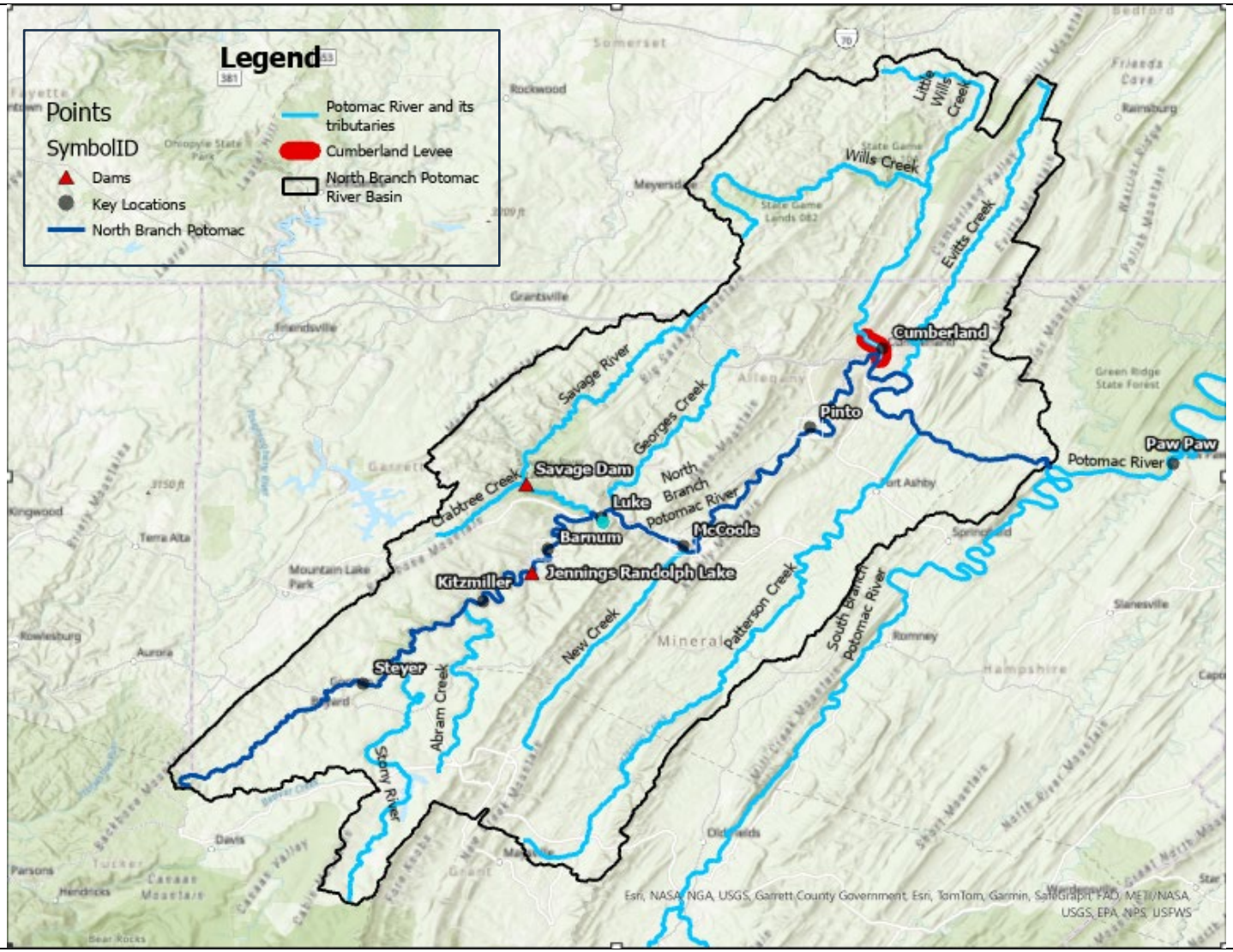
The drainage area upstream of Jennings Randolph Lake is 263 square miles, and the watershed is roughly rectangular in shape - about 23 miles long and 12 miles wide. The highest elevation, approximately 4,150 feet National Geodetic Vertical Datum of 1929 (NGVD29), is located on Cabin Mountain in the headwaters of Stony River. From its source at elevation 3,150 feet NGVD29, the North Branch Potomac River descends to an elevation of about 1,220 feet NGVD29 at the damsite, approximately 36 miles. Except for the first mile which is steep, slopes vary from 12 feet to 63 feet per mile. Within the reservoir, the slope is about 44 feet per mile. Valley storage of flood waters is small because of steep stream slope and narrow valleys. The watershed contains no natural lakes and only a few small marshy areas. Mount Storm Reservoir, owned by Dominion Energy, is located on Stony River, along with a drained and abandoned reservoir called Stony River Reservoir. The principal tributaries of the North Branch above Jennings Randolph Lake are Stony River and Abrams Creek.

Downstream of the dam, six major tributaries, Savage River; Georges Creek; New Creek; Wills Creek; Evitts Creek; and Patterson Creek, enter the NBPR before its confluence with the South

Branch Potomac River. Savage River enters about 7 miles downstream. Savage River Dam, located on Savage River about 5 miles upstream from its confluence with the NBPR, controls 105 of the 120 square mile drainage area of the Savage River. Georges Creek enters the NBPR at Westernport, MD, and New Creek enters at Keyser, WV. Wills Creek enters the NBPR at Cumberland, MD, and Evitts Creek and Patterson Creek enter the NBPR below Cumberland. At Oldtown, MD, the North and South Branches join to form the main stem of the Potomac River. The total drainage of the NBPR is 1,328 square miles (Baltimore District, 1997).

Figure 1.1 - Potomac Watershed Map

Figure 1.2 - Study Area: North Branch Potomac Basin



1.2.1. History of Jennings Randolph Reservoir

1.2.1.1. Project Description and Authorization

Jennings Randolph Lake, previously named Bloomington Lake, has a surface area of 915 acres and storage of 90,313 ac-ft at conservation pool and is owned and operated by USACE. The normal pool or conservation pool is typically defined as to the desired water level or storage capacity maintained in the reservoir under normal operating conditions. JRL was authorized by the Flood Control Act of 1962 - Public Law 874, 87th Congress, 2nd Session - for the purposes of flood control, domestic and industrial water supply, water quality, and recreation. The project was subsequently renamed Jennings Randolph Lake in May 1987 in honor of the longtime West Virginia senator. Whitewater recreation was added as a project purpose by the Water Resources Development Act of 1988. The primary flood control objective of the Jennings Randolph Lake project is to reduce flood crests and associated damages along the North Branch Potomac River (Baltimore District, 1997).

1.2.1.2. Related Projects

Jennings Randolph Lake is regulated in conjunction with the nearby Savage River Dam to provide flow augmentation at Luke, Maryland and to furnish the best possible water quality downstream throughout the year. Savage River Dam is owned and operated by the Upper Potomac River Commission (UPRC) but is regulated in conjunction with Randolph by USACE for downstream water management purposes (Baltimore District, 1997).

1.2.1.3. Original Reservoir Regulation Plan Development

The preliminary plan for project regulation, based on conditions and policies applicable at the time of authorization in 1962, was to maintain a minimum flow of at least 305 cfs at Luke, Maryland throughout the summer and fall months. As developed in the various design memoranda prepared during the 1960's, this value represented the combined dependable yield from Jennings Randolph and Savage River Reservoirs through the worst drought (1930-1931) in the period of record. The computations that derived 305 cfs as the dependable yield treated the entire 92,000 acre-feet of Jennings Randolph Lake's conservation storage as available for low flow augmentation.

During the 1970’s and 1980’s when JRL was being constructed, additional studies were performed concerning the water resources of the Potomac River Basin. Most notably, the Corps' Metropolitan Washington Area Water Supply Study demonstrated that significant increases in water supply availability could be achieved by wise use of storage in JRL, i.e., releases based on need rather than on satisfying a constant minimum flow target of 305

cfs. Analyses also indicated that acid mine drainage and municipal/industrial wastewater discharges were not degrading the water quality of the North Branch as much as previously thought, and Federal policies had changed to require treatment of pollution at its source rather than relying upon the assimilative capacity of the stream. Taken together, these factors led to a reassessment of the regulation criteria for Jennings Randolph Lake and Savage River Dam.

Additionally, the remaining storage (40,194 acre-feet) in Jennings Randolph Lake was purchased for water supply by the users in the Metropolitan Washington Area in 1982. The reassessment process performed over the course of several years, eventually resulted in an updated water control plan. The plan provides for varying the flow at Luke daily, if necessary, to meet established water quality standards at the location, but with a minimum target of 120 cfs. Releases from Randolph to achieve the water quality standards at Luke are taken from the project's water quality storage. Additional releases, which are requested by the Interstate Commission of Potomac River Basin (ICPRB) on behalf of downstream water suppliers, are taken from water supply storage. Further water quality improvements in the lake's inflow continue to occur as various agencies aggressively seek to reduce the adverse impacts of acid mine drainage from both active and inactive mines.

Following project completion, it was observed that water in the stilling basin and immediately downstream sometimes exhibited elevated saturation levels of dissolved oxygen during times of high releases. A severe fish kill occurred in May 1990 in the tailwater area, where MD DNR had fish rearing net pens located at the time, when the discharges approached 3,500 cfs. Further investigations concluded that dissolved oxygen saturation levels in the outflow approach the saturation point when the discharge reaches about 1,000 cfs and approaches 115% when the discharge reaches 2,500 cfs. At such levels, supersaturation causes damage to fish and ultimately results in fish kills if exposure is of an extended duration. As a result, current reservoir regulation procedures attempt to limit releases below 2,000 cfs, with 2,500 cfs as an upper limit, unless hydrologic conditions indicate that greater releases must be made. Even though the fish pens have since been removed, MD DNR still believes this limit is beneficial for the fish population in and near the tailwater area.

While the project’s regulation for flood control and water supply have remained mostly unchanged over the years, recreation and water quality needs have grown and changed. When the project was first completed, due to the poor water quality, no one foresaw recreation being a large component of Jennings Randolph Lake and the North Branch Potomac. Due to multiple watershed wide factors, the water quality has greatly improved, providing opportunities for recreation, both in-lake and downstream, to grow. Regulation of the project continues to be a balance between competing needs.

As the water quality has improved, the management strategy applied has changed from releasing as much volume as possible to using the water more strategically to provide the most benefits. The Baltimore District in coordination with MD DNR developed new regulation strategies. This includes managing the temperature and quantity of flow which helps maintain a cold-water fishery downstream. Using the intake tower's selective withdrawal system, a release temperature of 13+/- 0.5°C is maintained during the summer months when not superseded by other project purposes.

Through discussions with downstream users during the North Branch Potomac River Advisory Committee (NBPAC) meetings (created in 2005) flow ranges for different recreational activities were established. These ranges define the river conditions needed for wade fishing, float fishing, and whitewater recreation. These recreational benefits are considered when making release decisions and are balanced with the other competing needs, both in-lake and downstream (Baltimore District, 1997).

1.2.2. Existing Water Regulation Strategies

The Water Management section in the Baltimore District owns and operates Jennings Randolph and regulates the reservoir behind the dam. Numerous types of data, including flow, stage, reservoir elevation, reservoir temperature, and water quality data are collected within the watershed. A reservoir regulation manual prescribes how the reservoir is to be regulated for its authorized purposes. Releases are determined by the district Water Managers and Dam Operators based on current conditions, weather forecasts, and Corps Water Management System (CWMS) model forecasts to meet the project purposes of flood control, domestic and industrial water supply, water quality, and recreation.

Figure 1.3 illustrates the lake elevations and storage within the lake for which Jennings Randolph Lake is regulated. Jennings Randolph Lake has a conservation (or summer) pool of 1,466 ft PCD (Project Construction Datum). Storage above this elevation, and up to elevation 1,500 ft is available for storing excess water during high water events. The top of the dam is at elevation 1,514 ft PCD. The conservation pool (storage below 1466 ft) is allocated for water quality and water supply. Releases from the dam can be made from several different ways. The outlet works include both quality control and flood control gates. The quality control (QC) gates allow for smaller releases along with a selective withdrawal capability. Releases from various elevations of the reservoir can be made through this selective withdraw system to regulate releases primarily for water temperature, but also historically for parameters such as pH. The flood gates are larger gates which allow for higher releases to be made. Releases can also be made from the gated spillway but are typically only needed during very large and extreme flood events.

Figure 1.3 - Jennings Randolph Lake and Dam Diagram

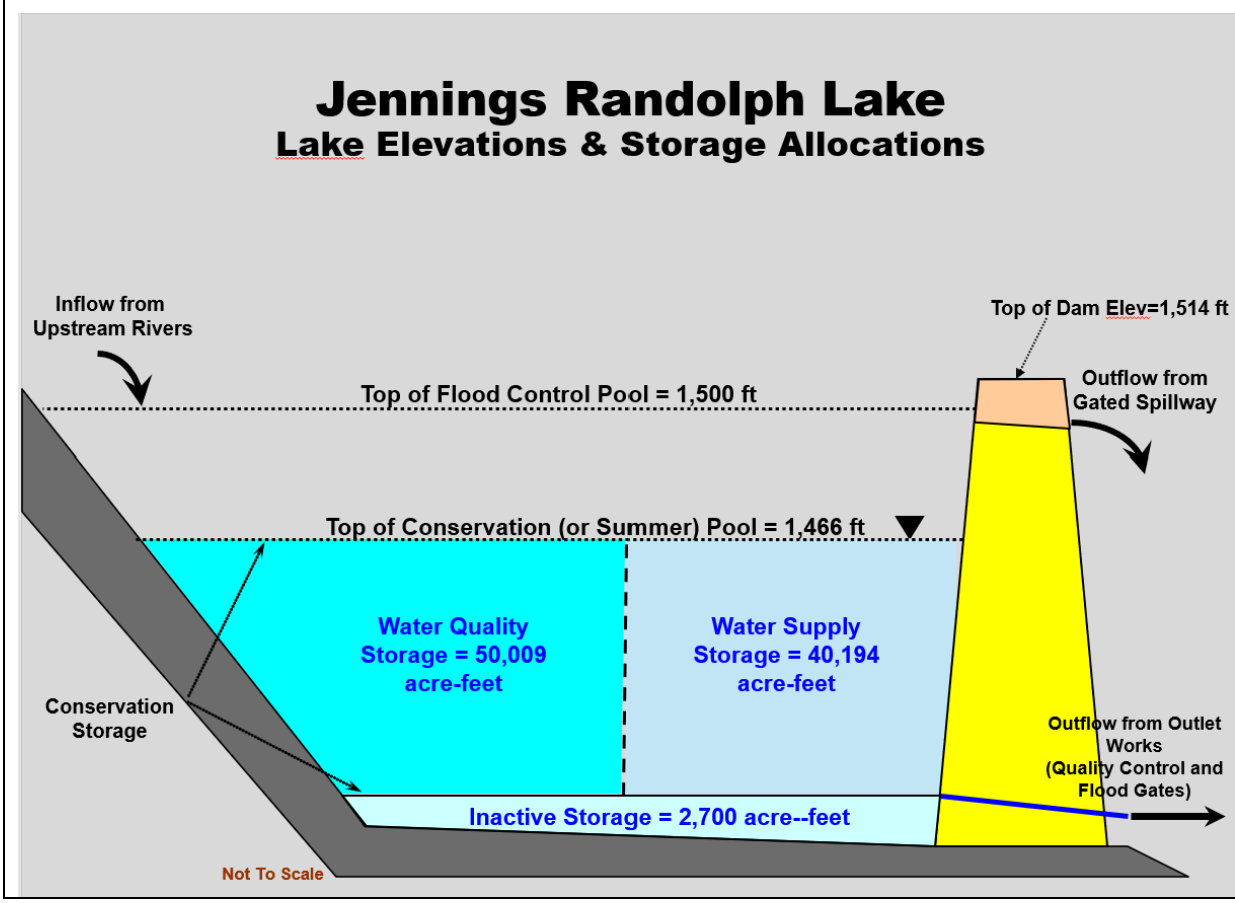
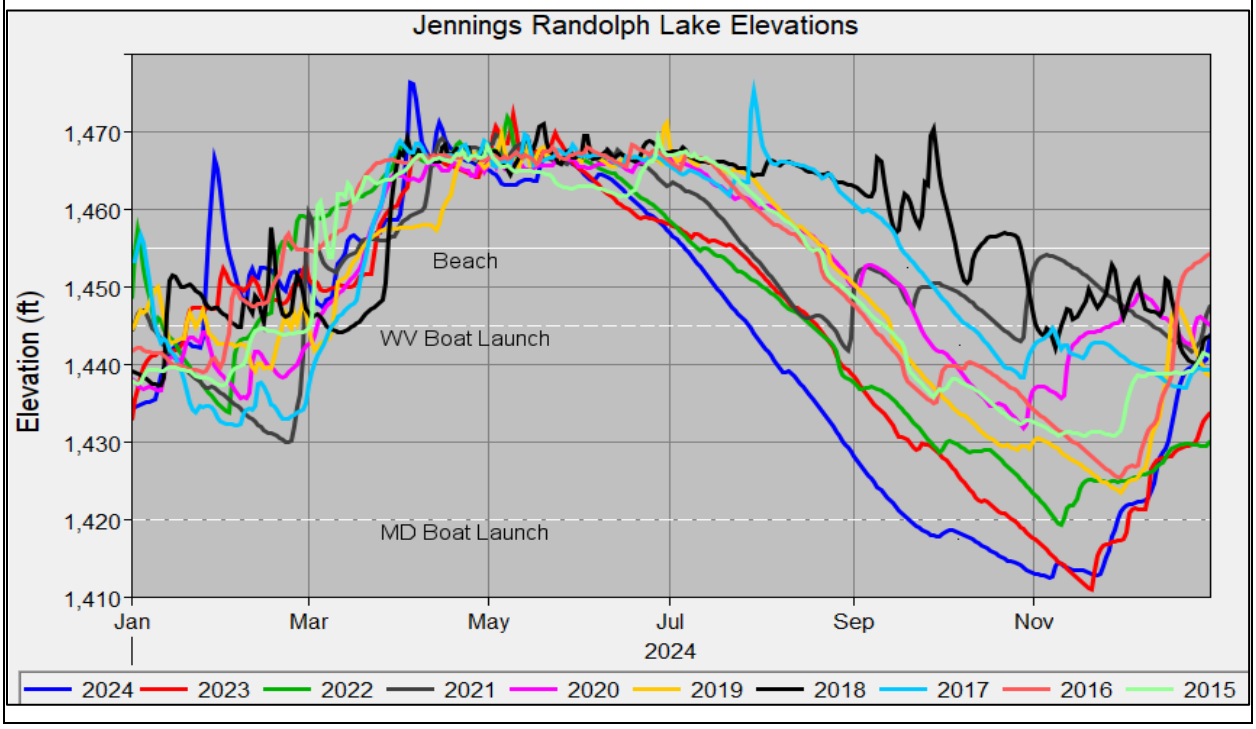


Figure 1.4 shows a plot of the lake elevation for Jennings Randolph over the course of a year over the past 10 years. While the conservation pool for Jennings Randolph Lake is at elevation 1,466 ft PCD, the lake elevation typically varies seasonally, particularly during the summer

months when the lake tends to have a falling trend while water quality releases are made that are greater than the flow coming into the lake; and then a rising trend in the spring when the lake is refilled from snowmelt and/or spring runoff events back to the conservation (or summer) pool.

Figure 1.4 - Jennings Randolph Lake Elevation 10 year Comparison Plot



1.2.3. Flood Risk Management

Jennings Randolph Lake is a multi-purpose project, with flood control as the highest priority mission. The objective of the JRL project for flood risk management is to store water during flood events, thereby reducing river stages (and thus associated damages) below the dam along the North Branch Potomac River during flood events. Communities benefiting from flood control regulation by JRL include Luke, Westernport, Piedmont, Keyser, Cumberland, Ridgeley, and Oldtown. Additionally, communities such as Paw Paw and Hancock further downstream along the main stem of the Potomac River receive some benefit from flood control regulation.

1.2.4. Water Quality

The water quality of the North Branch Potomac River at the time of project completion ranged from poor to good quality depending upon the river reaches. The major pollution

sources in the basin were AMD from abandoned deep and strip mines and, to a lesser extent, poorly treated wastes from cities, towns, and industries scattered throughout the basin.

Of JRL's total conservation storage, 51,019 acre-feet (55.4%) is allocated to water quality storage. The water quality storage has historically been used, to manage releases for pH, water temperature, and overall pollutants and improve downstream water quality conditions. Through careful regulation, the project has had a positive impact on the water quality downstream by following these general objectives for water quality control:

- Improve pH
- Manage water temperature
- Achieve uniformity in water quality releases
- Dilute wastewater discharges from downstream municipal and industrial users
- Augment North Branch flows during low flow seasons
- Maintain a healthy downstream aquatic habitat

As the downstream water quality improved, the aquatic habitat also improved and now hosts a highly valued trout population. Due to these improvements, the main focus for water quality management over recent years has transitioned from a focus on water quantity to improve water quality, to a more temperature-based focus in which release of the cold-water storage in the reservoir is more conservative. This theoretically allows for a longer availability of cold water through the summer and early fall. A temperature of around 13°C is released from the project in the spring, summer, and fall. These cold releases have helped build and maintain the cold-water fishery downstream over the years.

When the outflows from Jennings Randolph Lake have been maintained at low uniform rates for an extended period, suspended organic solids have been found to settle and accumulate on the riverbed. Such accumulations tend to deplete dissolved oxygen in the water, smother benthic organisms, and generally lower the quality of the aquatic habitat. The accumulations are a particular problem between Westernport and Cumberland. To help minimize the potential adverse effects during extended low flow periods, Artificially Varied Flow (AVF) releases are made. AVF is a regulation tool for removing accumulated organic sediments, thus improving the downstream aquatic environment. During an AVF release, the outflow from Randolph is varied at a higher rate. AVF releases were originally designed to be made periodically from June through September, depending upon weather and river conditions. However, most years they have only been made twice a year, in August and September.

1.2.5. Water Supply

A group of municipal water supply entities composed of the District of Columbia, the Washington Suburban Sanitary Commission, and the Fairfax County Water Authority (Users) have purchased 40,194 acre-feet of water supply storage in Jennings Randolph Lake. The users determine and request a rate of release whenever there is a need for additional water supply in the Metropolitan Washington Area. Currently, the Cooperative Water Supply Operations on the Potomac (CO-OP) in the Interstate Commission on the Potomac River Basin

(ICPRB) requests releases for the entire group. The release rate may be adjusted at their discretion and is made as requested, unless it is determined that the release will be detrimental to the downstream environment and not in the overall public interest.

Whenever the CO-OP requests a release from water supply storage, USACE monitors the water supply release at the U.S. Geological Survey stream gage located just below the dam at Barnum, WV. The users are notified via the ICPRB when there is 50% or less of the water supply storage remaining and when that storage will be depleted at the current rate of withdrawal. Should the purchased water supply storage be depleted, no additional release for water supply purposes will be made. As the lake is refilled after all or a portion of the water supply storage has been utilized, the amount of storage available for water supply will be regained at a rate proportional to the rate of filling. Due to contractual obligations to the users, the water supply storage will be utilized for purposes other than water supply only in the most extreme circumstances when it is determined that the alternate use is in the overall public interest. Examples of such circumstances are: (1) a major industrial accident downstream when there might be a need for a substantial release for dilution purposes and for the protection of life and property; or (2) a severe industrial accident upstream of the lake, when it would be dangerous to make releases from the project. When a release for water supply storage is requested, the total flow at Luke is composed of the water quality flow of 120 cfs plus the water supply release. The water quality flow will be released from JRL and Savage River Dam. The water control manager will decide how much each reservoir may release for water quality during the water supply release period. The decision is made based on available and projected storage and quality. The regulation objective is to maintain overall water quality at or near the pre-water supply release condition (Baltimore District, 1997).

1.2.6. Recreation

Recreation is a major component of Jennings Randolph Lake and the North Branch Potomac River. Recreation is a secondary use of the project's lands and waters. Recreational users are plentiful both in-lake and downstream where they enjoy boating, swimming, and fishing (Gallup 2019). In-lake recreation season typically extends from May – August when the summer pool has an elevation around 1,466 feet Project Construction Datum (PCD).

Jennings Randolph recreation operations involve whitewater recreational activity and fishing. Upon request by the Mineral County Department of Parks and Recreation, WV, whitewater releases are scheduled over four alternating weekends in April and May. About 800 cfs to 1,000 cfs are released for this purpose. Whitewater releases are made only when the reservoir exceeds elevation 1,463 feet in April and May. The whitewater releases may be cancelled if the pool elevation falls below 1,463 feet. When AVF releases are made in the summer and fall, whitewater enthusiasts may also utilize these releases. Releases of about 1,000 cfs are made for this purpose. A desirable flow for fishing downstream of the dam is around 400 cfs. If possible, daytime releases are held at or below 400 cfs to assist fisherman access (Baltimore District, 1997).

1.3. Knowledge Gaps

The study area does not have extensive and current studies available on environmental flow recommendations. The middle and lower Potomac River and estuary have been more heavily researched. Environmental flows have not been established for the North Branch. Drought and climate change studies are also lacking. A 2014 study of the Middle Potomac found, "Biological data collected in comparable ways in the largest river segments of the Potomac are presently insufficient for flow needs assessments. Independent development of Flow Alteration-Ecology (FA-E) relationships, such as those for macroinvertebrates in streams and small rivers, is not possible at this time" (U.S. Army Corps of Engineers et al., 2014). Additionally, knowledge gaps are listed in Table 1.1.

Table 1.1 - Knowledge Gaps

Flow effects in the North Branch Potomac	• Drought effects • Vegetation • Floodplain connectivity • Riparian forests • Ecological response to flow
Water quality	• Sedimentation in river • Improvements needed besides pH • DO concentrations in organic solid sediments on river bottom. • Changes since 2019 closure of Luke Paper Mill
Ecology	• Mussel/clam (bivalve) surveys • Effect of flows on invasive species
Groundwater	• Groundwater withdrawal effects • Groundwater dominant areas of the river, injecting cold-water

Chapter 2. Existing Conditions

2.1. History of Environmental Strategies in the North Branch Potomac River

The North Branch Potomac River watershed has a long history of environmental impairment. Natural resource extraction activities for coal and wood products have long impacted the North Branch of the Potomac River. Acid mine drainage lowered the river's pH which is detrimental to aquatic life, while effluent from a pulp and paper mill in Luke, Maryland and wastewater from Luke and Westernport in Maryland, and Piedmont, West Virginia contributed further to impairment. Water sampling in the late 1960s showed consistently low pH in the North Branch, with readings consistently below 4.5 and as low as 2.3, well below a healthy river's pH of around 6.5 (Hansen et. Al, 2010).

In 1987, West Virginia and Maryland signed a cooperative agreement to develop a plan for AMD abatement. One measure pursued was the installation of lime dosers on low pH systems, the first of which was installed in 1992. See Figure 4.2 for locations of the lime dosers. This coincided with the first whitewater boating releases from Jennings Randolph Lake. This also followed documented natural reproduction of trout in the lower North Branch Potomac in 1991. Fish populations rebounded through the 1990s. During the 2000s, tens of thousands of trout were stocked in the North Branch and by 2010 eight dosers were operational. Some improvement of the pH in the river system may also be attributed to reductions in acid deposition from precipitation as a result of the Clean Air Act (1970) and other regulations (ICPRB 2019). Other water quality improvement may be attributable to other AMD efforts and the federal Surface Mining Control and Reclamation Act.

AMD impairment was the primary water quality concern in the North Branch, but industrial discharge from the paper mill in Luke contributed to poor water quality. Since 1960, the Upper Potomac River Commission has operated the Westernport Wastewater Treatment Facility, treating the industrial waste from the paper mill as well as municipal sewage from Westernport, Luke, and Piedmont. Improvement to water treatment has been added over its operational life, including a secondary clarifier to settle out solids added at the paper mill in 1991. Production at the mill ceased on May 31, 2019. In April 2021, the owners of the closed mill agreed to clean up "black liquor", a byproduct of the paper pulping process, and contaminants associated coal ash waste that had entered the river.

Partly in response to water quality issues in the watershed, JRL was authorized to provide for water quality control in the North Branch Potomac River, as well as industrial and municipal water supply, flood control protection, and recreation. Similarly, the Savage River Dam (Savage), constructed in 1951 on the Savage River which enters the North Branch just above Luke, was constructed for the purposes of augmentation of stream flows for industrial supply and pollution abatement, flood control, and municipal water supply for Westernport.

Dam operations at Randolph have used various approaches to achieve the specific objectives identified in the Reservoir Regulation Manual. During high flow events, releases from JRL and nearby Savage River Dam are both limited to reduce the risk of flooding in downstream riverside towns, including Bloomington, Luke, Westernport, McCoole, and Cumberland in Maryland; and Piedmont, Keyser, and Ridgeley in West Virginia (Figure 1.2). During low flow periods, JRL releases increase downstream water levels and flows for the purpose of diluting downstream pollution. JRL stratifies thermally and chemically, and a selective withdrawal system with ports at different reservoir depths allows JRL dam operators to influence downstream river temperatures. During severe droughts, in coordination with the Interstate Commission on the Potomac River Basin (ICPRB), water is released from Randolph to help supply the drinking water needs of metropolitan Washington, D. C. The ICPRB represents the metro area water suppliers and channels requests for releases through the USACE's Water Management Section. These requests are infrequent, with the last two occurring in 2002 and 2010.

As mentioned previously, AVFs are a water quality strategy where releases are increased to a specified flow, usually 1,000 cfs, and held for 24 to 48 hours. Their purpose is to flush organic solids that settle to the river bottom. Historically, industrial activity was the primary source of the organic solids, but solids have been decreasing over time due to watershed improvements. In addition, AVF flows may flush out didymo algae (*Didymosphenia geminate*), an invasive species. Didymo coats the bottom of rivers and smothers the habitat and food supply of fish (Gallup, 2019).

In late May 1990, severe trout fish kills were observed in fish-rearing pens within the stilling basin at JRL. The project was discharging around 4,200 cfs at the time. It was first suspected that chemical toxicity due high aluminum and pH levels caused the fish kill, but it was later determined that the fish mortality resulted from prolonged exposure to supersaturated gas. External symptoms included exophthalmia or pop-eye, bubbles in the caudal fin, hemorrhage, and emboli in gill blood vessels. Operational changes were made to reduce gas supersaturation by controlling the discharge rate and frequency of high-volume discharges. The fish pens were removed in 2007, but supersaturation is still managed by keeping the discharge below 2,000 cfs when possible, with 25,000 cfs as an upper limit. If releases exceeding these amounts are needed, the duration is limited to the extent possible.

On December 2, 1993, Governor William Donald Schaefer of Maryland, Governor Gaston C. Caperton of West Virginia, and Chairperson Phyllis M. Cole of the Interstate Commission on the Potomac River Basin (ICPRB) entered into an agreement to form a task force for the improvement of water quality and recreational opportunities on the North Branch of the Potomac River. The Task Force included representatives of state and local governments, the U.S. Army Corps of Engineers, the ICPRB, and the private sector. The Flowing to the Future restoration plan for the North Branch was adopted in 1995 and revised in 1998. The plan envisioned the North Branch watershed ten years from the plan's adoption, with a vision of a healthy watershed supporting a high-quality nationally recognized trout fishery and providing diverse recreational opportunities.

Five goals with multiple objectives and action items helped guide actions through the plan's implementation.

In 2005, the North Branch Potomac River Advisory Committee was formed by agencies and stakeholders with mutual interest in the operations and management of the Jennings Randolph and Savage River Dams as well as the lands surrounding those reservoirs. The NBPAC meets to review operations and identify areas for improved management of the projects. The diverse stakeholder representation in the NBPAC helps operators maximize the use and quality of downstream resources while maintaining the project purposes of flood control, water quality improvement, and drinking water storage.

A series of stakeholder meetings in 2007 and 2008, facilitated by ICPRB, were convened to develop a shared vision for the North Branch watershed.

2.2. Fishery Profile

Since it was established, the trout fishery in the NBPR has continued to improve and grow in popularity. It has become a destination fishery for anglers hoping to catch numerous trout with the chance for trophy fish as well. Currently, there are several special trout management areas to provide a variety of angling opportunities, including Catch and Release - Artificial Lures Only, Put and Take, and Catch and Release - All Tackle (formerly Zero Creel) regulation areas. As the popularity of this fishery continues to grow, managers remain diligent in the monitoring of this resource to maintain the quality of the trout fishery and manage it to its full potential.

2.2.1. Trout Fishery

The trout fishery in the NBPR tailwater is managed in three ways: Put and Take, Put and Grow, and Wild Trout. Put and Take management areas are stocked annually in both Maryland and West Virginia with adult hatchery trout (browns and rainbows) for the purpose of catch and harvest. Areas managed as Put and Grow are stocked on multiple occasions annually with fingerling brown and rainbow trout that grow to adulthood in the river. Wild trout are self-sustaining and contribute to the fishery through natural reproduction. Each fishery is impacted by the annual flow regime in the NBPR, often benefiting from more stable conditions.

2.2.1.1. Put and Take

Put and Take management efforts are implemented for anglers to catch and harvest adult trout produced in state-owned aquaculture facilities. This is a renewable fishery, as fish are stocked annually. Although flow regimes can impact angler attitudes and success when targeting stocked adult trout, they are not necessary to maintain the fishery.

2.2.1.2. Put and Grow

Fingerling brown and rainbow trout are stocked into the NBPR annually to supplement the wild trout fishery. Put and grow fingerlings grow in the river to adult size, where they become part of the overall fishery, looking and behaving the same as wild trout. The survival of fingerling trout is not as closely tied to flow during earlier life stages, but stable conditions are beneficial for recruitment to adulthood.

2.2.1.3. Wild Trout

Brook, brown and rainbow trout all successfully reproduce in the NBPR. Brook and brown trout spawn during the fall, typically from mid-October through November. Rainbow trout spawn during late winter or early spring, sometimes as early as mid-February. All species spawn over cobble/gravel (marble to pea size) with upwelling flows and/or in shallow riffle areas with constant flow that keeps the eggs oxygenated and free of sediment.

Variations in annual flow regimes can have drastic impacts on the spawning success of wild trout populations in free-flowing rivers. Due to generally stable flow conditions in tailwaters, trout reproduction is often more consistent than in un-manipulated systems. Spring flooding, summer drought and extreme events during the spawn can have detrimental impacts to a cohort of trout. Likewise, flooding events during the fry life stage can have catastrophic impacts to a year-class. Work in the nearby Savage River tailwater wild trout fishery supports the body of scientific evidence that maintaining stable, elevated flows during spawning and egg incubation periods are strongly correlated with spawning success. These conditions provide the constant flow necessary to keep eggs oxygenated and free of sediment while avoiding the dewatering of redds. Likewise, the survival of fry is positively correlated with stable flows but negatively correlated with high water events (i.e. floods) due to stranding and/or physical injury.

The ability to store and strategically release cold water from Jennings Randolph Reservoir is paramount to the success and viability of the tailwater trout fisheries in the NBPR as well as the benthic macroinvertebrate communities that are the backbone of the trophic web. Efforts by the USACE have allowed for the effective management of trout fisheries as much as 30 miles downstream of the reservoir, while maintaining the multiple-use objectives of the Project. Working with Maryland Freshwater Fisheries and Hatcheries staff, further refinement of the temperature enhancement efforts have been accomplished, maximizing the cold-water fisheries in the river. Likewise, issues leading to trout mortality from gas supersaturation at releases above 2,000 cubic feet per second have largely been avoided (MDNR, 2025).

Chapter 3. Watershed Characteristics

3.1. Watershed Climate

Jennings Randolph Lake is within the North Branch Potomac River basin. This area is characterized by a temperate climate, with the average annual temperature in 2025 being between 45°F and 70°F and the average annual precipitation being between 26 and 76 inches (NOAA, n.d.). The greatest monthly precipitation in the basin occurs from May through August; the least occurs in the late fall and winter. The winters are not severe, but there usually is heavy snowfall (Gallup, 2019).

3.2. Physiography

Jennings Randolph Lake is located within the Central Appalachian level IV ecoregion which stretches from the Blue Ridge Mountains in the southern United States to the Allegheny Mountains of Virginia, West Virginia, and Maryland. It is composed of high, dissected, rugged plateau of sandstone, shale, conglomerate, and coal. Due to the rugged terrain, cool climate, and infertile soils agriculture is limited, and it is mostly forested land cover.

JRL is in the North Branch watershed and the Allegheny Plateau. The Allegheny Plateau is a high, deeply dissected plateau bounded by an eastward-facing escarpment known as the Allegheny Front. Elevation in the vicinity of JRL ranges from approximately 1,500 feet NGVD29 to 2,100 feet NGVD29.

The bedrock is sedimentary creating branched out streams. The valleys slope toward the center of the basin. The westernmost basin is drained by the Savage River, which is joined by the North Branch Potomac River, Georges Creek, and New Creek. The Jennings Randolph project lands have generally steep slopes, usually over 10% (Gallup, 2019).

3.3. Hydrology

Streamflow statistics were calculated in the Indicators for Hydrologic Alteration (IHA) program for the historical streamflow data for the following USGS gages on the North Branch Potomac: Bloomington, Kitzmiller, Barnum, Luke, Pinto, Steyer, and Cumberland (Figure 1.2). These gages were chosen due to being within the dam's reach of flow influence for the purposes of flood control and having robust and consistent data available. Flow quartile calculations were done for the aforementioned gages as well as Hancock and Paw Paw. Hancock and Paw Paw were not included in the IHA analysis due to the data set not being extensive enough for the purposes of this report.

The IHA is a software program that provides statistical analysis of daily flow data, to help understand the hydrologic impacts of human activities or to develop environmental flow recommendations for water managers (TNC, n.d). "The software program assesses 67 ecologically relevant statistics derived from daily hydrologic data. For instance, the IHA software can calculate the timing and maximum flow of each year's largest flood or lowest flows, then calculate the mean and variance of these values over some period of time. Comparative analysis can then help statistically describe how these patterns have changed for a particular river or lake, due to abrupt impacts such as dam construction or more gradual trends associated with land- and water-use changes." (TNC, n.d.)

3.3.1. USGS Gage Data

The USGS provides streamflow data from gages along the Potomac River. The following section discusses key gages on the NBPR.

Computed inflow and outflow data is available from 1984 onward for Jennings Randolph Dam, calculated by USACE. Daily stream flow dates for the other pertinent gages are in Table 3.1. There was a streamflow gage located approximately at the site of the current Jennings Randolph Dam called Bloomington that was discontinued. It provides the direct pre-dam inflow data from 1924 to 1948. The Kitzmiller gage is further upstream of the gage site and has data from 1949 to the present. The most immediate downstream gage from the dam is Barnum (1966 to the present). Table 3.1 contains the relevant gages and their periods of record.

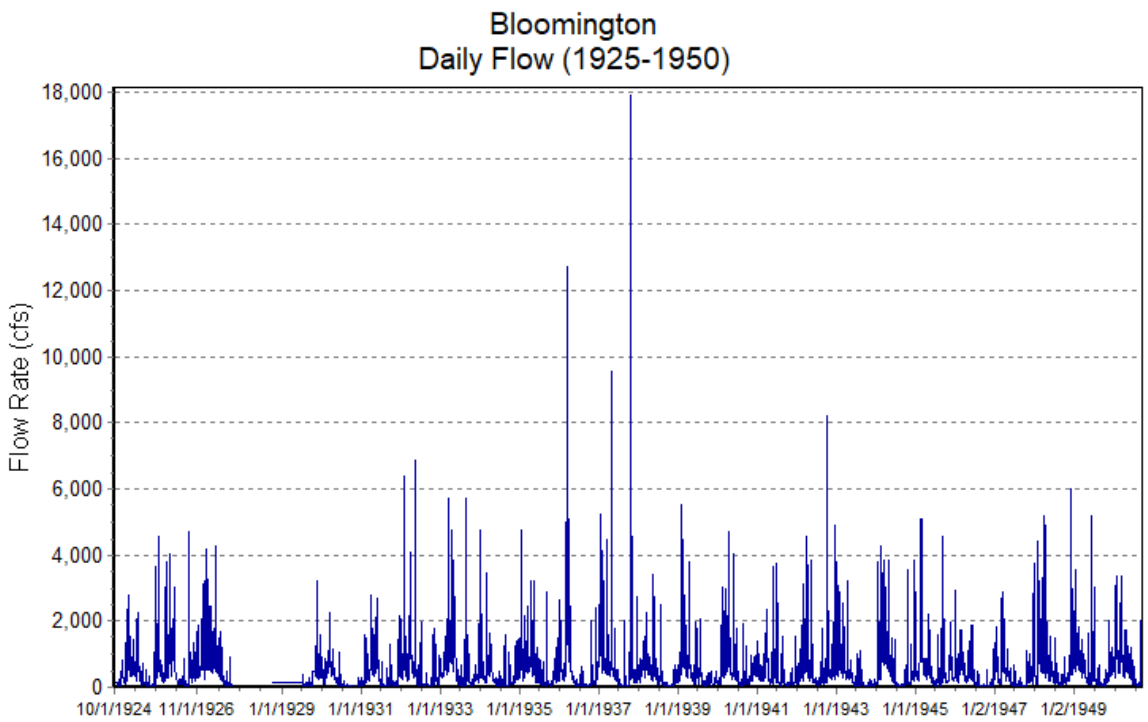
Table 3.1 - North Branch Potomac Gages

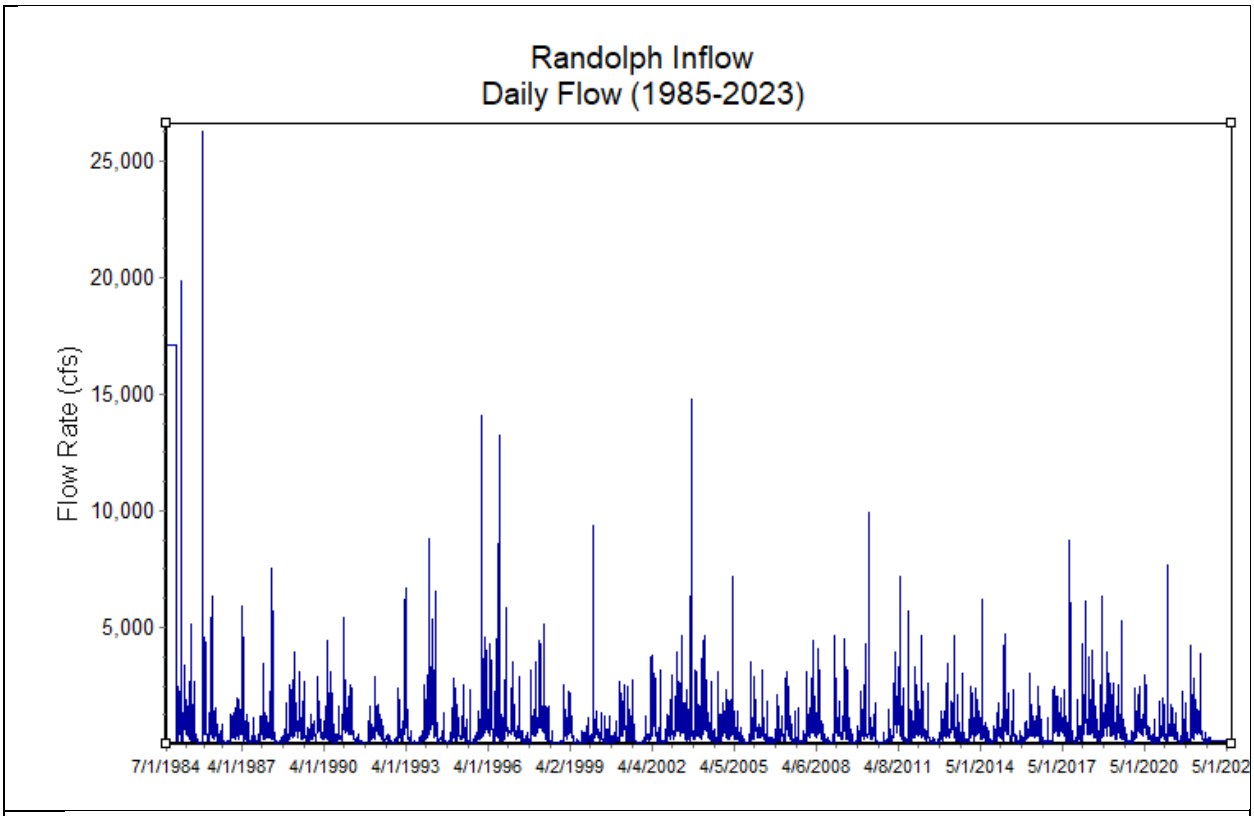
Gage	Location relative to JRL	USGS Gage ID	Period of record	Date of last data point used in analysis
Bloomington	Near current dam site	01596000	1 October 1924 to 22 December 1948	--------
Kitzmiller	Upstream	01595500	October 1949 to present	13 December 2022
Barnum	Downstream	01595800	2 July 1966 to present	13 December 2022
Luke	Downstream	01598500	27 June 1899 to 15 July 1906, 1 Oct 1949 to present	26 February 2023
Pinto	Downstream	01600000	2 October 1938 to 30 Sep 2008 *Intermittent data gaps in set	--------
Steyer	Upstream	01595000	1 July 1956 to present	26 February 2023
Cumberland	Downstream	01603000	24 May 1929 to present	24 December 2022

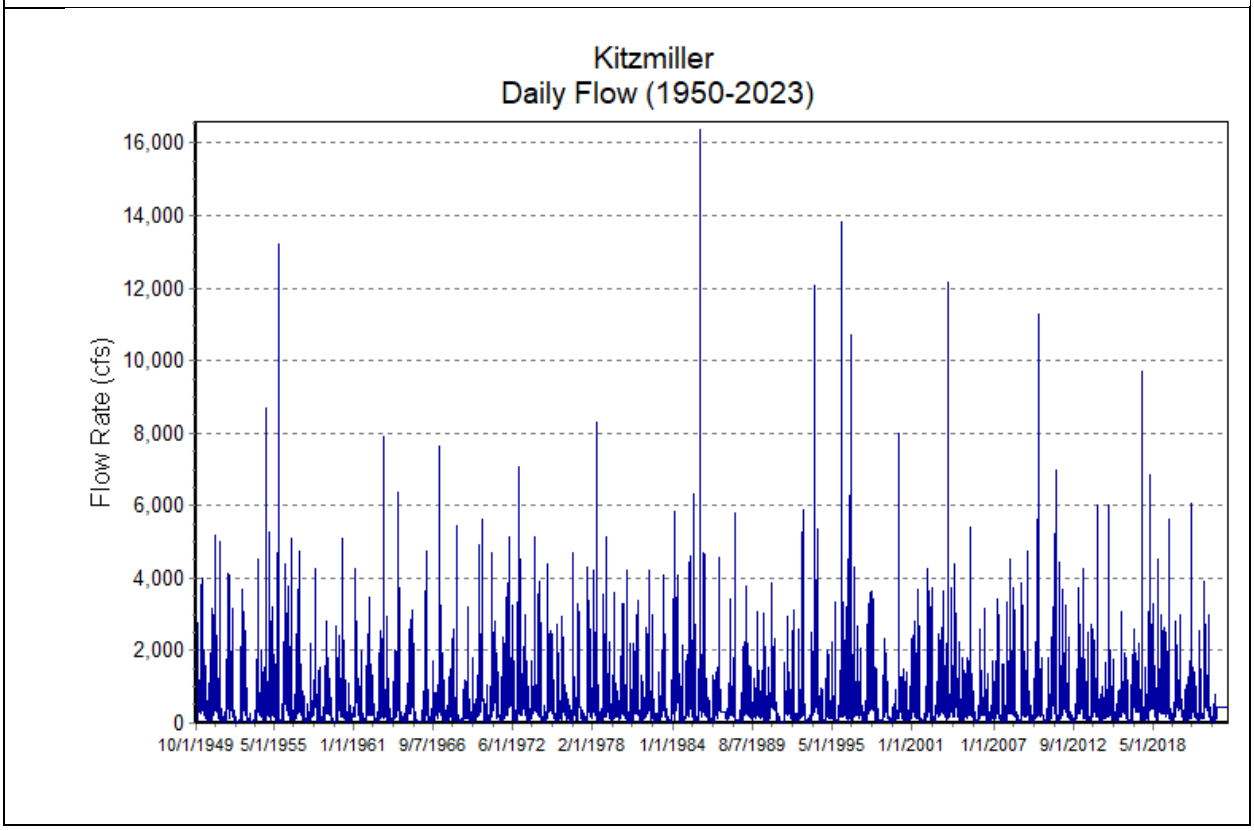
3.3.2. Daily Flows

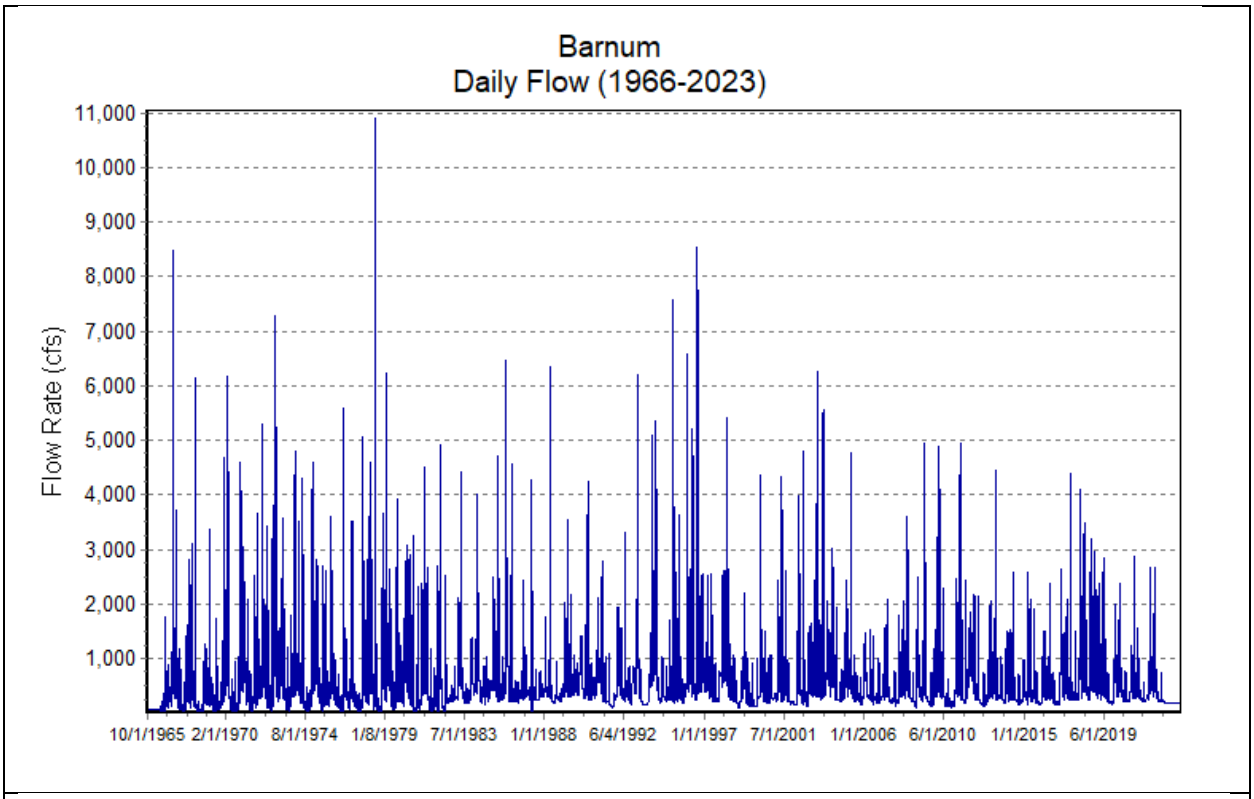
Figure 3.1 below, shows graphs of the daily flows for the study area gages.

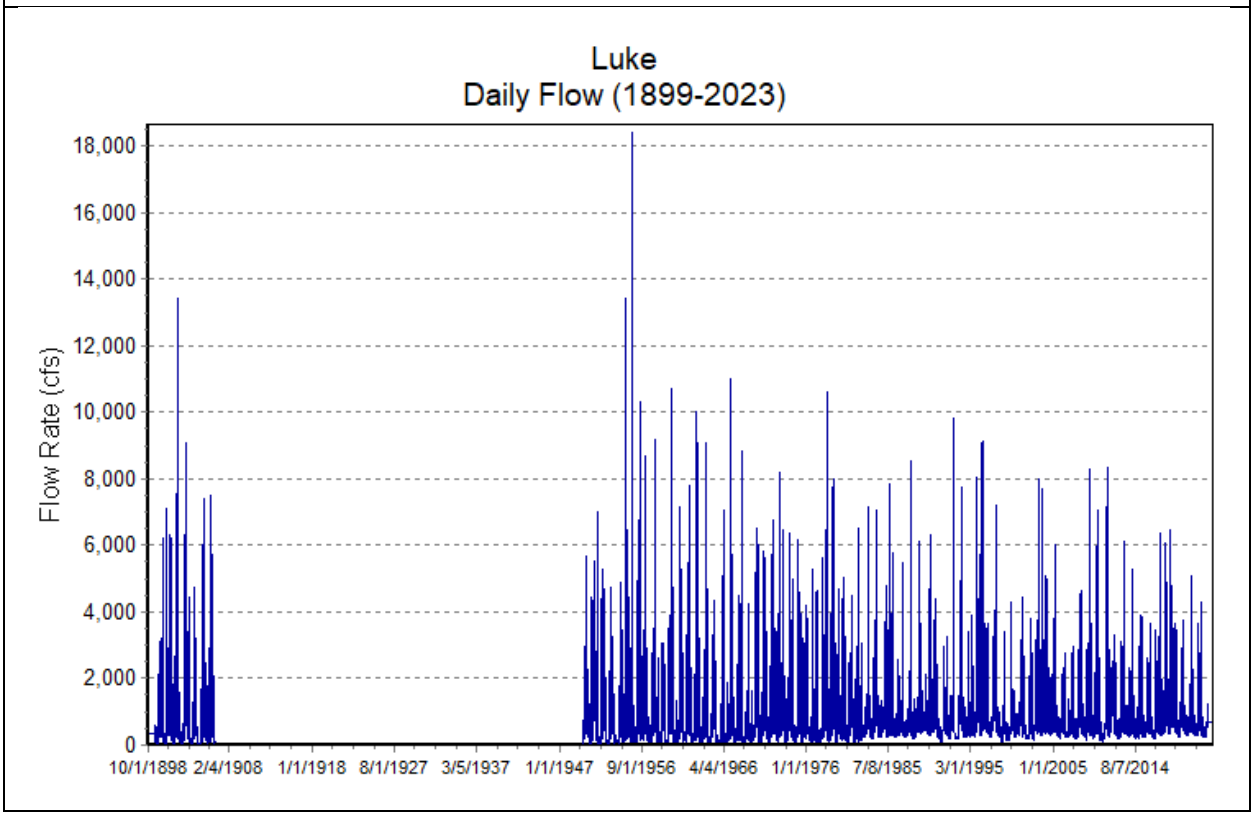
Figure 3.1 - Daily flows

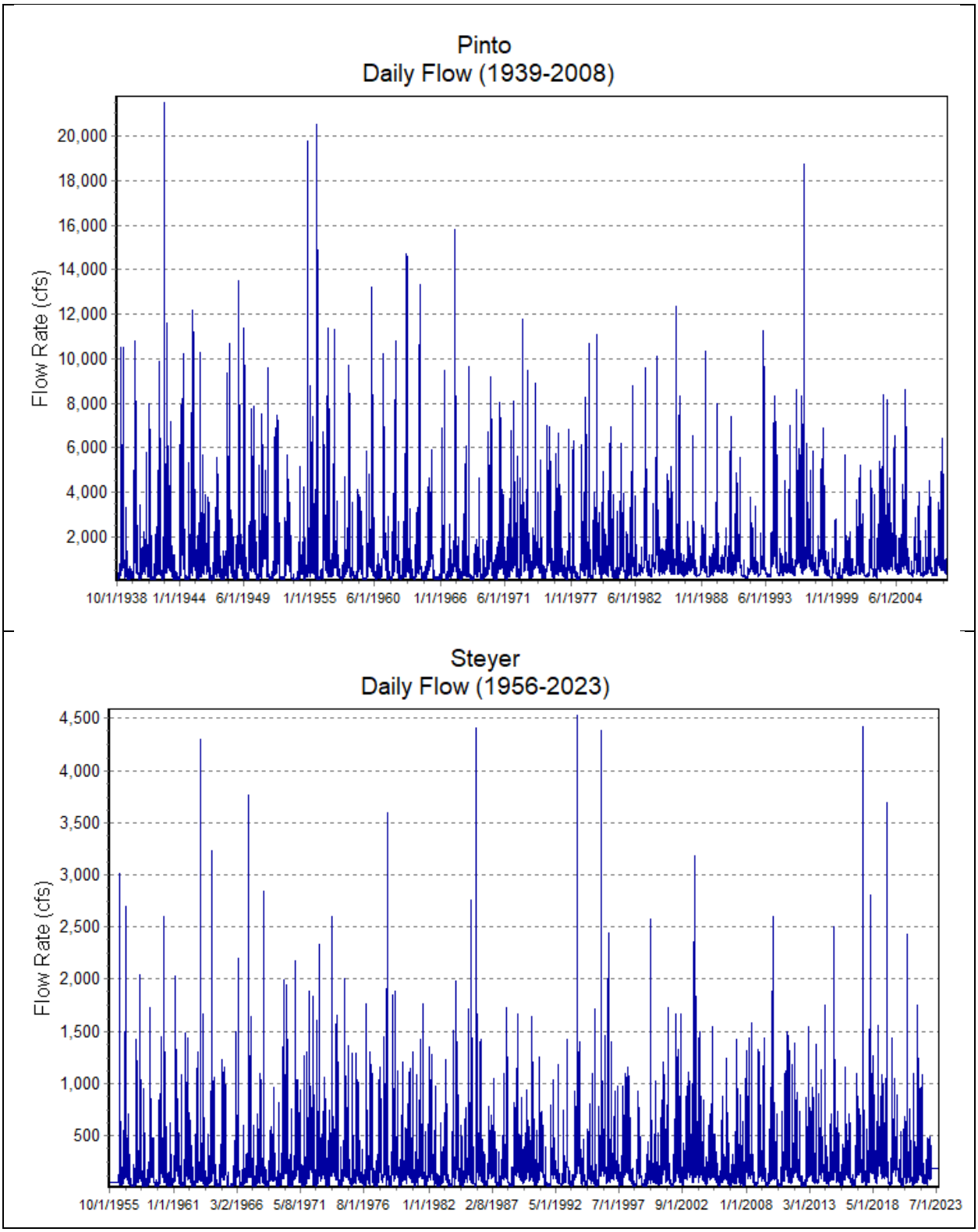


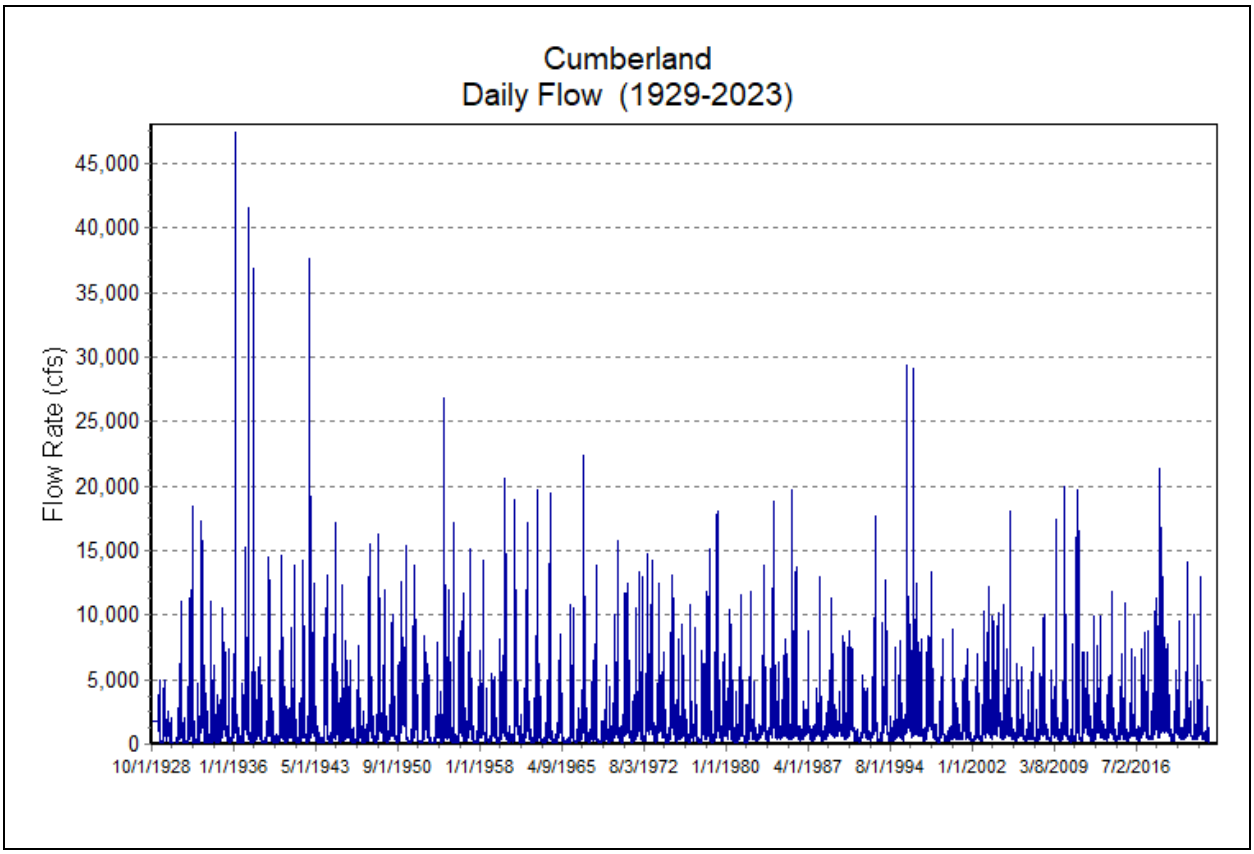

Randolph Inflow
Daily Flow (1985-2023)
Flow Rate (cfs)
25,000
20,000
15,000
10,000
5,000
7/1/1984 4/1/1987 4/1/1990 4/1/1993 4/1/1996 4/2/1999 4/4/2002 4/5/2005 4/6/2008 4/8/2011 5/1/2014 5/1/2017 5/1/2020 5/1/202


Kitzmiller
Daily Flow (1950-2023)
Flow Rate (cfs)
16,000
14,000
12,000
10,000
8,000
6,000
4,000
2,000
0
10/1/1949 5/1/1955 1/1/1961 9/7/1966 6/1/1972 2/1/1978 1/1/1984 8/7/1989 5/1/1995 1/1/2001 1/1/2007 9/1/2012 5/1/2018


Barnum
Daily Flow (1966-2023)
Flow Rate (cfs)
11,000
10,000
9,000
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
10/1/1965 2/1/1970 8/1/1974 1/8/1979 7/1/1983 1/1/1988 6/4/1992 1/1/1997 7/1/2001 1/1/2006 6/1/2010 1/1/2015 6/1/2019


Luke
Daily Flow (1899-2023)
Flow Rate (cfs)
18,000
16,000
14,000
12,000
10,000
8,000
6,000
4,000
2,000
0
10/1/1898 2/4/1908 1/1/1918 8/1/1927 3/5/1937 1/1/1947 9/1/1956 4/4/1966 1/1/1976 7/8/1985 3/1/1995 1/1/2005 8/7/2014


Pinto
Daily Flow (1939-2008)
Flow Rate (cfs)
20,000
18,000
16,000
14,000
12,000
10,000
8,000
6,000
4,000
2,000
10/1/1938 1/1/1944 6/1/1949 1/1/1955 6/1/1960 1/1/1966 6/1/1971 1/1/1977 6/1/1982 1/1/1988 6/1/1993 1/1/1999 6/1/2004
Steyer
Daily Flow (1956-2023)
Flow Rate (cfs)
4,500
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
10/1/1955 1/1/1961 3/2/1966 5/8/1971 8/1/1976 1/1/1982 2/8/1987 5/1/1992 7/1/1997 9/1/2002 1/1/2008 3/1/2013 5/1/2018 7/1/2023



3.3.3. Flood Frequency Pre- and Post- Dam

The general trend after the dam construction in the North Branch is that there are larger storm events, therefore larger flows have become more common. This section will compare changes to the frequencies in the time periods before and after the dam construction for Kitzmiller, Bloomington, Barnum, and Luke. For instance, Kitzmiller only exceeded 8,000 cfs 3 times in 35 years since 1949. However, after 1983 (39 years), flood peaks have exceeded 8,000 cfs 7 times, with increasing frequency.

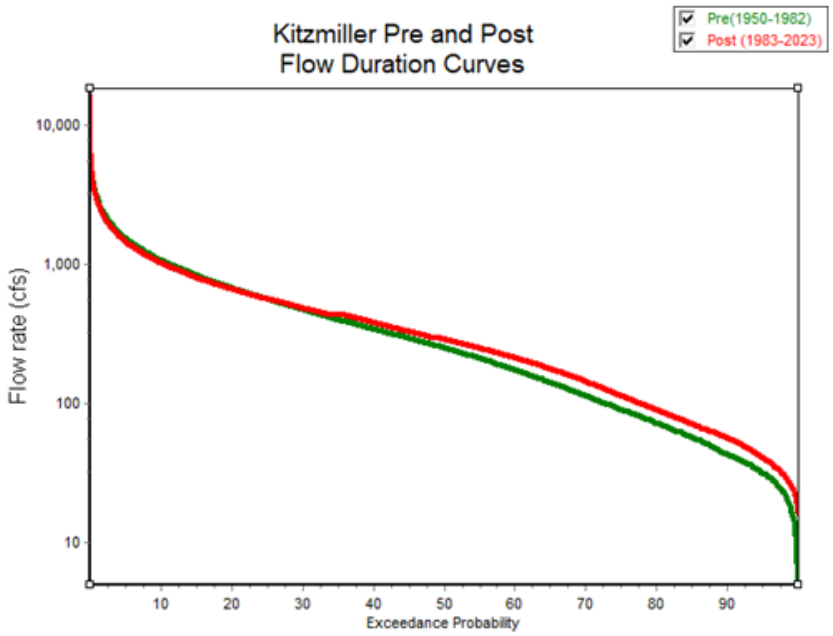
At Kitzmiller, pre and post dam frequencies did not change drastically which was expected because it is unregulated. The 100-yr. storm pre-dam is 3,070 cfs and post dam is 2,910 cfs. The greatest difference is in the 1,000-yr. storm from 5,250 cfs to 6,080 cfs respectively. At Bloomington the 100-yr. storm is 3,340 cfs and the 1,000-yr. storm is 6,000 cfs.

At Barnum, the largest change post-dam is in between the 1- and 2-yr. storm (50-100% chance exceedance). The more frequent storms became larger. For example, for the 1.25-yr. storm (80% chance exceedance) the flow pre-dam is 84 cfs and 216 cfs after. The 100-yr. storm pre-dam is 3,240 cfs and post-dam is 3,080 cfs. The 1,000-yr. storm pre-dam is 6,170 cfs and post-dam is 6,260 cfs. There is a similar trend at Luke, with the more frequent storms at times doubling in flow. The 100-yr. storm pre-dam is 4,720 cfs and post-dam is 4,170 cfs.

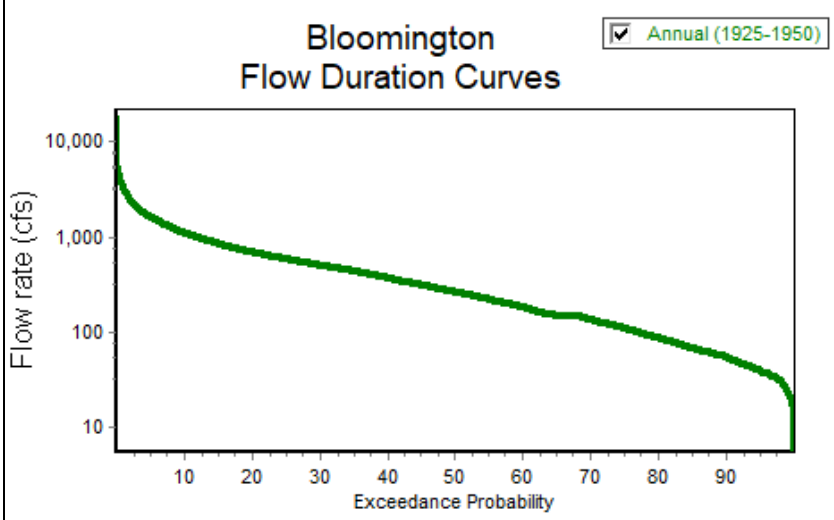
The 1,000-yr. storm pre-dam is 8,830 cfs and post-dam is 7,740 cfs. The 1.25-yr. storm (80%) the flow pre-dam is 128 cfs and 285 cfs after. These divergences in the downstream gages may be explained by dam operations releasing storm flow slowly over the course of several days rather than the flashier flows in an unregulated system.

Figure 3.2 illustrates plots and tables for storm frequencies at the various gage locations.

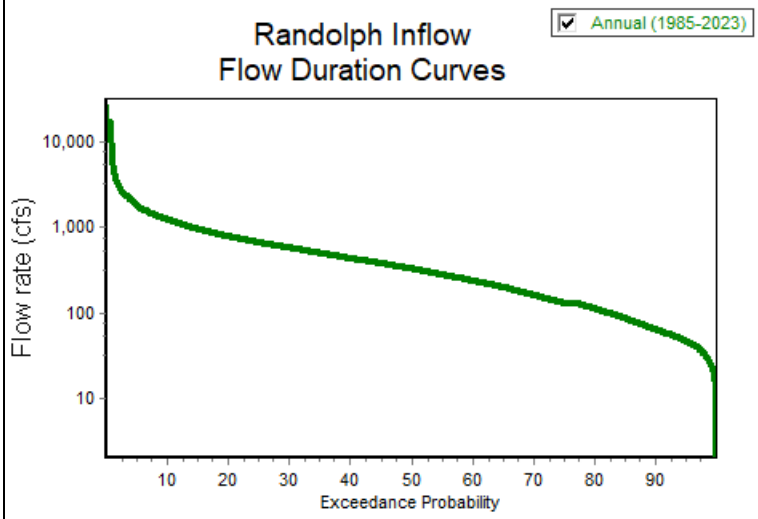
Figure 3.2 - Storm Frequency



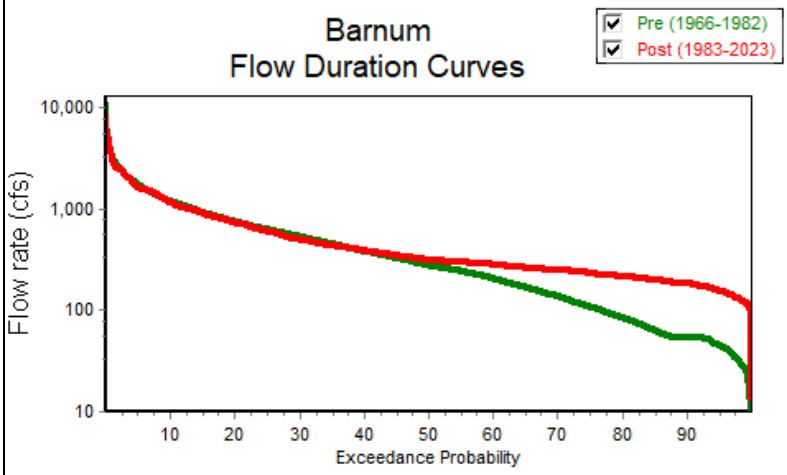
Kitzmiller Storm Frequency			
Yr.	*% Exceedance*	*Pre-dam Flow (cfs)*	*Post-dam Flow (cfs)*
1	100	5	0-16
2	50	250	289
5	20	671	662
10	10	1070	1020
25	4	1710	1610
50	2	2340	2170
100	1	3070	2910
500	0.2	4910	5080
1000	0.1	5250	6060



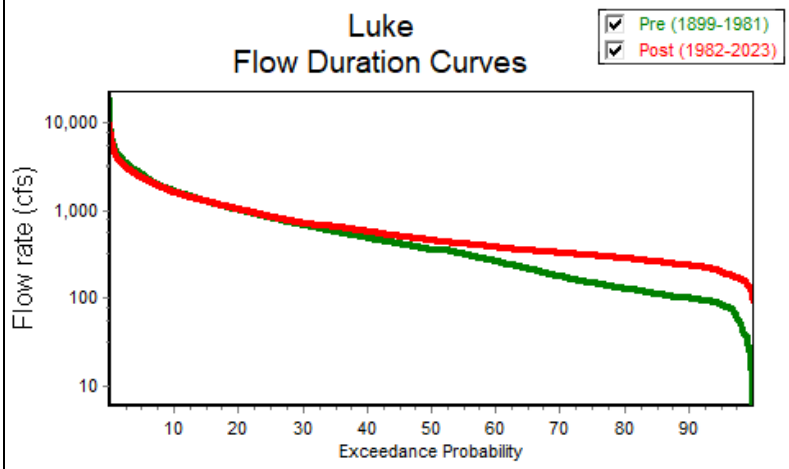
Bloomington Storm Frequency		
Yr.	*% Exceedance*	*Flow (cfs)*
1	100	5
2	50	260
5	20	690
10	10	1100
25	4	1800
50	2	2450
100	1	3340
500	0.2	5080
1000	0.1	6000



Randolph Inflow Storm Frequency		
Yr.	*% Exceedance*	*Flow (cfs)*
1	100	2-10
2	50	328
5	20	784
10	10	1,235
25	4	2,109
50	2	3,086
100	1	6,847
500	0.2	17,065
1000	0.1	17,065



Barnum Storm Frequency			
Yr.	*% Exceedance*	*Pre-Dam Flow (cfs)*	*Post-Dam Flow (cfs)*
1	100	10	13
2	50	279	315
5	20	750	731
10	10	1190	1160
25	4	1920	1880
50	2	2600	2506
100	1	3240	3080
500	0.2	5300	5421
1000	0.1	6170	6260



Luke Storm Frequency			
Yr.	*% Exceedance*	*Pre-Dam Flow (cfs)*	*Post-Dam Flow (cfs)*
1	100	6	95
2	50	358	457
5	20	1020	1030
10	10	1660	1610
25	4	2860	2610
50	2	3790	3350
100	1	4720	4170
500	0.2	7420	6990
100	0.1	8830	7740

3.3.4. Droughts

The driest months for the North Branch are July, August, September, October, and November. These are referred to as the JASON months. The most severe prolonged period of drought in the North Branch Potomac River occurred from May 1930 to February 1931. During the drought, the flow at Bloomington averaged 24 cfs in October and 19 cfs in November. Another severe drought occurred from June through December 1965. During this period, the average monthly daily flow at Kitzmiller was 24 cfs, 21 cfs, and 34 cfs for August, September, and October, respectively. A more recent drought was the summer of 1999 (from June to August). Data for Kitzmiller is not available for 1999. At Steyer, upstream of Kitzmiller, the average monthly flow was 31 cfs, 22 cfs, and 15 cfs for June, July, and August respectively. Due to climate change, the North Branch is forecast to have higher flows in the winter and spring and lower flows in the summer (Stagg & Moglen, 2017).

3.3.5. Land Use

The influence of land use on hydrology and water quality has been described extensively in literature. Normal flow conditions in the Potomac River watershed are considered those in heavily forested areas as this is the natural historic condition of the region (U.S. Army Corps of Engineers et al., 2014). Increased forest cover and lower levels of development and urbanization are generally related to lower stream flows, less overall hydrological alteration, higher stream biotic integrity, and better water quality (Cruise et al. 2010, Kennen et al. 2009).

The North Branch Potomac River watershed is notably less developed than other sub-basins of the Potomac River. About 73.6% of the watershed area is forested. Agricultural lands (pasture and crops) make up approximately 10.3% of the land use, primarily in lower-lying valleys east of the Allegheny Front, and natural succession land uses (6.5%) are concentrated in the southwestern portion of the watershed. Patterns of land use generally follow the northeast to southwest orientation of ridges and valleys, with large areas of contiguous forest in high elevations. See Table 3.2 below based on the USGS National Land Cover Database, 2018.

Table 3.2 - General Land Use	
General Land Use	**%**
Forest	73.6
Pasture/Hay & Cropland	10.3
Natural Succession	6.5
Impervious (Roads, Structures, Other)	2.0
Turf Grass	1.9
Tree Canopy over Turf Grass	1.4
Pervious Developed, Other	1.4
Tree Canopy, Other	1.3
Wetlands & Water	0.9
Extractive	0.3
Tree Canopy over Impervious	0.3
Source: Cruise et al. 2010, Kennen et al. 2009	

The largest population centers, and associated impervious surface concentrations, occur in the cities and towns of Cumberland, Frostburg, Westernport and Luke, in Maryland, and Keyser, and Ridgeley, in West Virginia. All of these population centers, except Frostburg, are located on the banks of the North Branch Potomac River. Land use immediately adjacent to the main stem river, therefore, is more developed and consists of a higher concentration of impervious surface compared to the watershed.

The influence of historic and current extractive practices plays a critical role in understanding how land use impacts water quality in the watershed. The North Branch Potomac River watershed has a long history of extractive uses including coal mining, oil and natural gas production, and logging. Water quality is still recovering from impacts of AMD and legacy sediments from forest clear-cutting. Additionally, much of the land use classified as natural succession is concentrated in areas with significant historic and present mining activity, potentially suggesting the influence of policies requiring mine reclamation and revegetation.

Chapter 4. Physical Processes and Conditions

4.1. Runoff

The channel beds in the North Branch are generally dominated by cobble and small boulders creating low permeability and flashiness in flow (Hitt et el., 2022).

Parts of the North Branch have karst terrain (limestone bedrock). Groundwater stabilizes stream flow and temperature because of recharge. In addition, increased rock contact area conducts heat and moderates the stormflow (Hitt et al., 2022). ICPRB started preliminary work in 2023 to identify groundwater driven areas in the North Branch.

4.2. In-Channel Habitats

Physical habitat attributes are generally categorized into six categories— 1) stream size and channel dimensions, 2) channel gradient, 3) channel substrate size and type, 4) habitat complexity and cover, 5) vegetation cover and structure in the riparian zone, and 6) channel-riparian interactions. Stream size and channel dimensions include measurements such as stream order, watershed area, thalweg depth, wetted width, and bank angle. Channel substrate size and type measurements include substrate embeddedness, % substrate class (fines to bedrock), % substrate type (fines, wood, detritus, etc.), and relative bed stability. Habitat complexity and cover measurements include % habitat types (falls, cascades, riffles, runs, pools, etc.) and aerial cover types (algae, macrophytes, woody debris, overhanging vegetation, artificial structures, etc.). Vegetation cover and structure in the riparian zone measurements include % canopy density and riparian cover (canopy, mid-layer, ground-layer, and bare ground).

Due to the steep, rocky topography of the North Branch Potomac River, habitats such as oxbows are not common. The stream bed consists of bedrock with abundant boulders and large cobbles. Due to this, riffles are quite common during both high and low flows. Pools typically occur along the stream banks near tree roots and other woody debris. Runs and glides occur less frequently due to the steep and rocky stream bed. There are some small patches of freshwater forested/shrub wetlands at Westernport and some forested sandbars according to the Fish and Wildlife Service's National Wetlands Inventory. **Refer to Section 4.5.1.1 for further information about wetlands.**

4.3. Water Quality

As previously noted, the water quality in the North Branch Potomac River watershed has been adversely impacted by acid mine drainage, industrial effluents, residential wastewater discharges, and legacy sediment from forest clear cutting. The watershed continues to recover from these effects and has shown notable improvements since the construction of Jennings Randolph Lake in 1982.

ICPRB published a technical report entitled "*The Influence of Jennings Randolph Lake and Dam Operations on River Flow and Water Quality in the North Branch Potomac River*," in December 2019. In their report, ICPRB extensively documented trends in the NBPR mainstem (Steyer to Cumberland, not including Savage River) and tributary water quality parameters, including pH, water temperature, specific conductance (SpCond), total dissolved solids (TDS), dissolved oxygen, turbidity, total suspended solids (TSS), and total alkalinity, from the 1960s through 2018, as data was available. A summary of key water quality results from the NBPR mainstem are discussed below:

4.3.1. pH

pH levels along the NBPR have significantly improved since the 1960s, rising from highly acidic values below 5.0 units to consistently above 6.5 units—the state minimum for Maryland and West Virginia. Jennings Randolph Lake maintains slightly more acidic than upstream and downstream suggesting the reservoir has minimal buffering influence. pH varies between tributaries, some of which are impacted by AMD. The USGS gage at Barnum has continuous monitoring and shows no violations of state criteria, though there are daily fluctuations due to photosynthesis and respiration cycles.

4.3.2. Water Temperature

Since the completion of Randolph dam, downstream water temperatures in the NBPR have become significantly cooler compared to upstream of the project during the summer months. Cold-water releases create favorable conditions for species like Brook Trout. Cooling effects diminish further downstream.

4.3.3. Specific Conductance

Maryland and West Virginia do not have water quality standards for Specific Conductance (SpCond), however, values over 500 µS/cm are considered harmful to aquatic life. SpCond is historically highly variable in the NBPR watershed and often exceeds 500 µS/cm. Prior to Randolph's construction, high SpCond was common below Kitzmiller. Post-construction, values have declined indicating that the lake may act as a sink. Downstream sites at Keyser/McCoole still frequently exceed 500 µS/cm, but statistical analyses suggest this is due to inflows from tributaries and industrial and municipal sources between Bloomington and McCoole.

4.3.4. Total Dissolved Solids

Total Dissolved Solids (TDS) correlates closely with SpCond and show similar patterns. Concentrations above Randolph typically reach 1,100 mg/L, while they stay below 400 mg/L

downstream. Median TDS drops from below the dam to Bloomington, then rises to 312 mg/L at Keyser/McCoole before dropping again.

4.3.5. Dissolved Oxygen

Historically, low dissolved oxygen (DO) concentrations were a concern for the NBPR, but data shows improvements over the years. Only nine out of 6,270 samples between 2000 and 2017 showed DO concentrations of less than 5 mg/L—the Maryland and West Virginia standard.

4.3.6. Turbidity

From 2000 to 2017, turbidity was reduced by roughly 15% in Jennings Randolph Lake indicating that it appears to be a weak sink for particles. Median turbidity values were 2.0 NTU at Kitzmiller and 1.7 NTU at Bloomington but increased downstream to 6.4 NTU at Keyser/McCoole. The increase was likely due to inputs from Georges Creek, industrial discharges, and the WWTP.

4.3.7. Total Suspended Solids

Total Suspended Solids (TSS) was highly variable across the NBPR watershed from 2000 to 2017. As water passed through Jennings Randolph Lake, TSS concentrations were reduced by approximately 25%. Significant increases were then seen between Bloomington and Keyser/McCoole, which suggests a major source of suspended solids, likely linked to Georges Creek and industrial activity. Further downstream of Keyser/McCoole, TSS was relatively stable.

4.3.8. Total Alkalinity

Alkalinity has generally increased across the NBPR since the 1980s. From 2000 to 2017, alkalinity had a median of 26 mg CaCO3/L at Kitzmiller and 19 mg CaCO3/L at Bloomington. This indicates settling of carbonate is likely occurring in Jennings Randolph Lake. Despite this drop, alkalinity concentrations increase between Bloomington and Keyser/McCoole and continue to rise further downstream; this indicates a strong buffering capacity in the lower ~~mainstream~~ mainstem.

Since the publishing of the ICPRB Report in 2019, the Luke Paper Mill shut down. Multiple sampling events at two stations downstream of the Mill near the confluence with Georges Creek show water temperatures have cooled significantly. From 2019 to 2023, water temperature has met Class III-P criterion four out of five years; in 2022, only a 10.11% exceedance of the 20°C threshold occurred (Cutler, 2024). Additional data is needed to evaluate other parameters since the closure of the mill. Figures 4.1 and 4.2 show the location of the mill and known lime dosers, accordingly.

The State of Maryland has and is in the process of re-evaluating the designation of segments of the river above and below JRL to Class III-P. Class III-P is generally defined as nontidal cold water and public water supply streams. While the State water quality standard for Class III-P is that water temperature must fall below 20°C or the ambient temperature of the surface waters, whichever is greater, releases from JRL are typically between 13°C and 15°C. These releases can have water temperature impacts as far down as Keyser/McCoole.

Figure 4.1 - Luke Paper Mill Map

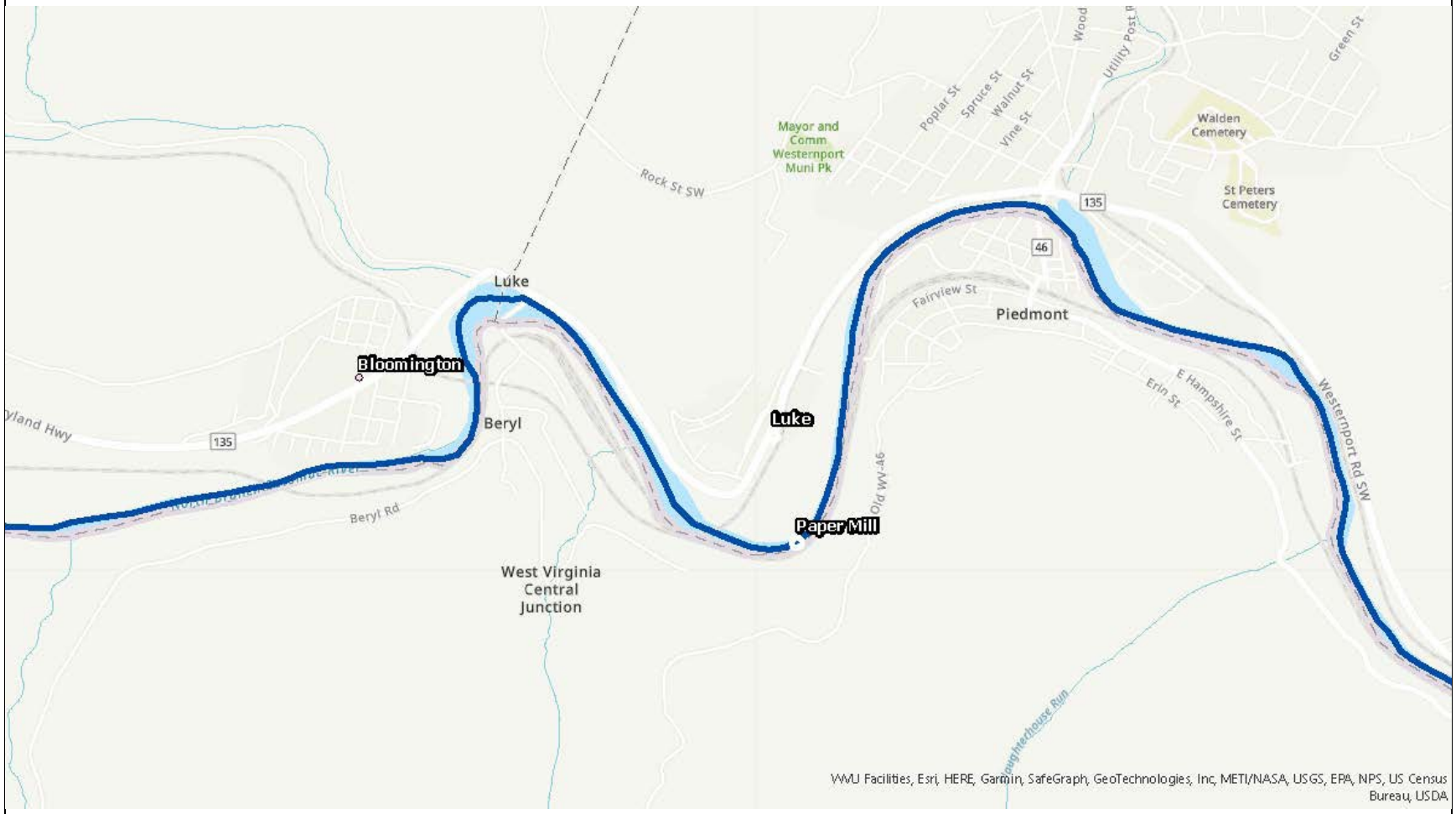
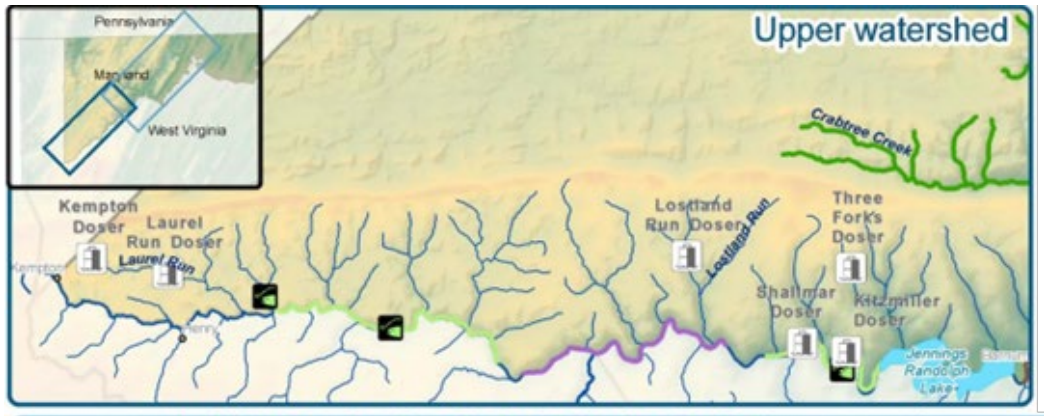
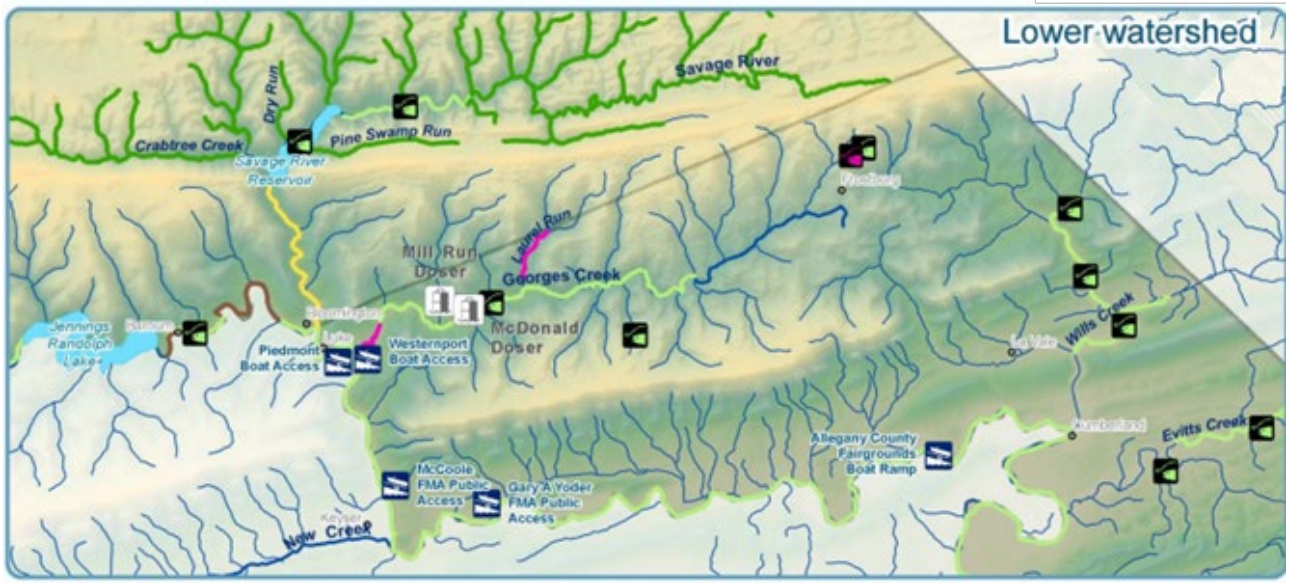
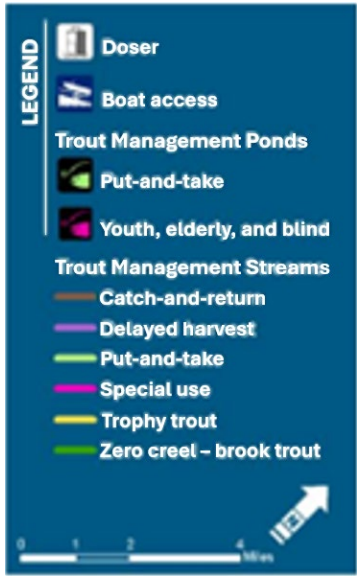


Figure 4.2 - Location of Lime Dosers



Doser	Year in service
Kitzmiller	1993
Lostland Run	1994
Laurel Run	1994
Three Forks	1996
Kempton	2000
McDonald	2003
Shallmar	2006
Mill Run	2011
Total	





Source: (Hansen et al.,2010)

4.4. Erosion and Sedimentation

4.4.1. North Branch Potomac

Further research is needed on erosion and sedimentation in the North Branch Potomac, but generally due to the watershed being forested and relatively undeveloped erosion is not a major concern in the stream channels. See Section 4.3.6. for turbidity information and Section 5.4.11 for flow alteration's effects on erosion and sediment.

4.4.2. Randolph

Currently, sedimentation is not a concern at JRL but may be in the future. Stagge & Moglen project by 2040: "WMA (Washington, DC metropolitan area) reservoirs are projected to lose 7–15% of their usable storage volume because of sedimentation between 2010 and 2040, causing an increase in storage failures… By 2040, the effects of sedimentation alone will begin to cause occasional storage failures in the Jennings Randolph…" (2017).

Hydrographic and bathymetric surveys were performed and compared with composite data sets to determine the sediment deposits in Jennings Randolph Lake. Sediment deposition reduces the useful life of reservoirs.

See Table 4.1 below for Randolph in-lake sedimentation data.

Table 4.1 - Jennings Randolph Sedimentation	
Total Original Conservation Storage Capacity	94,707 ac-ft
2013 Survey - Conservation Storage Capacity	90,313 ac-ft 1964 to 2013 (pre-impoundment survey to most recent): 4.6% loss (4,394 ac-ft)

4.5. Biological and Ecological Conditions

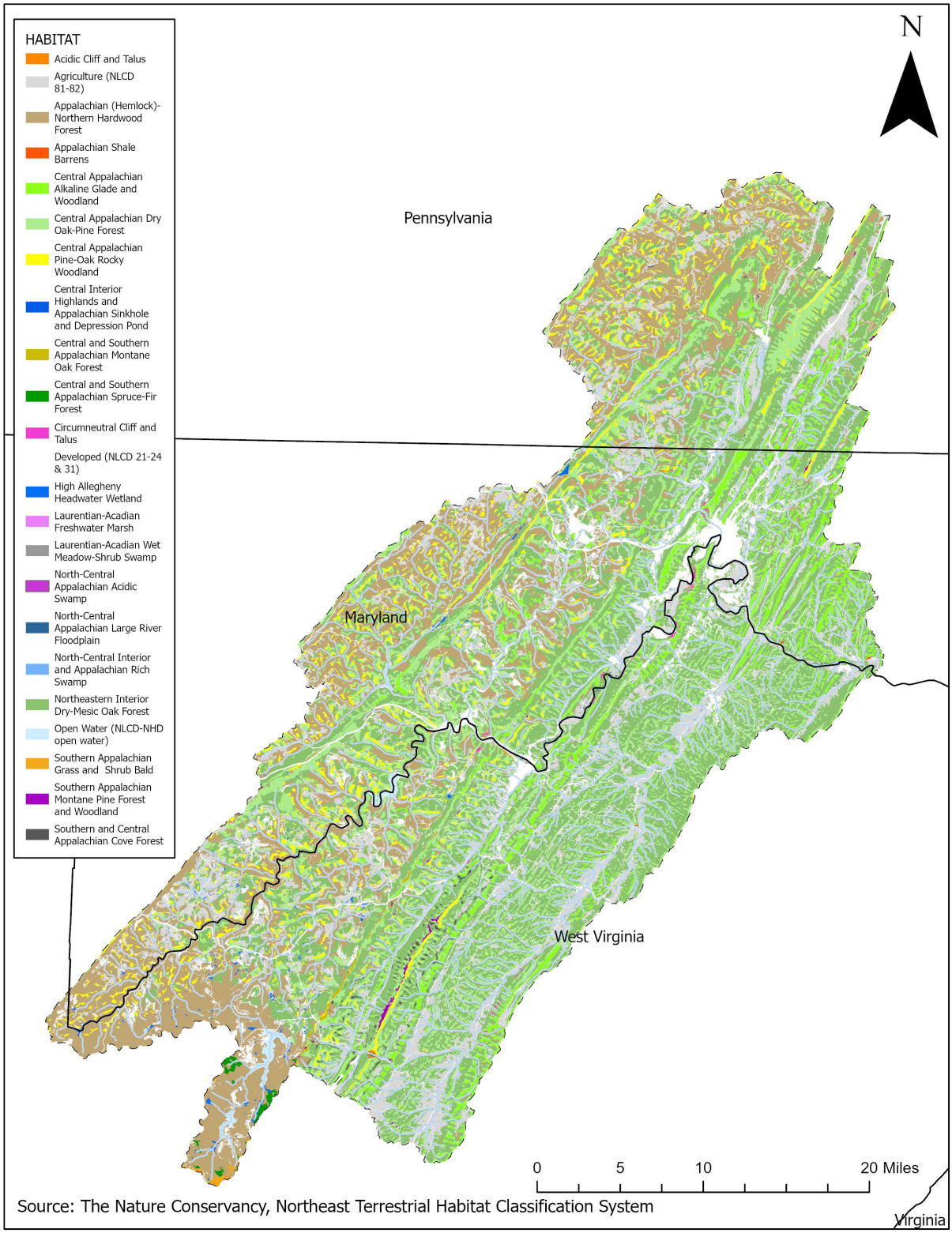
In addition to considering watershed characteristics and water management, here we present general information about aquatic and riparian biological and ecological conditions in the North Branch Potomac watershed.

4.5.1. Habitat Types

Terrestrial and aquatic habitat types in the NBPR watershed can be generalized by the watershed's two physiographic provinces – the Allegheny Plateau to the west and the Ridge

& Valley Province to the east, divided by the Allegheny Front. Much of the land in the watershed is contiguously forested with rocky hemlock and dry oak or oak-pine forests in the Allegheny Plateau and dry oak or oak-pine forests and glade and woodland habitat in the Ridge & Valley Province. The Ridge & Valley Province also features a higher density of developed and agriculture lands than the Allegheny Plateau. Compared to the rest of the watershed, the terrestrial areas immediately adjacent to the NBPR mainstem have higher densities of developed and agricultural land, but otherwise generally follow the habitat types described for each physiographic province. The lands around the headwaters of the NBPR are mostly contiguous hemlock forest interspersed with pine-oak rocky woodlands. Patches of near-vertical cliffs can be found adjacent to the NBPR mainstem further downstream, especially in the area roughly parallel to Route 220 between Luke and Cumberland.

Figure 4.3 - Habitat Types



Source: The Nature Conservancy, Northeast Terrestrial Habitat Classification System

4.5.1.1. Wetlands

Wetland systems in the NBPR watershed along the Appalachian plateau of Maryland are primarily characterized as palustrine wetlands (forested wetlands and shrubs), with some riverine wetlands near streams. Along the Valley and Ridge physiographic province of West Virginia, wetlands are mostly forested (USFWS).

Near Randolph, emergent wetlands have become established downstream of the dam as a result of the dam construction and are fed by incidental seepage and runoff. Generally, wetlands located on the project lands are limited by the steep topography and are relatively flat, low-lying areas, along the lake at the mouth of tributary systems. These wetlands are also found in the seepage basins, along the river, and on the face of the emergency spillway. There are fifteen wetlands on the JRL project lands, totaling approximately 10 acres (USACE, 2019).

4.5.2. Biological Communities

4.5.2.1. Fish

The study team attempted to summarize fish species which could be expected to occur in the mainstem of the NBPR to identify fish potentially impacted by changing flows. While fish populations in tributary streams of the NBPR have been well-surveyed through programs such as the Maryland Biological Stream Survey, data on mainstem NBPR fish communities (especially non-game fish) are limited. Therefore, fish discussed in this section are those documented to occur anywhere in the NBPR watershed (e.g., mainstem or tributary streams), even though any environmental flow recommendations would likely only affect mainstem populations. Sources of fish data include Maryland Biological Stream Survey (MBSS), USGS, Stauffer et al., 2007, as well as other online records.

Eleven species in need of conservation, including those ranked as vulnerable to critically imperiled/statewide rare in Maryland and/or West Virginia, have ranges overlapping with the NBPR watershed. As many as 23 fish species potentially found in NBPR are non-native/introduced. Table 4.2 lists fish species thought to occur in the NBPR watershed, their Native or Introduced Status, and state and global conservation rank/status. Rank/status definitions are included in Table 4.2.

In addition, the NBPR, especially in the area immediately downstream of Jennings Randolph Lake to Luke/Westernport, is a destination sport fishery, and there is a statewide focus in Maryland and West Virginia on developing the full recreational potential of the river. Trout, including brown trout and rainbow trout, are stocked regularly in the mainstem NBPR by Maryland and West Virginia state natural resources departments. In addition to brown and rainbow trout, brook trout are stocked in Randolph by the state of West Virginia. Other species potentially occurring in the North Branch Potomac watershed are listed in Table 4.2.

Table 4.2 - Fish Rank and Status						
Taxonomic Order	**Common Name**	**Native Status**	**Ranks**			**Recreational Importance**
			MD	**WV**	**Global**	
O. Anguilliformes						
F. Anguillidae - Freshwater Eels						
Anguilla rostrata	American Eel	N		S2	G4	
O. Salmoniformes						
F. Salmondiae - trouts						
Salmo trutta	Brown Trout	I				Gamefish
Oncorhynchus mykiss	Rainbow Trout	I				Gamefish
Salvelinus fontinalis	Brook Trout	N	S3/S4	S5	G5	Gamefish
F. Esocidae – pikes						
Esox americanus	Redfin Pickerel	N				
Esox niger	Chain Pickerel	N				Gamefish
Esox masquinogy	Muskellunge	I				Gamefish
O. Cypriniformes						
F. Cyprinidae - Carps and Minnows						
Campostoma anomalum	Central Stoneroller	N				
Cyprinus carpio	Common Carp	I				Gamefish
Carassius auratus	Goldfish	I				
Ericymba buccata	Silverjaw Minnow	N?				
Exoglossum maxillinqua	Cutlips Minow	N		S4	G5	
Nocomis micorpogon	River Chub	N				
Notemigonus crysoleucas	Golden Shiner	N				
Clinostomus funduloides	Rosyside Dace	N				
Notropis amoenus	Comely Shiner	N	S3	S3	G5	
Notropis rubellus	Rosyface Shiner	N				
Notropis hudsonius	Spottailed Shiner	N				
Notropis atherinoides	Emerald Shiner	I - (Jennings Randolph)				
Cyprinella aspilopterus	Spotfin Shiner	N				
Luxilus cornutus	Common Shiner	N		S1/S2	G5	
Luxilus chrysocephalus	Striped Shiner	I	S1/S2		G5	
Pimpephales promelas	Fathead Minnow	I				
Pimphales notatus	Bluntnose Minnow	I?				
Rhinichthys atratuls	Blacknose Dace	N				
Rhinichthys cataractae	Longnose Dace	N				
Semotilus atromaculatus	Creek Chub	N				

Semotilus corporalis	Fallfish	N				
Margariscus margarita	Pearl Dace	N	S2/S3		G5	
F. Catostomidae - Suckers						
Carpiodes cyprinus	Quillback Sucker	N				
Erimyzon oblongus	Creek Chubsucker	N		S3	G5	
Catostomus commersoni	White Sucker	N				
Hypentelium nigricans	Northern Hog Sucker	N				
Moxostoma erythrurum	Golden Redhorse	I?				
O. Siluriformes						
F. Ictaluridae - Bullhead Catfishes						
Ameiurus natalis	Yellow Bullhead Catfish	N				Gamefish
Ameiurus nebulosus	Brown Bullhead Catfish	N		S2	G5	Gamefish
Ictalurus punctatus	Channel Catfish	I				Gamefish
Noturus insignis	Margined Madtom	N				
F. Cyprinodontidae - Killifishes						
Fundulus diaphanus	Banded Killifish	N		S2	G5	
O. Perciformes						
F. Centrachidae - Sunfishes						
Ambloplites rupestris	Rock Bass	I				
Chaenobryttus gulosus	Warmouth Sunfish	I?				
Lepomis cynellus	Green Sunfish	I?				
Lepomis macrochirus	Bluegill Sunfish	I				
Lepomis auratus	Redbreast Sunfish	N				
Lepomis meglatois	Longear Sunfish	I				
Lepomis gibbosus	Pumpkinseed Sunfish	N				
Micropterus salmoides	Largemouth (Black) Bass	I				Gamefish
Micropterus dolumieui	Smallmouth (Black) Bass	I				Gamefish
Pomoxis nigromaculatus	Black Crappie	I?				
Pomoxis annularis	White Crappie	I				
F. Percidae - perches						
Etheostoma blennioides	Greensided Darter	I?				
Etheostoma flabellare	Fantailed Darter	N				
Etheostoma olmstedi	Tesselated Darter	N		S1/S2	G5	
Etheostoma caeruleum	Rainbow Darter	I?				

Perca flavescens	Yellow Perch	N				Gamefish
Sander vitreus	Walleye	I?				Gamefish
F. Cottidae - sculpins						
Cottus caeruleomentum	Blue Ridge Sculpin	N				
Cottus bairdi	Mottled Sculpin	N				
Cottus girardi	Potomac Sculpin	N				

Table 4.3 - Rank Definitions	
Global	
G3	Vulnerable/Watchlist - At moderate risk of extinction or extirpation due to a fairly restricted range, relatively few populations or occurrences, recent and widespread declines, threats, or other factors. Typically occurring in 21-80 populations
G4	Apparently Secure - At fairly low risk of extinction or extirpation due to an extensive range and/or many populations or occurrences, but with possible cause for some concern as a result of local recent declines, threats, or other factors
G5	Demonstrably Secure - At very low risk of extinction or extirpation due to a very extensive range, abundant populations or occurrences, and little to no concern from declines or threats.
Maryland	
S1	Critically imperiled/ highly state rare - At very high risk of extinction or extirpation due to very restricted range, very few populations or occurrences, very steep declines, very severe threats, or other factors. Typically occurring in five or fewer populations
S2	Imperiled/State Rare - At high risk of extinction or extirpation due to restricted range, few populations or occurrences, steep declines, severe threats, or other factors. Typically occurring in 6-20 populations
S3	Vulnerable/Watchlist - At moderate risk of extinction or extirpation due to a fairly restricted range, relatively few populations or occurrences, recent and widespread declines, threats, or other factors. Typically occurring in 21-80 populations
S4	Apparently Secure - At fairly low risk of extinction or extirpation due to an extensive range and/or many populations or occurrences, but with possible cause for some concern as a result of local recent declines, threats, or other factors
West Virginia	
S1	Five or fewer documented occurrences, or very few remaining individuals within the state. Extremely rare and critically imperiled; or because of some factor(s) making it especially vulnerable to extirpation.
S2	Six to 20 documented occurrences, or few remaining individuals within the state. Very rare and imperiled; or because of some factor(s) making it vulnerable to extirpation.
S3	Twenty-one to 100 documented occurrences. May be somewhat vulnerable to extirpation
S4	Common and apparently secure with more than 100 occurrences. *
S5	Very common and demonstrably secure. *
*S4/S5 species included in WV list if WV is responsible for a significant portion of the population (10%) or species range (25%), and if any one of the following criteria is met: It is listed by a regional or taxa working group as a species of concern; There is a recent threat or downward population trend not yet captured in S-ranking; or, It received a climate change vulnerability index (CCVI) score of extremely or highly vulnerable.	

USACE incorporated a fish management plan, developed with cooperation from the Maryland Department of Natural Resources and the West Virginia Division of Natural Resources, into its 1997 Master Plan update and environmental impact statement. Goals of the fish management plan not only included maintaining and improving fisheries in JRL, but also the use of selective withdrawals and AVFs from the reservoir to augment summer low flows, improve downstream water quality, and maintain outflow temperatures at 13°C as much as possible; all which extend temperature and other water quality conditions favored by trout further downstream.

MD DNR has noted an expansion of the breeding populations of brown and rainbow trout along with the more recent expansion of improved water quality and temperature conditions in the North Branch Potomac, including in areas historically limited by water quality. The existing Use III stream classification extends from Jennings Randolph dam downstream to the confluence with Laurel Run, with proposed expansion approximately to Luke, Maryland. Conditions meeting Use III classification, including presence of breeding trout populations and cold-water obligate macroinvertebrates, likely now exist downstream to at least Pinto, Maryland, if not as far downstream as Keyser, West Virginia. The small mouth bass population, which prefers warmer water temperatures, is simultaneously contracting and may soon face extirpation in the main stem of the NBPR.

4.5.2.2. Macroinvertebrates

MD DNR sampled macroinvertebrate populations downstream of Randolph in 2021 to determine if Maryland's cold water obligate species, including *Tallaperla* and *Sweltsa spp*, have begun to colonize the region. Results from this effort are included in the "North Branch Potomac River from Jennings Randolph Lake to Cumberland" section of the MD DNR 2021 Federal Aid Report. One specimen of *Tallaperla maria* was collected at the Tri-towns sampling station, and no other cold water obligates were collected during the survey. The report suggests the presence of this specimen may indicate that recent water quality improvements have created an opportunity for cold water obligate macroinvertebrates to move into the mainstem North Branch Potomac, but that it could take time for those populations to become fully established. Temperature statistics generated from MD DNR Sentinel Sites for *Tallaperla* and *Sweltsa* are recorded in Table 4.4.

Table 4.4 - Macroinvertebrates Temperature Statistics

Genus	Order	Average Daily Mean Temperature (°C)	Daily Mean Temperature Below Which There Is A 99% Probability of Taxa Being Observed (°C)
Tallaperla	Plecoptera	16.53	21.37
Sweltsa	Plecoptera	16.95	21.66

Source: Maryland Biological Stream Survey Sentinel Site Report, 2010.

MD DNR also evaluated macroinvertebrate taxa richness, Ephemeroptera, Plecoptera, Trichoptera (EPT) richness, Shannon-Weaver diversity indices, and Hillsenhoff Biotic Index (HBI) and compared values to 2016 and 2018 data at four locations. Taxa richness measures biodiversity using the total number of unique taxa observed, EPT richness measures richness specifically for pollution-sensitive species, and the Shannon-Weaver index provides a value calculated using richness as well as evenness. HBI scores decrease with increasing populations of pollution-sensitive taxa. All four sampling locations showed improving scores for taxa richness, EPT richness, and HBI, and even or decreasing Shannon-Weaver diversity indices. At the three locations where Shannon-Weaver scores decreased, the decrease is thought to be driven by population dominance of a single family, Brachycentridae. Dominance shifted from Hydropshidae to Brachycentridae, a more pollutant-intolerant family, from the 2016/2018 surveys at two locations, which may be an indication of improving water quality. Improving richness and HBI scores are also evidence that macroinvertebrate locations at these four locations are minimally affected by poor water quality.

4.5.2.3. Bivalves

Over 75% of North American freshwater mussel species are threatened or endangered with declining populations due to altered hydrology, pollution, and invasive species (USGS 2015). 13 species of bivalves are recorded in the North Branch Potomac watershed according to the MD Department of Natural Resource's 2016 Manual of the Freshwater Bivalves of Maryland; however, records in the NBPR upstream of Cumberland are generally lacking due to low survey effort in this region. Of the 13 NBPR watershed species, 11 species are native and 2 are non-native. All native species, except the Alewife floater (*Alasmidonta implicata*), are ranked as highly state-rare, state-rare/imperiled, vulnerable, or unknown in Maryland and/or West Virginia. The non-native plain pocketbook (*Lampsils cardium*) is ranked in West Virginia's Statewide Action Plan as somewhat vulnerable to extirpation (it is non-native in Maryland but native and critically imperiled in other portions of its North American range). 3 species are globally ranked as vulnerable, including the brook floater (*Alasmidonta varicosa*), yellow lampmussel (*Lampsilis cariosa*), and green floater (*Lampsilis subviridis*).

The Asian clam (*Corbicula fluminea*) is a non-native invasive species with an established population in the NBPR since at least 1990 (USGS, n.d). Impacts from Asian clam populations include fouling of irrigation canals and electric generating plants and competition for food and other resources with native species (Smithsonian, n.d.)

Table 4.5 includes a list of bivalve species potentially occurring in the North Branch Potomac along with their Maryland, West Virginia, and global status ranking.

Table 4.5 - Bivalve Rank and Status					
Scientific Name	**Common Name**	**Native Status**	**Rank**		
			MD	**WV**	**Global**
Alasmidonta undulata	Triangle Floater	Y	S1/S2	S1	G4
Alasmidonta varicosa	Brook Floater	Y	S1/S2	S2	G3
Alasmidonta implicata	Alewife Floater	Y			
Elliptio complanata	Eastern Elliptio	Y		S2	G5
Elliptio porducta	Atlantic Spike	Y	S2/S3		
Lampsils cardium	Plain Pocketbook	N		S3	G5
Lampsilis cariosa	Yellow Lampmussel	Y	SU	S2	G3/G4
Lampsilis radiata	Eastern Lampmussel	Y	SU	S1	G4
Lampsilis subviridis	Green Floater	Y	S1	S2	G3
Pyganodan cataracta	Eastern Floater	Y		S2	G5
Strophitus undulatus	Creeper	Y	S2/S3		
Utterbacia imbecillis	Paper Pondshell	Y	S3		
Corbicula fluminea	Asian Clam	N			

4.6. Recreation

Recreation is a major component of Jennings Randolph Lake and the North Branch Potomac River, but recreation is a secondary use of the project's lands and waters. JRL is authorized for flood control and water quality which heavily affects recreational activities. Flows affect fish populations, recreation areas, reservoir access, and the water levels for boaters and swimmers. In-reservoir recreation season typically extends from May – August when the summer pool has an elevation around 1,466 feet Project Construction Datum.

4.6.1. Reservoir Recreation

In-reservoir recreation includes boating, swimming, and fishing. The lake is comprised of deep and shallow areas and small coves, created by tributaries, which provides diverse opportunities for all types of boaters (Gallup, 2016). Jennings Randolph Lake is a premier fishing destination in the region. Fish species composition in-reservoir is diverse which allows fisherman to target warm-water species such as bass, bluegill, and catfish. Fishermen can also target cold-water species such as trout, walleye, and yellow perch. Jennings Randolph also contains a 200-foot swim beach known as Shaw Beach. Typically, throughout the summer the lake elevation will fall as releases are held higher than the inflow which results in these recreation facilities to be no longer accessible. This result of inaccessibility is delayed as long as possible while balancing other release needs.

4.6.2. Downstream Recreation

Downstream of the reservoir, the North Branch Potomac River supports a world-class trout fishery in an area known as Barnum, WV. The Water Resources Development Act of 1988 added downstream recreation as an authorized project purpose which supports kayaking/whitewater rafting and trout fishing. The Barnum Whitewater Recreation Area (BWRA) is owned by the U.S. Army Corps of Engineers and leased by Mineral County Parks and Recreation (MCPR). The area consists of approximately 40 acres operated and maintained through an outgrant with MCPR.

Barnum Whitewater Recreation Area is a popular trout fishing destination. The North Branch is stocked by West Virginia and Maryland Department of Natural Resources. Brown, brook, rainbow, and golden trout are stocked on a regular basis by both state agencies. In addition to fishing, the BWRA is a whitewater rafting location. With seven miles of Class I, II, and III rapids, this section of the NBPR is heavily used during Jennings Randolph Lake's Whitewater Releases and AVF releases. While AVFs are a water quality strategy, flows during these releases are very amenable for whitewater rafting and boaters take advantage of these scheduled releases.

Chapter 5. Identifying Ecosystem Flow Alteration Needs

5.1. Possible Options

An objective of the SRP is to determine the ecological goals for the flows within the NBPR, downstream of Jennings Randolph Dam. Potential ecological needs may include species life cycle support (fish, macro-invertebrate, mussels), sedimentation, combating invasive species, temperature mitigation, reduction of algae (Didymo specifically), and acid mine drainage mitigation. There may be additional benefits to other areas such as recreation, to include boating and fishing.

5.1.1. Flow Variability

Due to our limited information on ecology-flow relationships, we suggest focusing on the formation of natural habitat structures through variable flow to provide habitat for a variety of species because flow variability is a fundamental feature of river systems and their ecological functioning (Zeiringer et al., 2018). The movement of debris, sediment and water form natural structures that provide habitat to a variety of species. Dams reduce flow variability or abstract flows, therefore limiting the available habitat that would otherwise form in a natural system (Zeiringer et al., 2018). Dam releases are often held constant for days if not weeks, while a natural system would have variable flows even on a sub-daily time scale. Constant flows do not allow for the same movement and deposition as variable flows. See Figure 5.1 for a flow abstraction diagram. Further information on physical processes may be found in Chapter 4.

We recommend developing variable flows based on statistical analysis of pre-dam and calculated "natural" flows. Various hydrographs would be developed based on the IHA statistical timing, rate of rise, peak flows, flow thresholds, and rate of fall for each type of IHA flow regime (extreme low flows, low flows, high-flow pulses, small floods, and large floods). These would then be tied to key times of the year for ecological lifecycles or to strategically mimic seasonal rainstorms**. See Appendix A2 for IHA statistical results for the North Branch Potomac.**

Figure 5.1 – Flow Abstraction

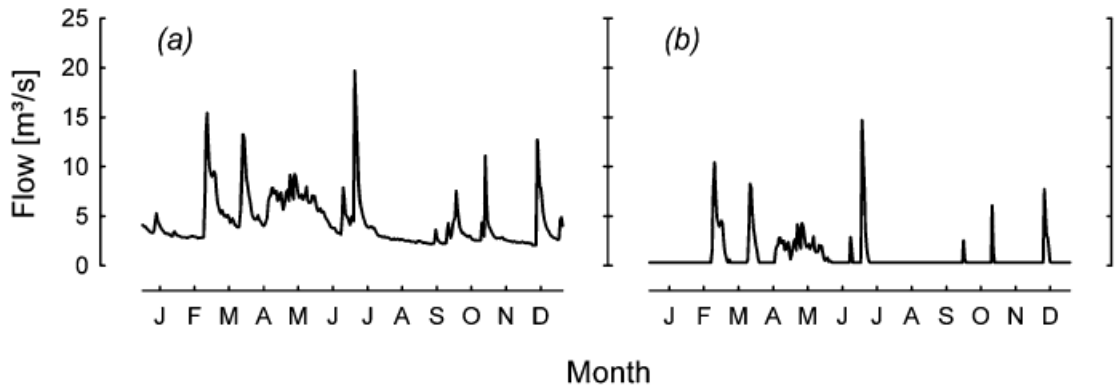


Fig. 4.8 Hydrological effects of water abstraction, (**a**) natural hydrograph, and (**b**) reduced and moderated flow in the residual flow section at the HPP Reichenau/River Schwarza (adapted from Zeiringer 2008a)

Source: Zeiringer et al., 2018

5.1.2. Sedimentation

Historically it had been observed that thick layers of sedimentation would form on the rocks downstream of the reservoir when releases would be held low and steady throughout the summer months. AVF releases are intended to remove sedimentation and algae after periods of low flows in the summer. In recent years, before and after the release, basic monitoring and testing is done to evaluate if the need for the release is still there. This effort could be an opportunity to increase testing and monitoring to fully observe the benefits of AVF releases.

5.1.3. Seasonal

According to Stagge and Moglen's 2017 paper on the optimization of the operating rules of Jennings Randolph Dam, JRL should make more conservative releases in March, April, and May to offset the dry months, July-November (JASON). In the JASON months releases can then be increased due to the increased storage from the spring. This was found by balancing storage, recreation season, flow by, and environmental flows (reducing low flows). As discussed in Chapter 3, Randolph's releases are currently about 4 times higher in the summer then they were pre-dam construction. So, there may be limited additional benefits through this strategy.

5.1.4. Species Based

Another option is to specify flows during key stages in the life cycle of native or endangered Potomac species. At the time of this paper, flows have not been specified for any North Branch species. The wood turtle, native trout and other species may be highlighted at the workshop and expanded on by subject matter experts.

5.1.5. Temperature

The NBPR downstream of the dam to Luke is designated as Class III-P and has cold water temperature standards. Recent water quality surveys indicate more cold-water tolerant macroinvertebrate species are being detected within this reach. The Corps has an opportunity to fine-tune cold-water releases at JRL.

5.1.6. Invasive Species

Invasive species have not been identified as a major issue in the North Branch Potomac River at this time. AVF releases may also be aiding in flushing out invasive species, such as didymo, that are increasingly becoming a concern.

According to the National Park Service "Didymo has extraordinary capacity to impact river and stream ecosystems. It attaches to submerged substrate on streambeds by excreting a stalk that is resistant to degradation by bacteria and fungi. Under nuisance bloom conditions, it produces numerous extracellular stalks that attach to rocks and plants…

Extensive blooms have been known to cover several miles of waterways, making it nearly impossible to eradicate once it's established. The mats, which can be over 8 inches thick, are capable of completely covering the substrate, engulfing the stream bottom, smothering native plants, insects, and mollusks causing extensive damage to the ecosystem." (NPS, 2019)

In addition to physical transport, low nutrient levels such as very low soluble reactive phosphorus (SRP; below ~2 ppb) may cause didymo blooms as well (Bothwell et al., 2014). One of the main sources of phosphorus is soil erosion.

5.2. <u>Introduction to Ecosystem Flow Alteration</u>

Environmental flows are defined as the quantity, timing, and quality of water flows required to sustain ecosystems. For reservoir operations, environmental flows manifest as management decisions that manipulate water and land-water interactions to achieve ecological or environmental goals. Environmental flows are the seasonal and inter-annually variable flows of water that sustain healthy river ecosystems and the goods and services that people derive from them. Aquatic species and natural communities have evolved in concert with naturally variable flows, and the ecological health of a river system depends on an intact hydrologic regime.

Flow-ecology hypotheses are designed to describe how specific taxa and ecological processes are expected to respond to changes in the flow regime. Related hypotheses can be aggregated based on similar timing, flow-sensitive life stages, and ecosystem function into a set of related flow needs that combine one or more responses of a group of taxa expected to respond similarly to a changed in flow conditions. In an aquatic ecosystem, species composition, relative abundance, and biological condition of the aquatic community are influenced by stream depth, width, velocity, substrate, habitat cover, turbidity, temperature, and chemical composition of the water (Gallup, 2019).

5.3. Water Quality Restoration Needs

Water quality has improved greatly in recent years. Previously, the major characteristics of mine drainage were the presence of sulfuric acid, heavy metals, and high dissolved solids. At present, approximately 40 miles of the NBPR and 100 miles of tributary streams are still affected by AMD. Measures being employed include waste treatment, reclamation of abandoned strip mines, lime treatment at Mount Storm Reservoir, and lime dosers. These measures have improved the water quality in this reach of the North Branch Potomac River to a pH of 6.0 or more. Since 1987, the water quality at Randolph has improved to an average pH of 6.0 and has stayed relatively uniform. The recent water quality improvements are believed to be the result of mine reclamation efforts and state-sponsored water treatment stations upstream of the lake. These improvements have significantly reduced the quantity and toxicity of the mine runoff reaching the lake (Gallup, 2019).

As discussed previously in Section 4.4, pH improvements are mostly due to state and federal regulation and lime dosers. Although pH has improved, due to the lime dosers, there is increased SpCond in the stream. SpCond above 500 µS/cm is harmful to freshwater species. Doser sites as well as former and current mine sites already breach 500 µS/cm. Secondary sources of increased conductivity maybe from coal, logging, the Luke Paper Mill, and the Westernport WWTP. Further investigation is needed. The trapping of metals within sediment in Randolph may also contribute to conductivity (Buchanan & Selckmann, 2019).

Other water quality improvements, aside from improvements in acidity, are unknown.

5.4. Flow Alteration and Ecological Response

In this section life history theory will be discussed. Life history theory explains environmental change and species survival strategies under cyclical or stable environmental conditions. Trade off in spawning season duration, fecundity, and parental characteristics for opportunistic, periodic, and equilibrium strategies of species in the Potomac River were studied in a 2022 paper (Hitt et al., 2022). It is important to note most species exhibit multiple strategies during their life cycle.

Hitt et al. (2002) found opportunistic strategies increased in low-permeability soils (flashy runoff) and decreased with more permeable karst terrain, because groundwater stabilizes stream flow

and temperature. In contrast, due to the stability of karst terrains equilibrium strategists increased.

Unfortunately, as mentioned previously, the ecological response to flow is a knowledge gap for the North Branch Potomac.

Below the terms and findings are summarized in Table 5.1.

Table 5.1 - Life History Strategies	
Strategies	• Opportunistic • Periodic • Equilibrium
Environmental Conditions	• Stable: constant • Cyclical: daily, seasonal, etc. • Stochastic: random or unpredictable

5.4.1. Flow Sensitive Vegetation

Limited information was found on the flow dynamics on vegetation.

5.4.2. Invasive Species

As mentioned in Section 5.1.6 didymo is an invasive species that thrives in rocky channels like the North Branch which can be 50-70% bedrock and boulders. It is mainly transported by human activity such as swimming, boating, and fishing. If river equipment such as waders are not washed off properly, didymo can spread further upstream throughout the summer recreation season. Higher flows can prevent further upstream establishment especially during dry months (NPS, 2019). During low summer flows didymo may be encouraged to bloom due to the lack of runoff.

5.4.3. Riparian Forests

No information was found on the flow dynamics of riparian systems in the North Branch. In a 1998 study, between Hancock, Maryland, and Harpers Ferry, West Virginia, a positive correlation was found in tree species richness and topographic complexity. Land use and bedrock geology were not found to have significant correlation. There is a negative relationship between tree species richness and the width of the riparian zone (river channel). Narrow sections had the best richness, possibly due to the edge effect of more species being able to access an increased light gradient and stable, well-drained soil (Everson, 1998).

5.4.4. Water Temperature

The tower at Jennings Randolph has two independent wet wells with five pairs of portals at center line elevations 1342, 1375, 1400, 1426, and 1449 feet PCD to provide releases for

water quality control. Each wet well has a low flow control gate referred to as the Quality Control (QC) gates. This design gives more control over not only the released flow but also water quality parameters including temperature. The cold-water storage is a finite resource used over the summer months which gets released but is not replenished until into the fall/winter. Hydrologic conditions dictate how that resource is used, and several scenarios can deplete that resource faster in some years than others. During prolonged periods of drought as well as high water events can impact the length of time the cold-water storage can last. This impacts the trout fishery and the natural populations of wild brook trout and brown trout as well as some small quantities of rainbow trout. Other fish species also inhabit the lake and river, including small mouth bass, lake trout, brown trout, rainbow trout, white sucker, and walleye (Gallup, 2019).

5.4.5. Floodplain Connectivity

The North Branch is steep and mostly forested, therefore lacking significant wetlands. Besides the Cumberland levee, there are not many obstructions in the flood plain. Floodplain connectivity is an area that needs to be further studied.

5.4.6. Flow Sensitive Wildlife

Limited information was found on the flow dynamics on wildlife. One possible area of improvement is with the wood turtle. The wood turtle is native to the Potomac and prefers clear cool waters in forested areas. Heavy sedimentation from soil runoff is shrinking their habitat. It is a threatened species in Virginia and an endangered species on the International Union for the Conservation of Nature (IUCN) Red List.

Much of the flow information on wood turtles is preliminary. Different populations seem to succeed in a wider variety of conditions so conclusive flow recommendations are not available. Generally, wood turtles are a sign of healthy stream conditions. During the summer nesting season, they live on land. Beginning in November, wood turtles rest on stream bottoms with sand, gravel, or rock substrate for the duration of winter. They use large wood like log jams or tree root systems as their habitat in stream (Jones et al., 2018).

Floods have been observed to have both negative and positive effects for wood turtle populations. Severe flooding disrupts the channel, altering out-of-water nesting areas. In stream, during the fall and winter, the turtles may be drowned or injured. On the other hand, this displacement may be beneficial in increasing the habitat range of the surviving turtles. Minimum flow for displacement was found to be 14.5 times the average daily flow in a study in Western Massachusetts. Another threat to wood turtles is the invasive species Japanese knotweed. Japanese knotweed is propagated during floods and crowds out nesting areas such as sandbars (Jones & Wiley, 2021b).

According to the Northeast Association of Fish and Wildlife Agencies Guide to Habitat Management for Wood Turtles:

"Dams … influence Wood Turtles by (1) flooding upstream areas and turning low-gradient stream habitat into unsuitable reservoirs, and (2) altering the downstream flow regime, which degrades nesting habitat and/or floods nests near the river. Region-wide recommendations for dam management include:

1.Minimize large water releases between late May and the estimated date of nest emergence (generally in August) on rivers with Wood Turtles and known or suspected low-lying nesting areas.

2. Allow high flows during early spring, before nesting, to encourage natural scouring of vegetation and redistribution of sand and gravel sediments. During dam re-permitting near Wood Turtle streams, managers should map essential resource areas and key features and determine whether nest-site creation or management is necessary as a result of the dam-induced flow regime." (Jones & Wiley, 2021a).

5.4.7. Fish

Extremely low streamflow can diminish salmonid species abundance due to habitat fragmentation, increased competition for food, and higher risk of predation. Brook trout (*Salvelinus fontinalis*) are affected by changes in flow regime, especially young of year (YOY) fish. Brook trout spawn in the fall, therefore, insufficient streamflow in the fall can limit spawning movement or result in dewatering of gravel nests, known as redds. Excessive high flows in winter and spring can also reduce abundance by scouring gravel nests. Bed scouring occurs under short-duration, high-flow events. During the spring, heavy streamflow can exceed swimming velocities of brook trout fry after emergence (Blum et al. 2018). In other salmonid species, high winter streamflow is strongly associated with reduced YOY abundance the following summer. In the fall, low streamflow can decrease spawning habitat availability by increasing the occurrence of gravel nest superimposition (the reuse of the same spawning locations by different individuals), increase the dewatering of redds, and restrict access to suitable spawning sites (Blum et al. 2018).

5.4.8. Macroinvertebrates

Benthic macroinvertebrates are key bioindicators of stream health in all running waters due to their tolerances of external stressors. As a result, these species are used as an index to evaluate stream health world-wide. Benthic macroinvertebrates have been shown to be negatively affected by altered streamflow and changes in temperature. As bottom-dwelling organisms, streamflow plays a large role in their life cycles. Daily and seasonal patterns of temperature fluctuation can affect their metabolism, growth, and reproduction. Three main orders of benthic macroinvertebrates are Ephemeroptera (mayfly), Plecoptera (stonefly), and Trichoptera (caddisfly). These three groups are key indicators of stream health and showcase a variety of habitat needs directly affected by streamflow. These species utilize various niches within the stream from high velocity areas to stream bottoms.

Water temperature also plays a role in macroinvertebrate assemblages. The emergence of many aquatic insects is influenced by water temperature. In most cases, warmer temperatures lead to earlier emergence. Therefore, life cycles are adapted to changes in water temperatures. For example, mayfly eggs hatch when temperatures reach a certain level in the spring. Shredder insects, stoneflies, and caddisflies will wait until fall to hatch, when there is an abundance of leaf litter to feed on.

Substrates provide sources of food and shelter for macroinvertebrates. Degradation of the substrate can be devastating to a macroinvertebrate community even if water quality remains suitable. Rocks, pebbles, and sand offer different types of niches for aquatic insects in fast flowing sections of streams. Leaf litter, algae, and aquatic plants provide habitat for macroinvertebrates in slow moving sections of streams. Macroinvertebrates have developed adaptations to live in these environments.

In 1991, a study was conducted after the construction of a hydropower dam in Olney, Maryland, where benthic macroinvertebrate assemblages were found to be positively affected by higher flows when substrate was fully underwater. Benthic macroinvertebrates were negatively affected by extreme high flows as these flows can dislodge invertebrates from their habitat and place them adrift. Extreme high flows affected the gravel substrate and fine sediment habitat that benthic macroinvertebrates rely on (Morgan et al. 1991). Extreme high flows have negative impacts on caddisfly (*Cheumatopsyche sp.*) feeding capabilities.

The common stonefly (Plecoptera) is sensitive to low dissolved oxygen levels; therefore, it thrives in cold, swift-moving streams. This species is commonly found in brisk streambed currents. Stoneflies are poor swimmers and rely on rocky substrate to complete their life cycles. Caddisflies (Trichoptera) rely on silt, sand, and small substrate. Caddisfly larvae build hollow cases for protection and pupation, from sand, silt, leaf particles, and small twigs all found in the substrate. Caddisflies spin nets to capture food. During extreme high flows, these nets may be washed away or destroyed. Mayflies (Ephemeroptera) thrive in cold, swift flowing water. Mayfly nymphs live in the substrate where they utilize rocks for protection and feed on decaying organic matter.

5.4.9. Mussels

There is insufficient information on mussels in the North Branch.

5.4.10. Flow Sensitive Habitat Types

There is insufficient information on flow sensitivity of habitats in the North Branch.

5.4.11. Flow Alteration and Erosion and Sediment

Generally, dams contain sediment that would have naturally migrated down the river. As discussed previously improvements in Sections 4.3.6 and 4.3.7, turbidity and total suspended solids (TSS) drop downstream of Jennings Randolph Dam. These sediments settle on the river bottom providing many functions for aquatic organisms, but these sediments can be harmful if the sources of sediments are polluted, such as from the Luke Paper Mill, mining, or sewage. Downstream of the dam the main source of sediment in recent years has been from Georges Creek. Turbidity over 25 NTU starts to negatively affect aquatic species (Buchanan, 2019).

During extreme low flow conditions AVFs are scheduled to flush out sedimentation and prevent low DO creating dead zones. There are typically two AVF releases every summer from Jennings Randolph Dam. However, more quantitative data needs to be taken to provide a clearer link to DO, as monitoring from 2000-2017 found the North Branch to be maintaining water quality standards (Buchanan & Selckmann, 2019). Qualitative analysis has found lower sedimentation on the stream bed after AVF releases.

The most common collected metrics of flow alteration measured for the North Branch are bank stability, channel alteration, embeddedness, riparian vegetation, epifaunal substrate. Scores in each of these metrics and the mean of all the metrics are ranked as follows: 16 and above are optimal, 15-16 are acceptable, and 13-14 are suboptimal.

Downstream of the Jennings Randolph Dam overall is suboptimal at 13.6. The poorest metric was epifaunal substrate. Epifaunal substrate is the relative quantity and variety of natural structures (e.g., cobble, large rocks, fallen trees, logs and branches, undercut banks) that are available as refugia, feeding, or spawning and nursery sites. Bank stability is acceptable at 15.4 and Epifaunal substrate is suboptimal at 12.3. By comparison, upstream of the dam overall is 14.8, with a 15.3 epifaunal substrate score. The Savage tributary is in the best condition in the North Branch at 17.0 average and 16.9 epifaunal substrate (Buchanan & Selckmann, 2019). Therefore, bank erosion is not a large concern in the North Branch, but epifaunal substrate can be improved.

5.4.12. Other Dam Operations - Temperature

In terms of temperature, JRL releases cool water from the selective withdraw system to meet the 4-20 °C standard for stocked trout. In contrast, the river upstream at Kitzmiller does not meet this range in the summer. JRL's operations may be expanding the habitat, migration and range for trout and other cold-water species (Buchanan & Selckmann, 2019).

Chapter 6. Flow Alteration and Hydrology - Pre and Post Dam

6.1. IHA Flow Regimes (Environmental Flow Components)

Streamflow statistics were calculated in the Indicators for Hydrologic Alteration program for the historical streamflow data for the following USGS gages on the North Branch Potomac: Bloomington, Kitzmiller, Barnum, Luke, Pinto, Steyer, and Cumberland. These gages were chosen due to being within the dam's reach of flow influence and having available robust and consistent data. Flow quartile calculations were done for the aforementioned gages and Hancock and Paw Paw. Hancock and Paw Paw were not included in the IHA analysis due to the data set not being extensive enough for the purposes of this report. Cumberland is the furthest point downstream that the dam has an impact on flow within standard operation. IHA defines terms different flow regimes as "environmental flow components" or EFC.

IHA also uses the Range of Variability Approach (RVA). For the purposes of the report only RVA boundaries for monthly median flows were used. The RVA boundaries are the statistically expected ranges in the following parameters: monthly median flows, minimum/maximum flows, dates of flow minimums/maximums, pulse counts/durations, rate of rise/fall, and number of flow direction reversals. Below is a detailed explanation from the IHA manual.

"When analyzing the change between two time periods (or comparing two Hydro Data files), the IHA software enables users to implement the Range of Variability Approach described in Richter et al. (1997). The RVA uses the pre-development natural variation of IHA parameter values as a reference for defining the extent to which natural flow regimes have been altered. The pre-development variation can also be used as a basis for defining initial environmental flow goals" (The Nature Conservancy, 2009).

"Each IHA parameter is analyzed to determine the frequency with which it falls into one of three RVA categories (Low, Middle, High), as defined by the RVA Category Boundaries. The change in frequency from pre-impact to post-impact of each IHA parameter in each of the three RVA categories is reported as the Hydrologic Alteration (HA) for that parameter. The factors are calculated by the equation: (observed frequency – expected frequency) / expected frequency, where "expected frequency" is the frequency expected in the post-impact period if it followed the same pattern as the pre-impact period, and "observed frequency" is the frequency actually observed in the post-impact period. A Hydrologic Alteration factor greater than zero indicates increased frequency (from pre- to post-), whereas HA less than zero corresponds to decreased frequency" (The Nature Conservancy, 2009).

IHA flow regime definitions
EFC-Environmental Flow Components- IHA defines five major components of flow that are ecologically important: extreme low flows, low flows, high flow pulses, small floods, and large floods. Environmental Flow Components are only available for models with a daily timestep. **Low flows** – This is the dominant flow condition in most rivers. In natural rivers, after a rainfall event or snowmelt period has passed and associated surface runoff from the catchment has subsided, the river returns to its base- or low-flow level. These low-flow levels are sustained by groundwater discharge into the river. The seasonally varying low-flow levels in a river impose a fundamental constraint on a river's aquatic communities because it determines the amount of aquatic habitat available for most of the year. This has a strong influence on the diversity and number of organisms that can live in the river. **Extreme low flows** – During drought periods, rivers drop to very low levels that can be stressful for many organisms but may provide necessary conditions for other species. Water chemistry, temperature, and dissolved oxygen availability can become highly stressful to many organisms during extreme low flows, to the point that these conditions can cause considerable mortality. On the other hand, extreme low flows may concentrate aquatic prey for some species or may be necessary to dry out low-lying floodplain areas and enable certain species of plants to regenerate. **High-flow pulses** – During rainstorms or brief periods of snowmelt, a river will rise above its low-flow level. As defined here, high-flow pulses include any water rises that do not overtop the channel banks. These pulses provide important and necessary disruptions in low flows. Even a small or brief flush of fresh water can provide much-needed relief from higher water temperatures or low oxygen conditions that typify low-flow periods and deliver a nourishing subsidy of organic material or other food to support the aquatic food web. High-flow pulses also provide fish and other mobile creatures with increased access to up- and downstream areas. **Small floods** – During floods, fish and other mobile organisms are able to move upstream, downstream, and out into floodplains or flooded wetlands to access additional habitats such as secondary channels, backwaters, sloughs, and shallow flooded areas. These usually inaccessible areas can provide substantial food resources. Shallow flooded areas are typically warmer than the main channel and full of nutrients and insects that fuel rapid growth in aquatic organisms. As used here, a "small flood" includes all river rises that overtop the main channel but does not include more extreme, and less frequent, floods. **Large floods** – Extreme floods will typically re-arrange both the biological and physical structure of a river and its floodplain. These large floods can literally flush away many organisms, thereby depleting some populations but in many cases also creating new competitive advantages for some species. Extreme floods may also be important in forming key habitats such as oxbow lakes and floodplain wetlands.
Source: IHA Manual (The Nature Conservancy, 2009)

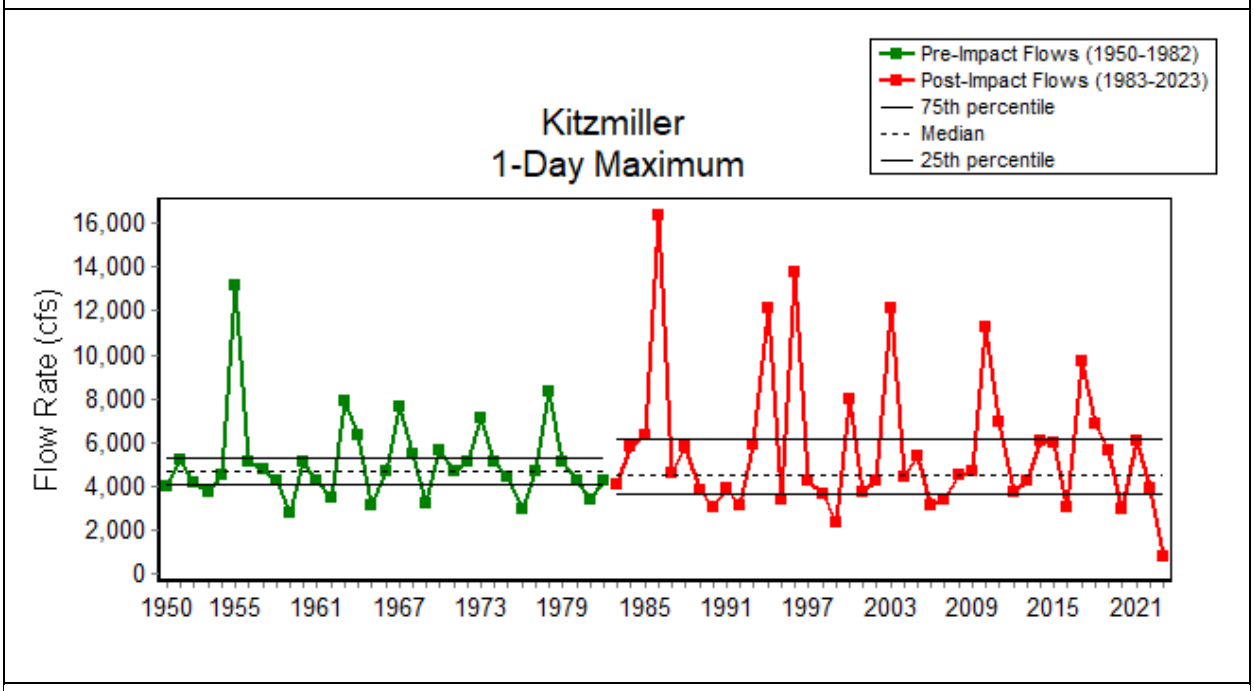
6.2. IHA Streamflow Results

See Section 3.3 for daily stream flows and flow frequencies.

6.2.1. Upstream Gages - Bloomington and Kitzmiller

Bloomington and Kitzmiller are upstream of the dam. Mount Storm Dam sporadically makes large releases which directly affect the flow at Kitzmiller. Therefore, the Kitzmiller gage cannot be used to fully model pre-dam conditions, therefore the Bloomington gage is included in this analysis. The Bloomington gage was discontinued before Mount Storm Dam was built in 1965. Kitzmiller's median monthly flows match pre-dam flows generally but are lower in March possibly due to retention from Mount Storm. Post-dam Kitzmiller has greater 1-day maximum flow and greater 1-day minimums. As stated, earlier Bloomington only has historical data up to 1949. The gage was able to capture the 1930-1931 drought of record, with flows less than 20 cfs. See Figures 6.1 and 6.2 below.

Figure 6.1 - Kitzmiller, Maximum and Minimum Flows



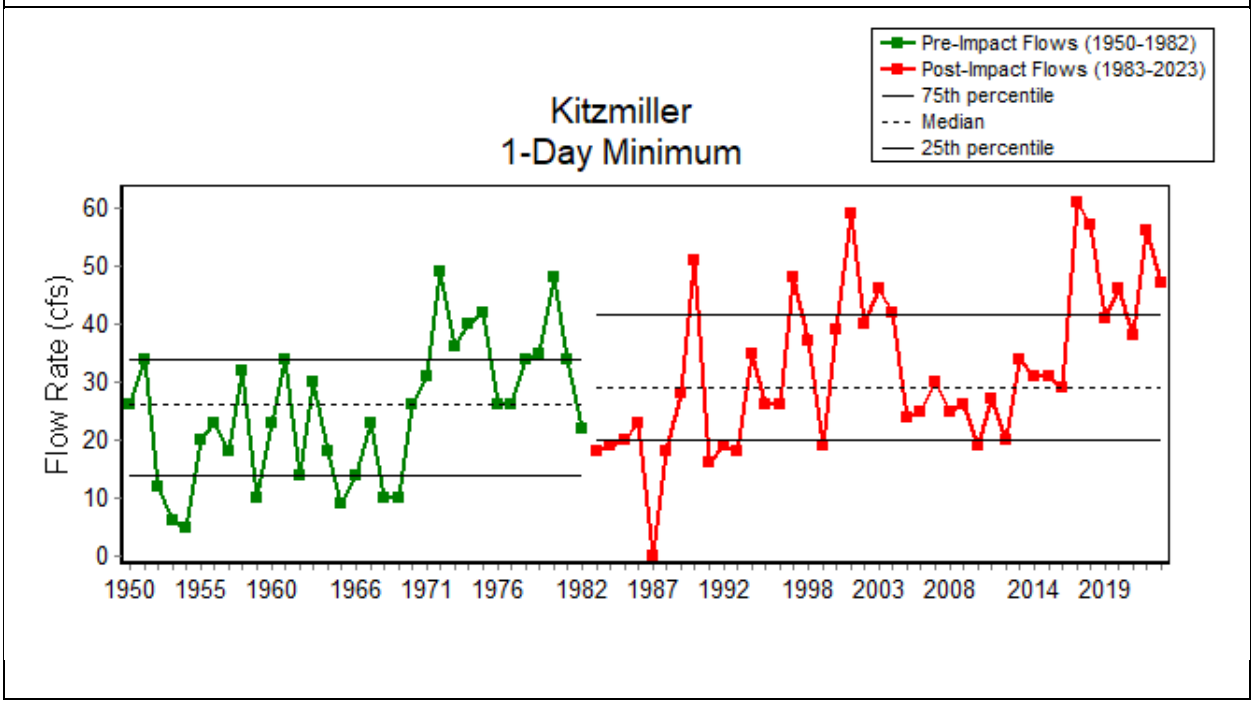
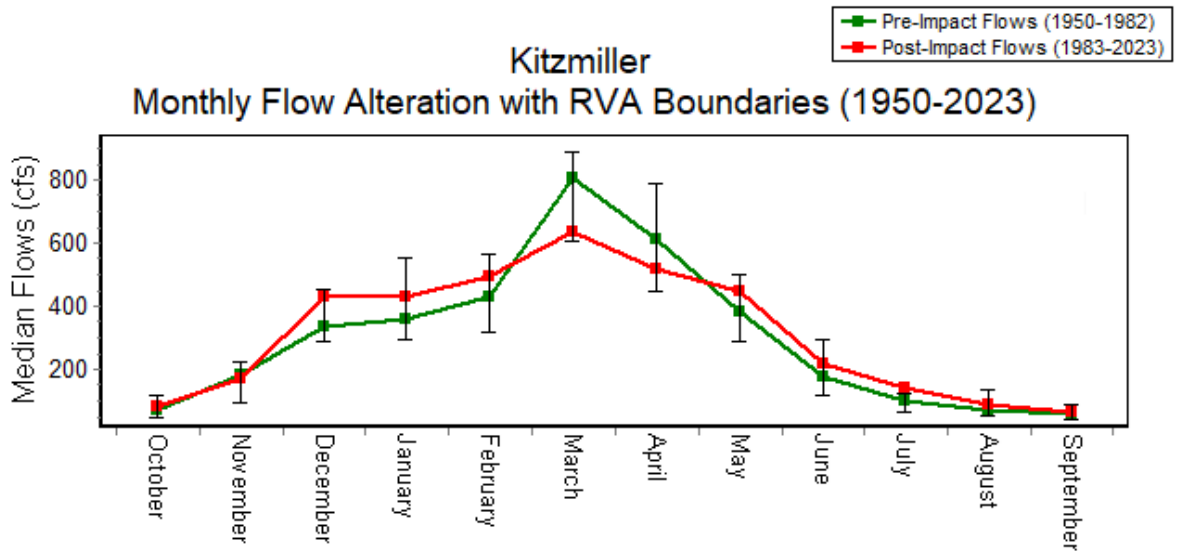


Figure 6.2 - Kitzmiller, Monthly Median Flows



The IHA analysis for Bloomington helps a establish baseline "natural' flow for Jennings Randolph's location. The different IHA flow regimes are extreme low flows, low flows, high-flow pulses, small floods, and large floods. See appendices for the technical definitions (low flows are not defined as the driest conditions etc.).

See Table 6.1 for a comparison of the Bloomington gage flows and current operations at JRL.

Table 6.1 - Historical Inflow vs Randolph Operations

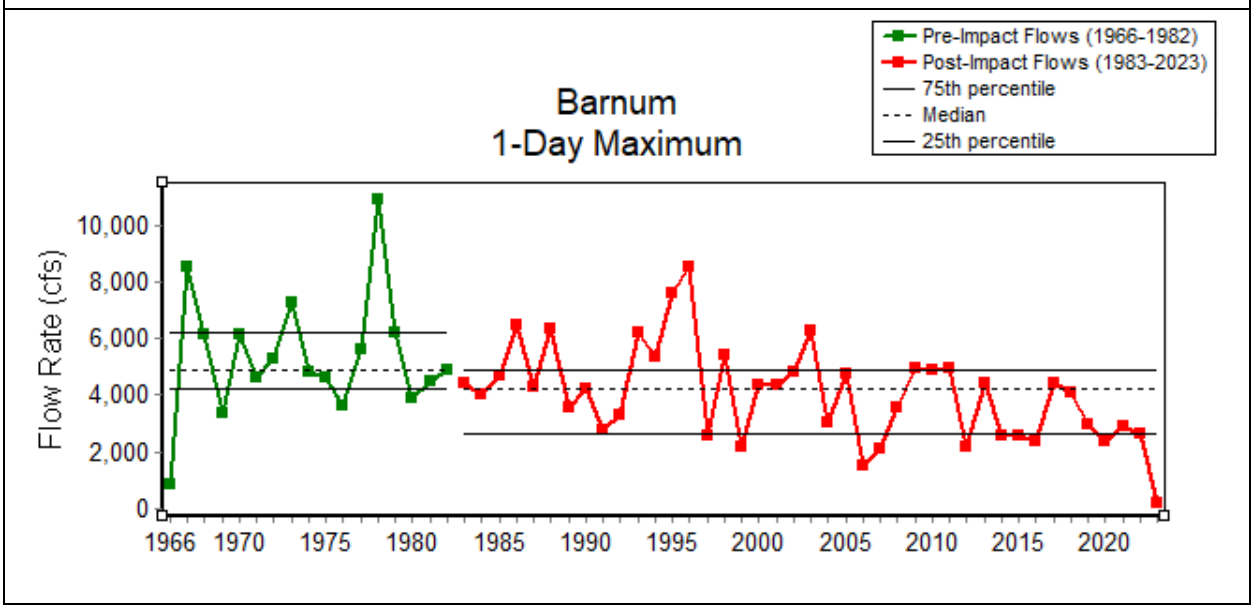
	Definition	Bloomington (pre-dam) (cfs)	Randolph releases (cfs)	Discussion
EFC low flow threshold:	Normal conditions	260	250-600	The 25%-75% percentile of Randolph releases are 250-600 cfs. The Reservoir Regulation Manual specifies 250-300 cfs outflow during the winter lake refill.
EFC high flow threshold:	Normal precipitation or snow melt	580	600	600cfs is ~75% percentile of recorded outflow

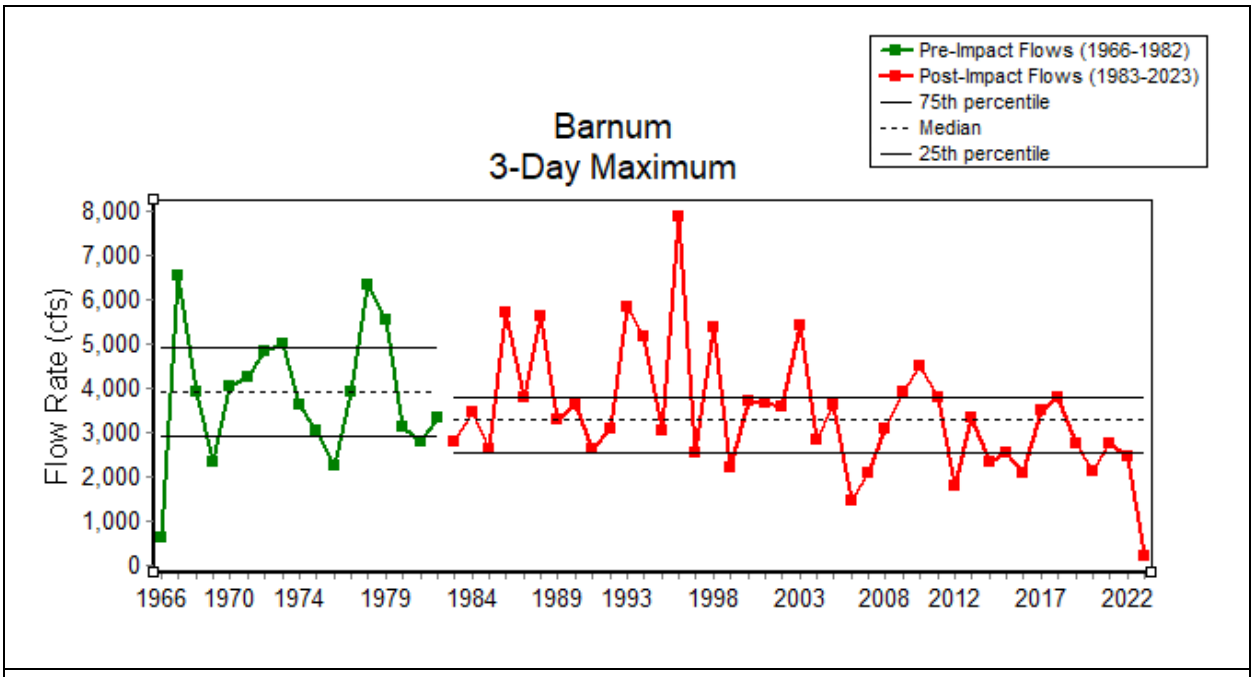
EFC extreme low flow threshold:	Drought, driest conditions	53	50	The Reservoir Regulation Manual sets a 50 cfs min release *if conditions allow* for Randolph, but 94 cfs is the lowest recorded outflow due to target flow downstream at Luke is 120 cfs in the summer.
EFC small flood minimum peak flow:	Heavy rain Exceeding banks	4,710	2,500	2,500 cfs is typically the highest release during non-flood conditions due to fish kill issues caused by supersaturation.
EFC large flood minimum peak flow:	Extreme flooding:	10,800	5,900	~5,900 cfs is the largest recorded release at Randolph.

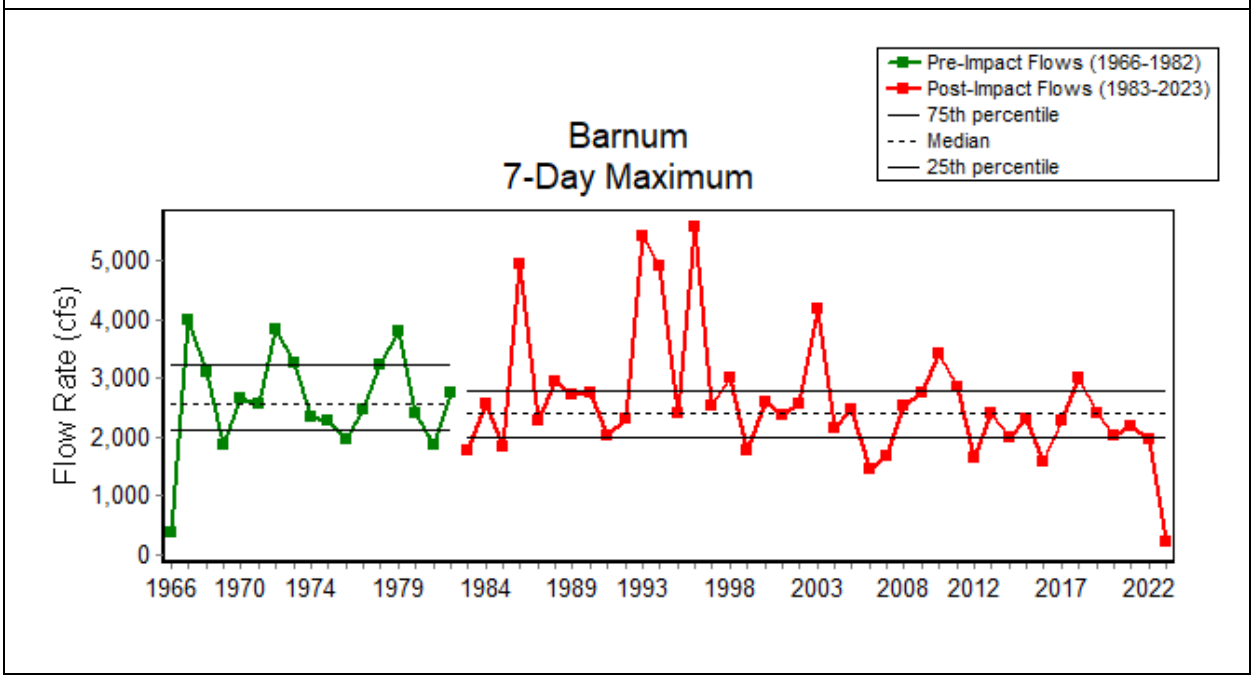
6.2.2. Downstream Gage - Barnum

Daily stream flows have been altered at Barnum since the construction of Jennings Randolph Dam. Peak daily stream flows have been reduced post dam (1983). The multi-day average peak (3 and & 7-day averages) Barnum flows medians are lower pre-1983, but there are few events that the flow magnitude is much larger than pre-1983. This may be explained by the reservoir absorbing flow form large storms then releasing it over several days. See Figure 6.3 below.

Figure 6.3 - Barnum, Peak Flows



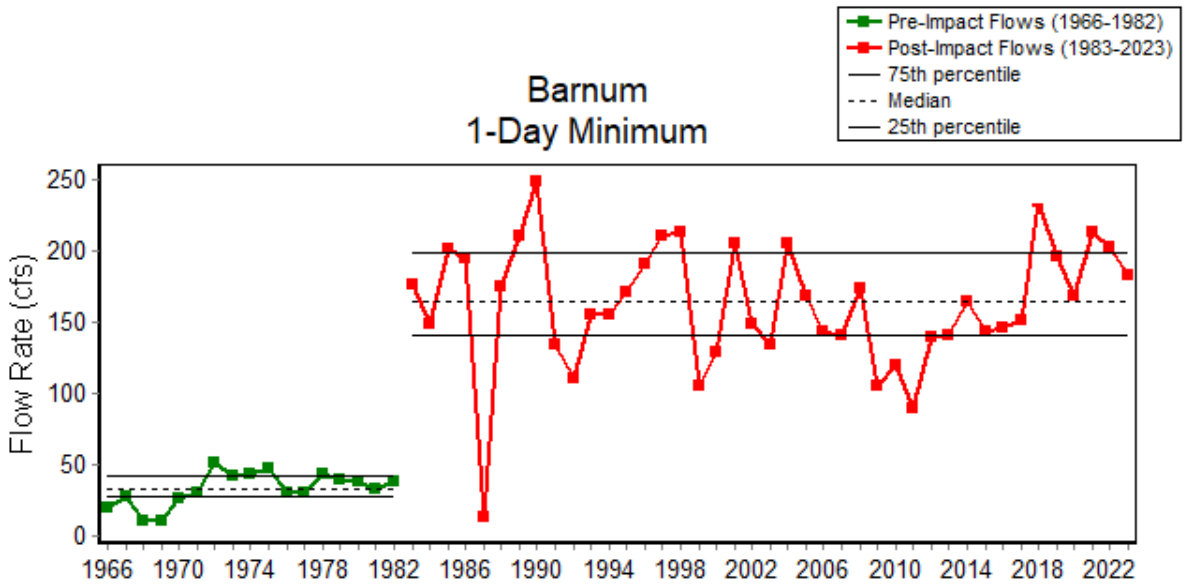


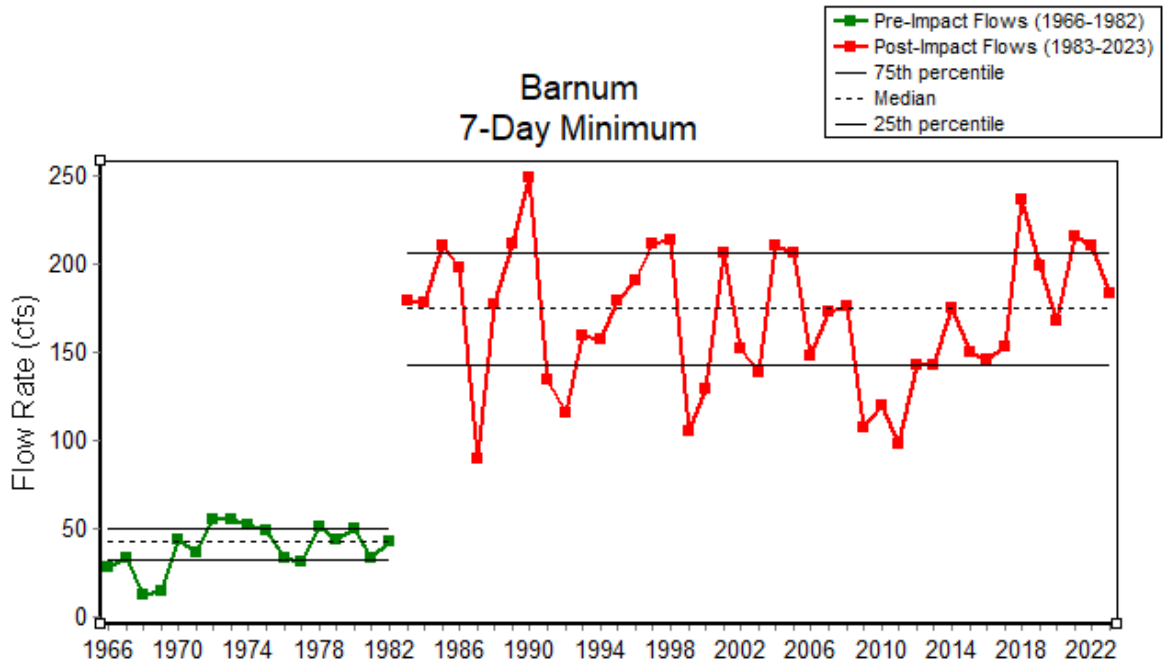


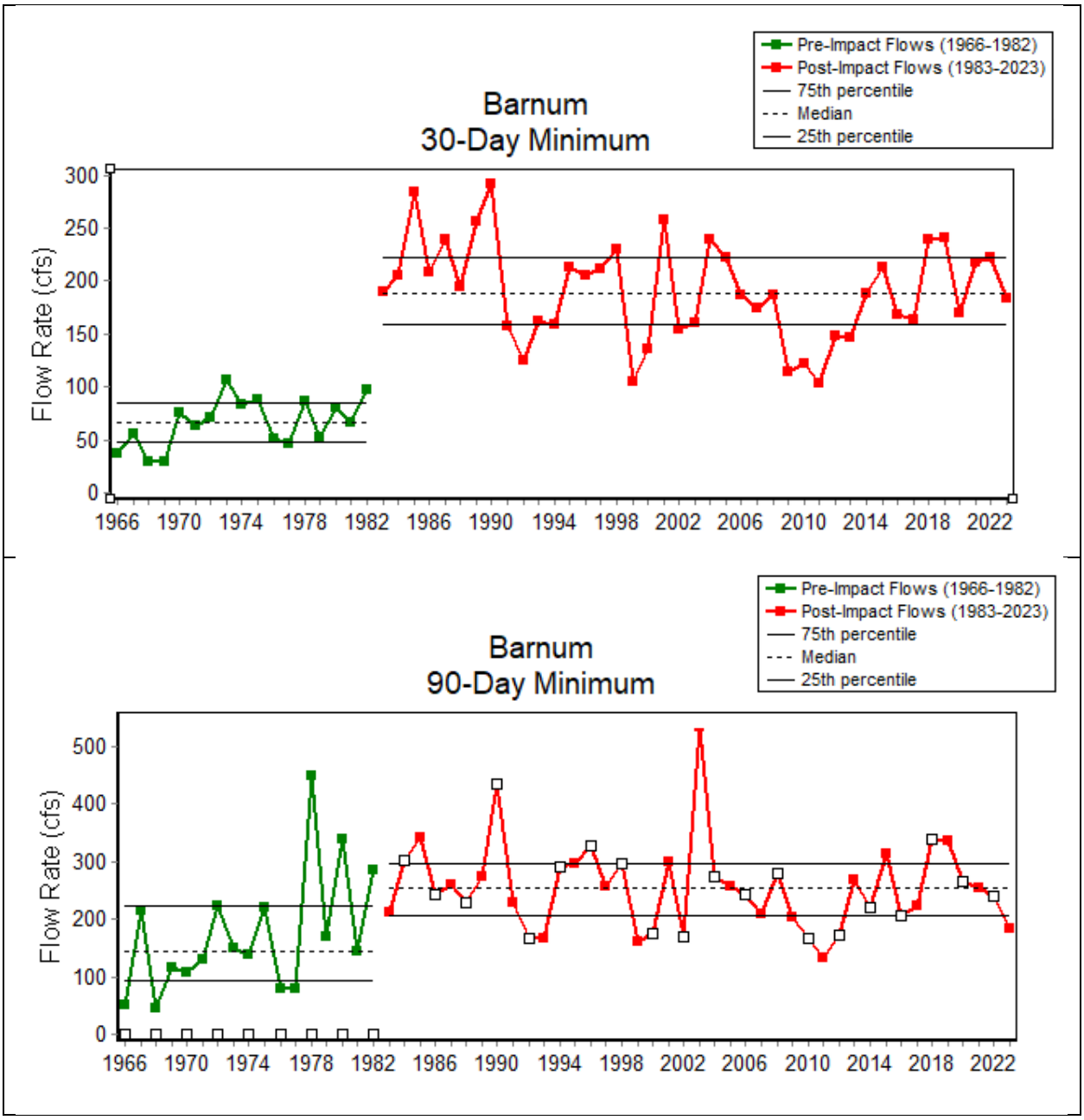
Minimum Flows

In terms of minimum flows, Barnum post-dam flows are consistently significantly higher than pre-dam flows for 1, 3, 7, and 30-day average minimums. The 90-day average minimum approaches pre-dam flow but is still significantly higher than pre-dam flow. See Figure 6.4 below.

Figure 6.4 - Barnum, Minimum Flows

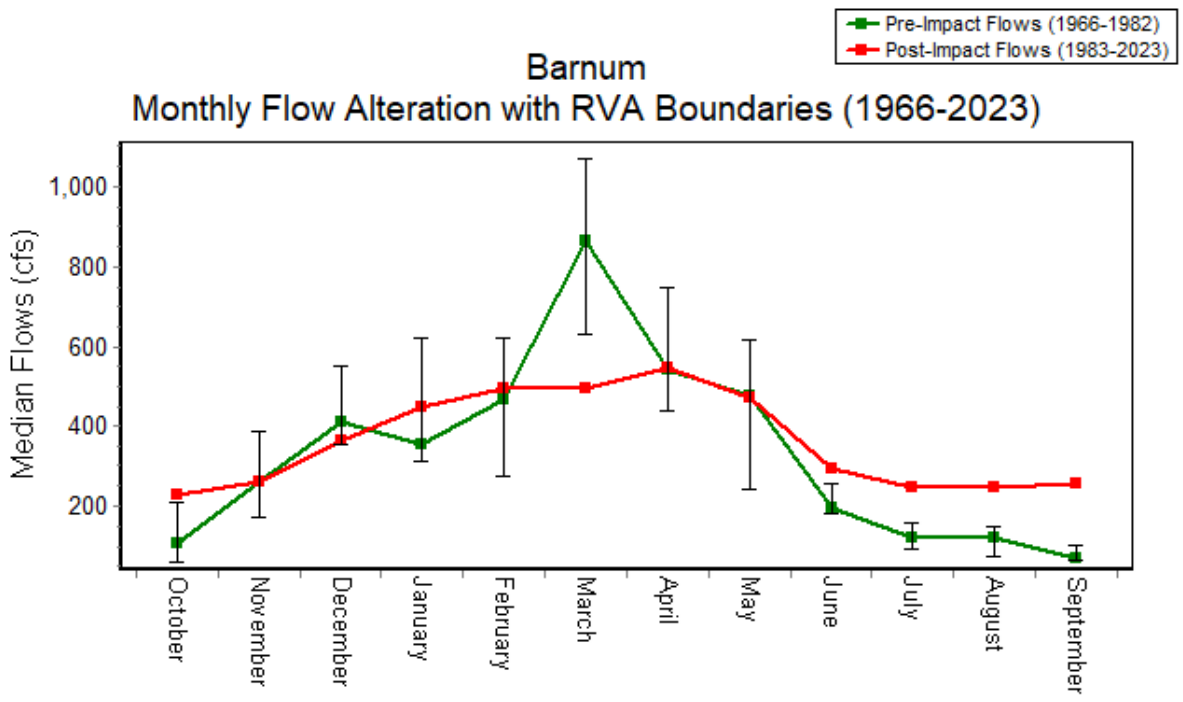






During the dry months (June-November) the median flow post 1983 is higher than before 1983. For the wet months (December through May) flow is equivalent except for March. The March median is significantly lower after 1983, due to flood control operations retaining storm flow. See Figure 6.5 below.

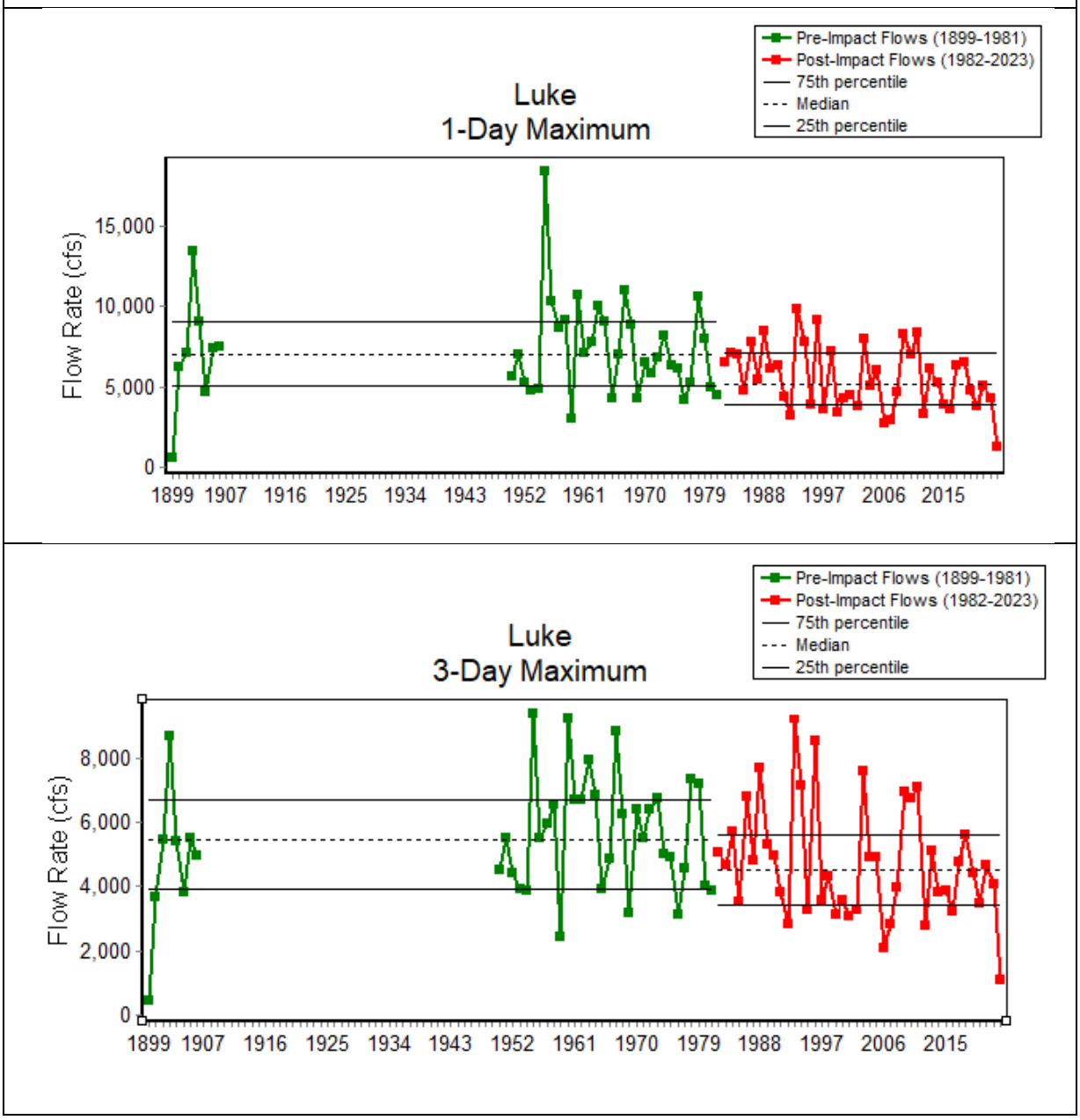
Figure 6.5 - Barnum, Median Flows

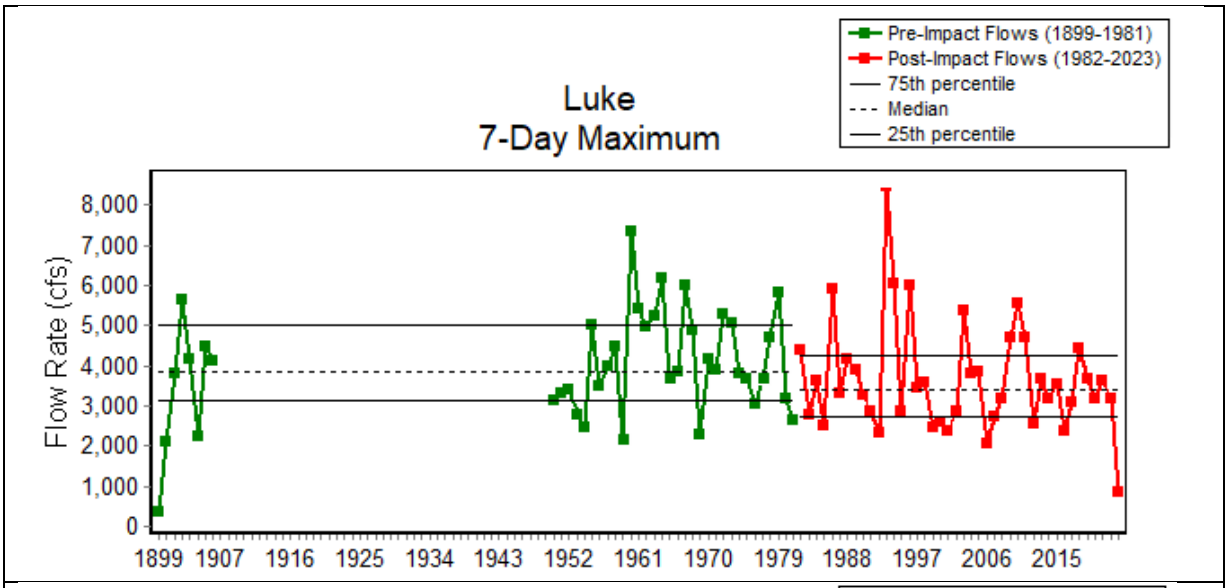


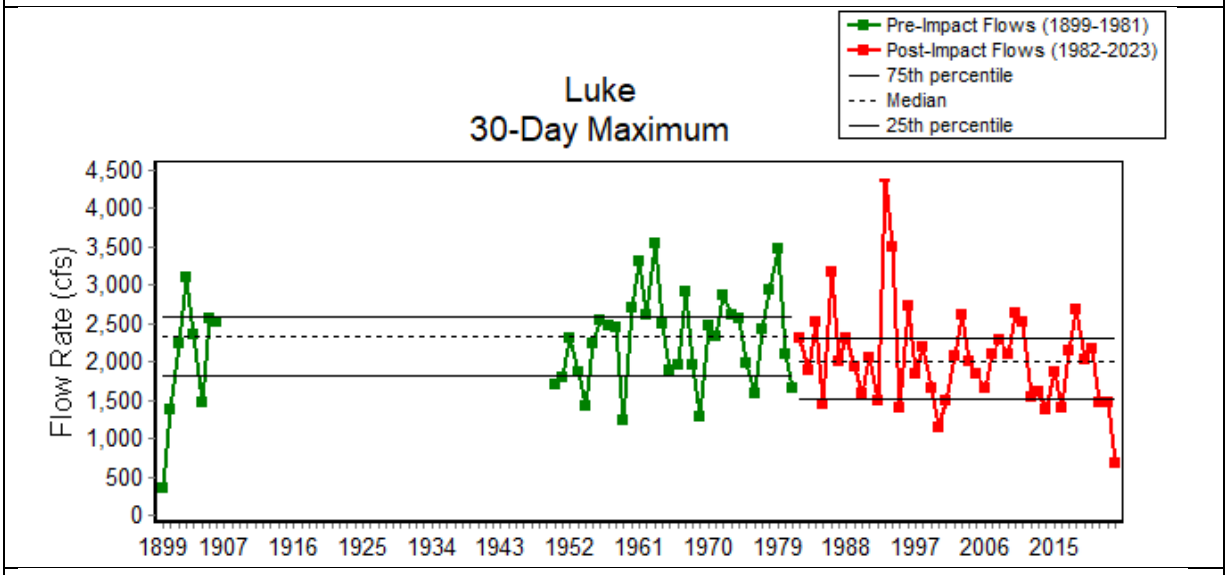
6.2.3. Downstream Gage - Luke

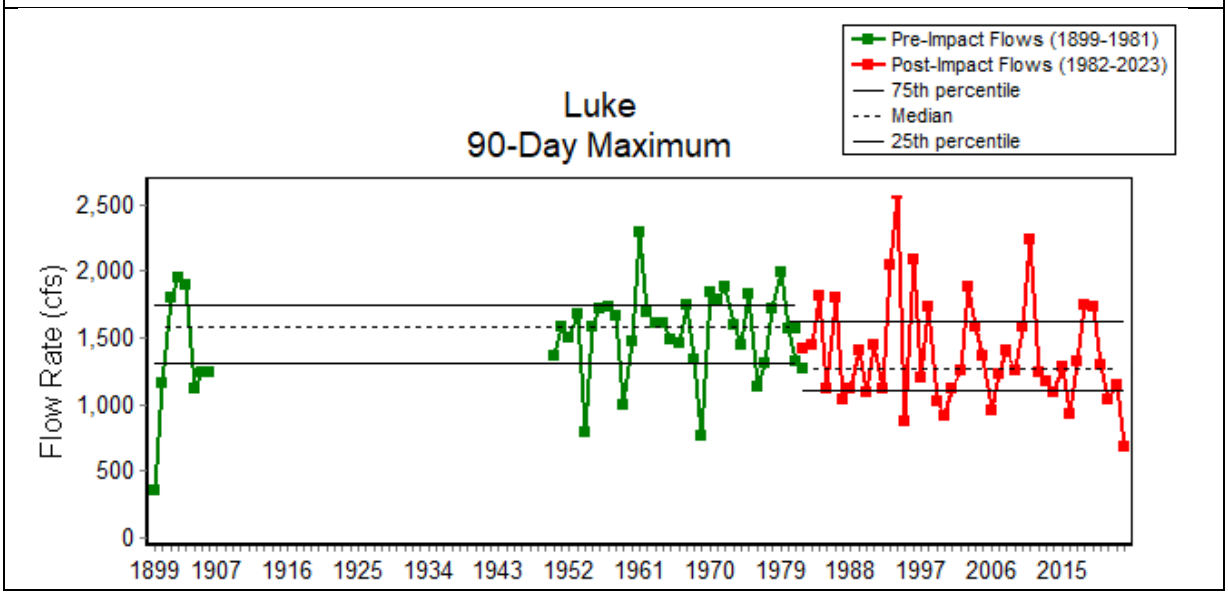
Daily streamflow has been altered at Luke since the construction of Jennings Randolph Dam. Peak daily stream flows have been reduced post dam (1983). In contrast, the multi-day average peak (3 and & 7-day averages). Luke flows are equivalent or higher than pre-1983. See Figure 6.6 below.

Figure 6.6 - Luke, Maximum Flows




Luke
7-Day Maximum
Pre-Impact Flows (1899-1981)
Post-Impact Flows (1982-2023)
75th percentile
Median
25th percentile
Flow Rate (cfs)
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
1899 1907 1916 1925 1934 1943 1952 1961 1970 1979 1988 1997 2006 2015

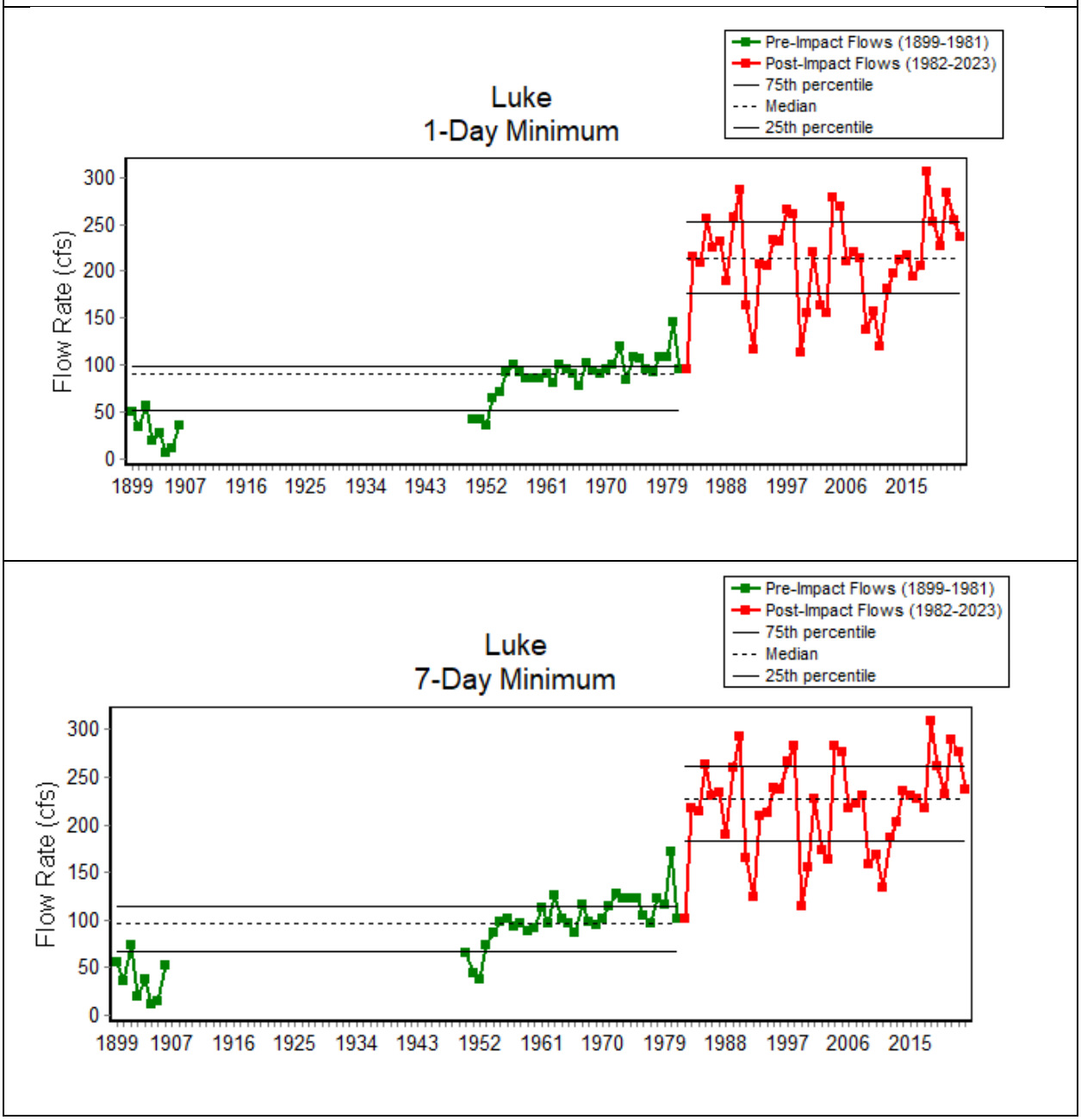

Luke
30-Day Maximum
Pre-Impact Flows (1899-1981)
Post-Impact Flows (1982-2023)
75th percentile
Median
25th percentile
Flow Rate (cfs)
4,500
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
0
1899 1907 1916 1925 1934 1943 1952 1961 1970 1979 1988 1997 2006 2015

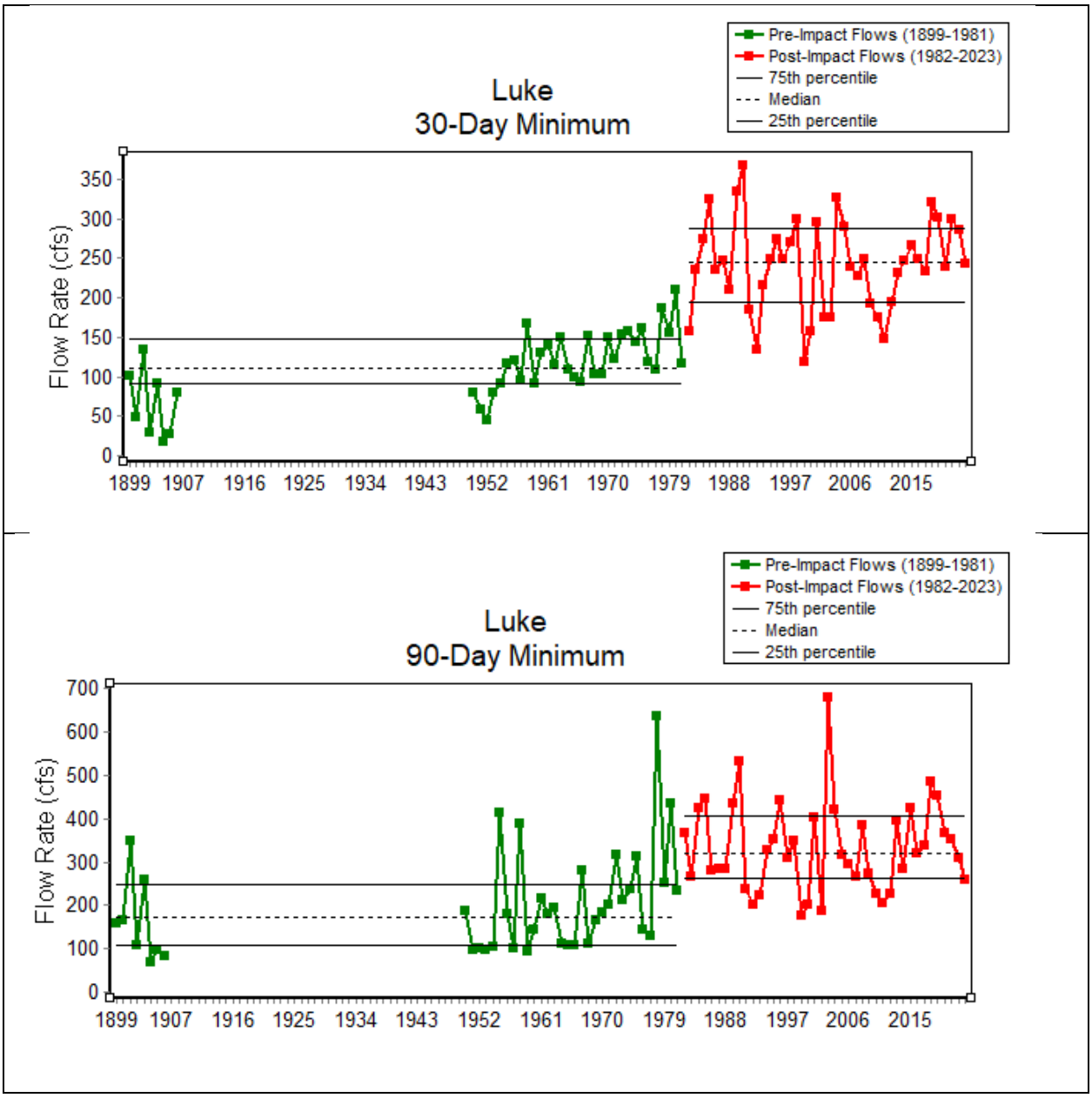

Luke
90-Day Maximum
Pre-Impact Flows (1899-1981)
Post-Impact Flows (1982-2023)
75th percentile
Median
25th percentile
Flow Rate (cfs)
2,500
2,000
1,500
1,000
500
0
1899 1907 1916 1925 1934 1943 1952 1961 1970 1979 1988 1997 2006 2015

Minimum Flows

In terms of Luke minimum flows post dam flows are consistently significantly higher than pre-dam flows for 1, 3, 7, and 30-day average minimums. The 90-day average minimum for Luke approaches pre-dam flow but is still significantly higher than pre-dam flow. See Figure 6.7 below.

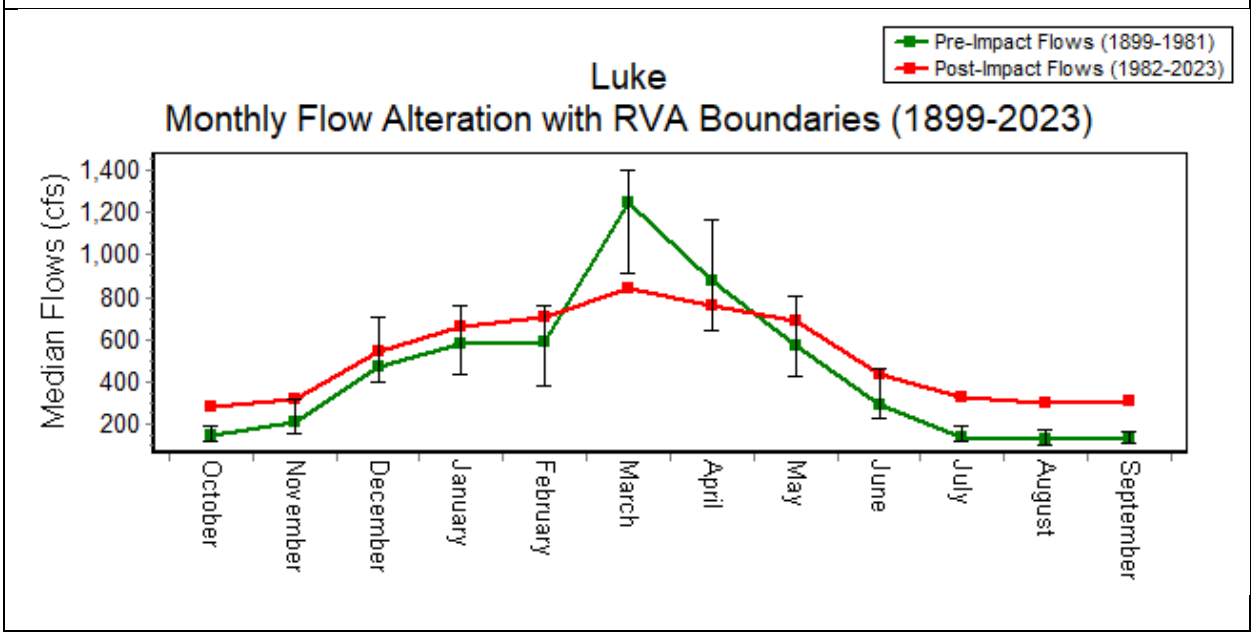
Figure 6.7 - Luke, Minimum Flows





During the dry months (June-November) the median flow post 1983 is higher than before 1983. For the wet months (December through May) flow is higher except for March. The March median is significantly lower after 1983. See Figure 6.8 below.

Figure 6.8 - Luke, Median Flows



6.3. Flood Frequencies

The general trend in the North Branch is that there are larger storm events, therefore larger flows have become more common. For instance, Kitzmiller exceeded 8,000 cfs only 3 times in 35 years since 1949. After 1983 (39 years), flood peaks have exceeded 8,000 cfs 7 times, with increasing frequency.

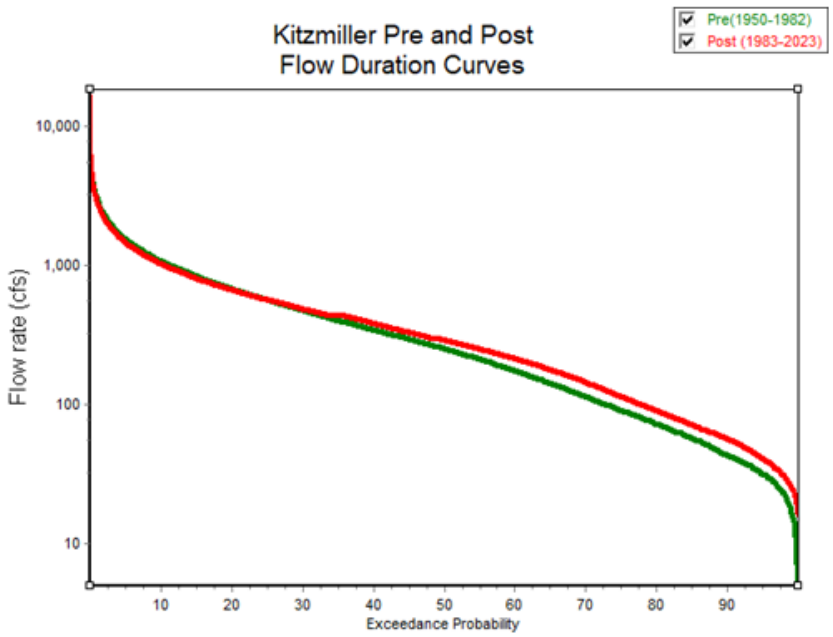
This section will discuss Randolph inflows, Kitzmiller, Bloomington, Barnum, and Luke. Please note inflows into Jennings Randolph Lake are computed values based on a USGS lake elevation gage. At Kitzmiller pre and post dam frequencies did not change drastically. The 100-yr. storm pre-dam is 3,070 cfs and post dam is 2,910 cfs. The greatest difference is in the 1,000-yr. storm from 5,250 cfs to 6,080 cfs respectively. The Bloomington gage was located at the current site of Jennings Randolph Dam from 1924 to 1948. Randolph inflow data is from 1984 to 2022. At Bloomington the 100-year storm is 3,340 cfs and the 1,000-yr. storm is 6,000 cfs. Typically flow into JRL is about 1.2 times the flow at Kitzmiller. Randolph inflows are computed based on observed lake elevation changes. At JRL the 100-yr. storm is 6,847 cfs and the 1,000-yr. storm is 17,065 cfs. JRL's 100-yr. and 1,000-yr. storms are more than 200% greater in comparison to the historical Bloomington flow gage.

At Barnum the largest change post-dam is in between the 1- and 2-yr. storm (50-100% chance exceedance). The more frequent storms became larger. For example, for 1.25-year storm (80%) the flow pre-dam is 84 cfs and 216 cfs after. The 100-yr. storm pre-dam is 3,240 cfs and post-dam is 3,080 cfs. The 1000-yr. storm pre-dam is 6,170 cfs and post is 6,260 cfs. At Luke there is a similar trend as Barnum, with the more frequent storms at times doubling in flow. The 100-yr. storm

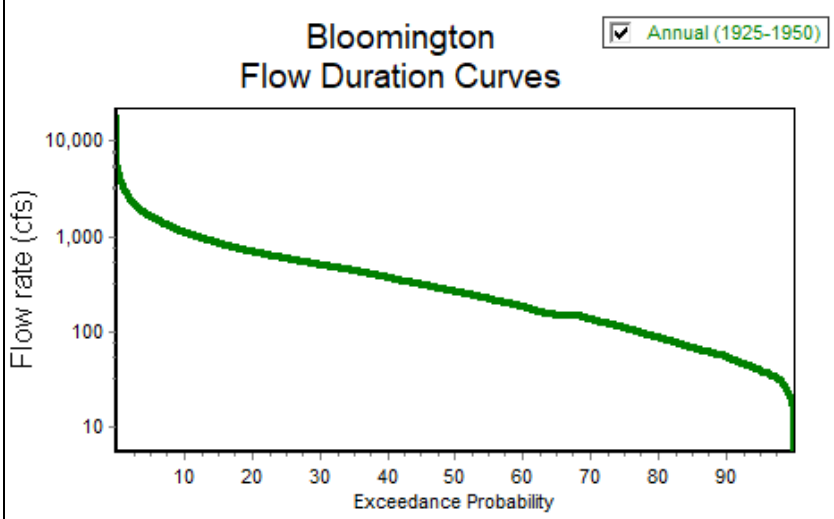
pre-dam is 4,720 cfs and post-dam is 4,170 cfs. The 1000-yr. storm pre-dam is 8,830 cfs and post-dam is 7,740 cfs. The 1.25-yr. storm (80%) the flow pre-dam is 128 cfs and 285 cfs after.

The following section has frequency plots and tables.

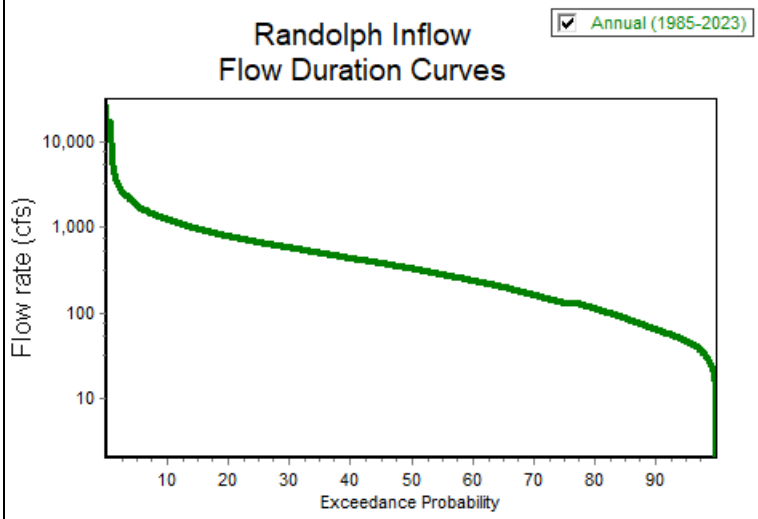
Figure 6.9 - Storm Frequency



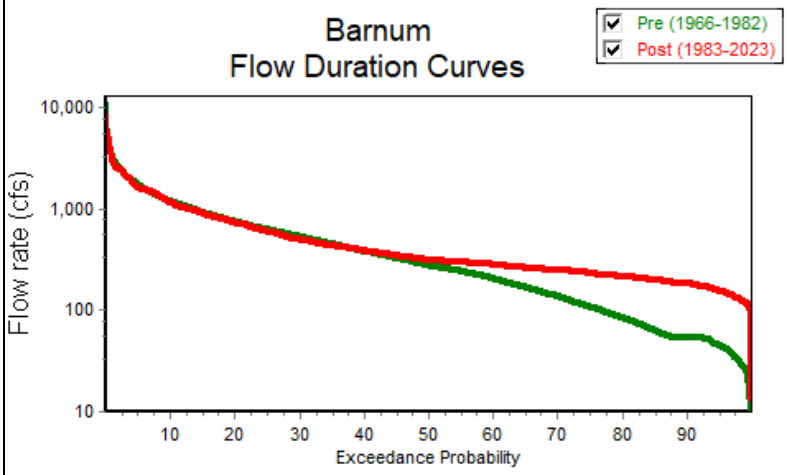
Kitzmiller Storm Frequency			
Yr.	*% Exceedance*	*Pre-dam Flow (cfs)*	*Post-dam Flow (cfs)*
1	100	5	0-16
2	50	250	289
5	20	671	662
10	10	1070	1020
25	4	1710	1610
50	2	2340	2170
100	1	3070	2910
500	0.2	4910	5080
1000	0.1	5250	6060



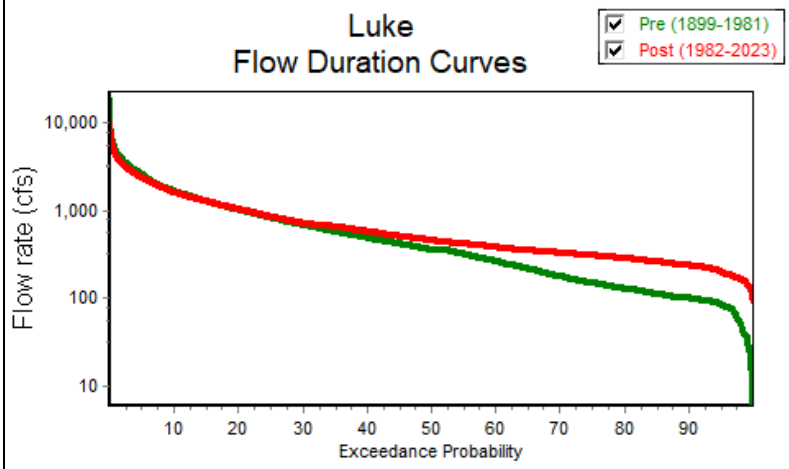
Bloomington Storm Frequency		
Yr.	*% Exceedance*	*Flow (cfs)*
1	100	5
2	50	260
5	20	690
10	10	1100
25	4	1800
50	2	2450
100	1	3340
500	0.2	5080
1000	0.1	6000



Randolph Inflow Storm Frequency		
Yr.	*% Exceedance*	*Flow (cfs)*
1	100	2-10
2	50	328
5	20	784
10	10	1,235
25	4	2,109
50	2	3,086
100	1	6,847
500	0.2	17,065
1000	0.1	17,065



Barnum Storm Frequency			
Yr.	*% Exceedance*	*Pre-Dam Flow (cfs)*	*Post-Dam Flow (cfs)*
1	100	10	13
2	50	279	315
5	20	750	731
10	10	1190	1160
25	4	1920	1880
50	2	2600	2506
100	1	3240	3080
500	0.2	5300	5421
1000	0.1	6170	6260



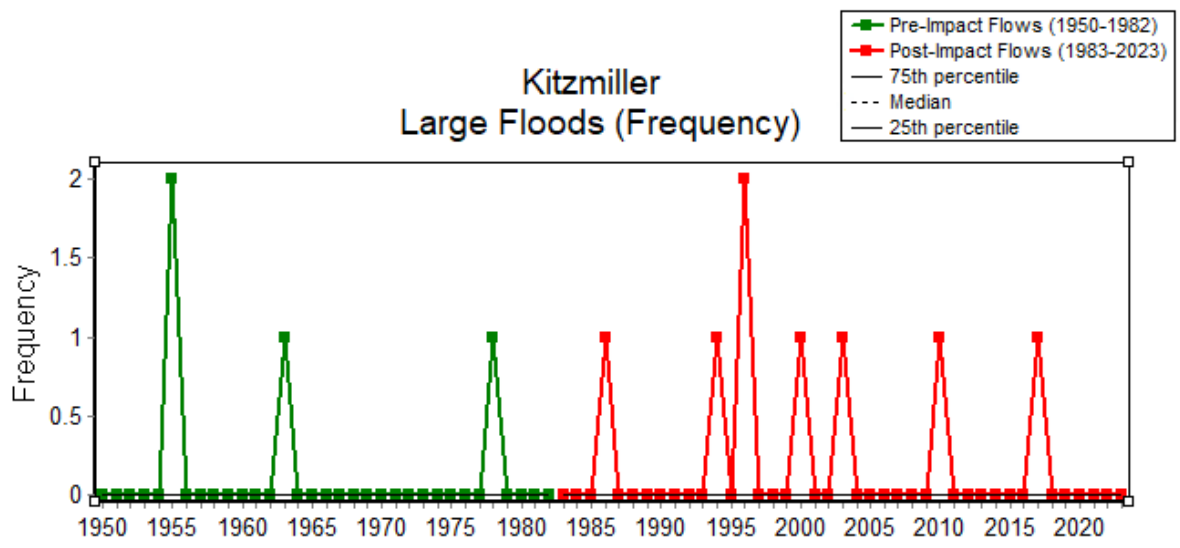
Luke Storm Frequency			
Yr.	*% Exceedance*	*Pre-Dam Flow (cfs)*	*Post-Dam Flow (cfs)*
1	100	6	95
2	50	358	457
5	20	1020	1030
10	10	1660	1610
25	4	2860	2610
50	2	3790	3350
100	1	4720	4170
500	0.2	7420	6990
100	0.1	8830	7740

6.3.1. Peak Flows

6.3.1.1. Kitzmiller

Kitzmiller peak inflow magnitude and frequency (more high flow days) have increased after dam construction. Pre-dam the peak flow was around 11,000 cfs and post-dam the peak flow was around 17,000 cfs. See Figure 6.10 below.

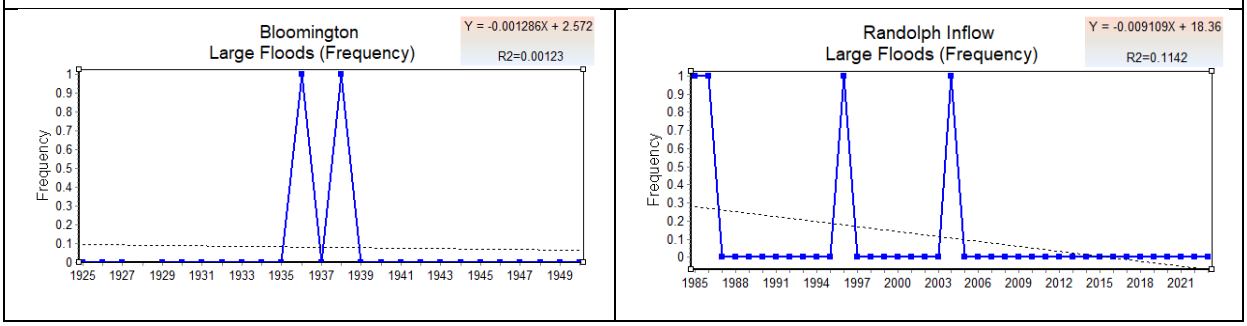
Figure 6.10 - Kitzmiller, Large Floods (Freq)



6.3.1.2. Dam Site- Bloomington and Randolph

Since the Bloomington gage was located at the current site of Jennings Randolph Dam it will be used to approximate natural or pre-dam flows. Peak inflows magnitude and frequency (more high flow days) have increased after dam construction. Pre-dam the peak flow was around 18,000 cfs and post-dam the peak flow was around 26,000 cfs.

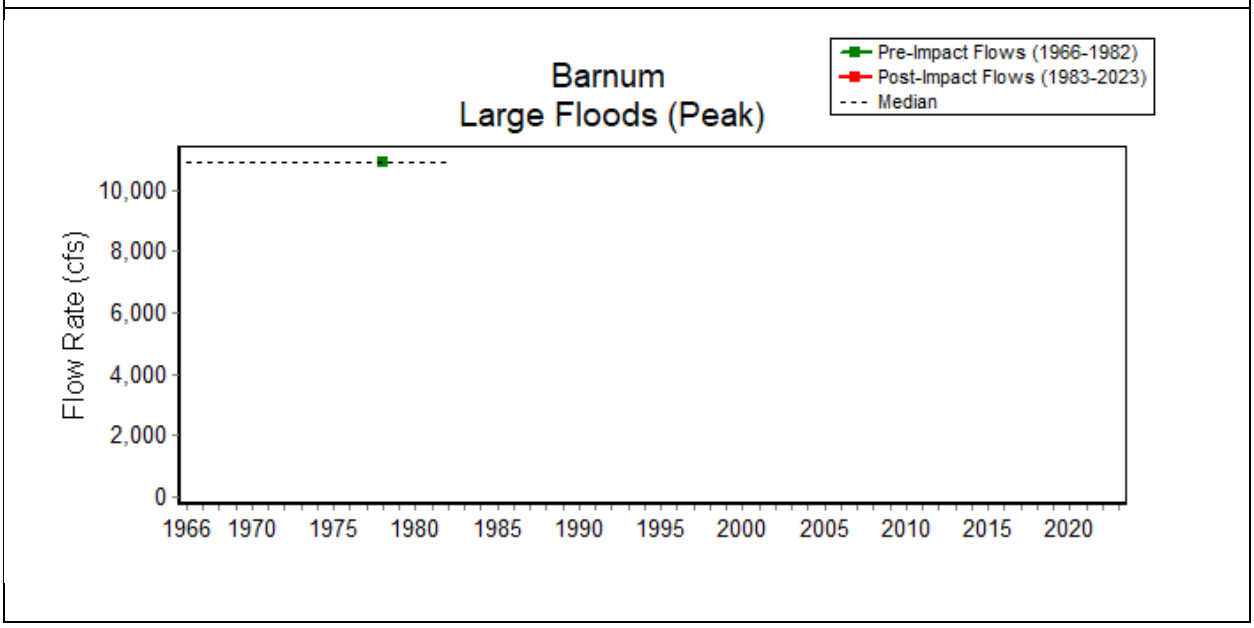
Figure 6.11 - Bloomington and Randolph Inflow, Large Floods (Freq)

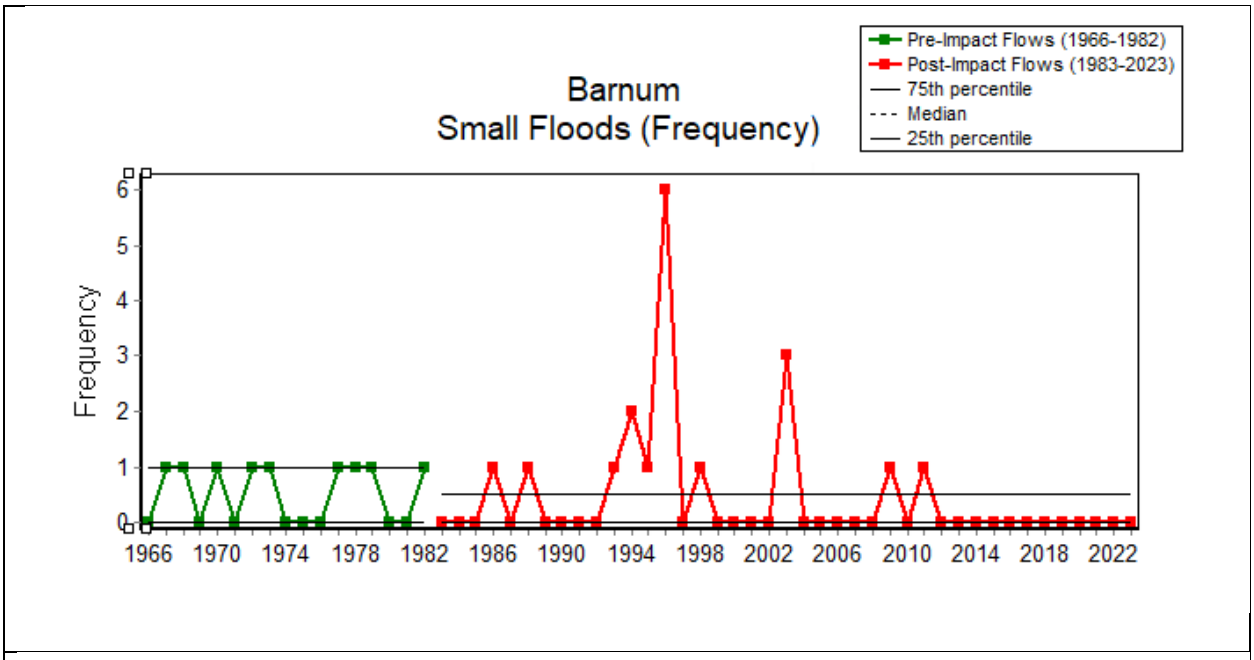


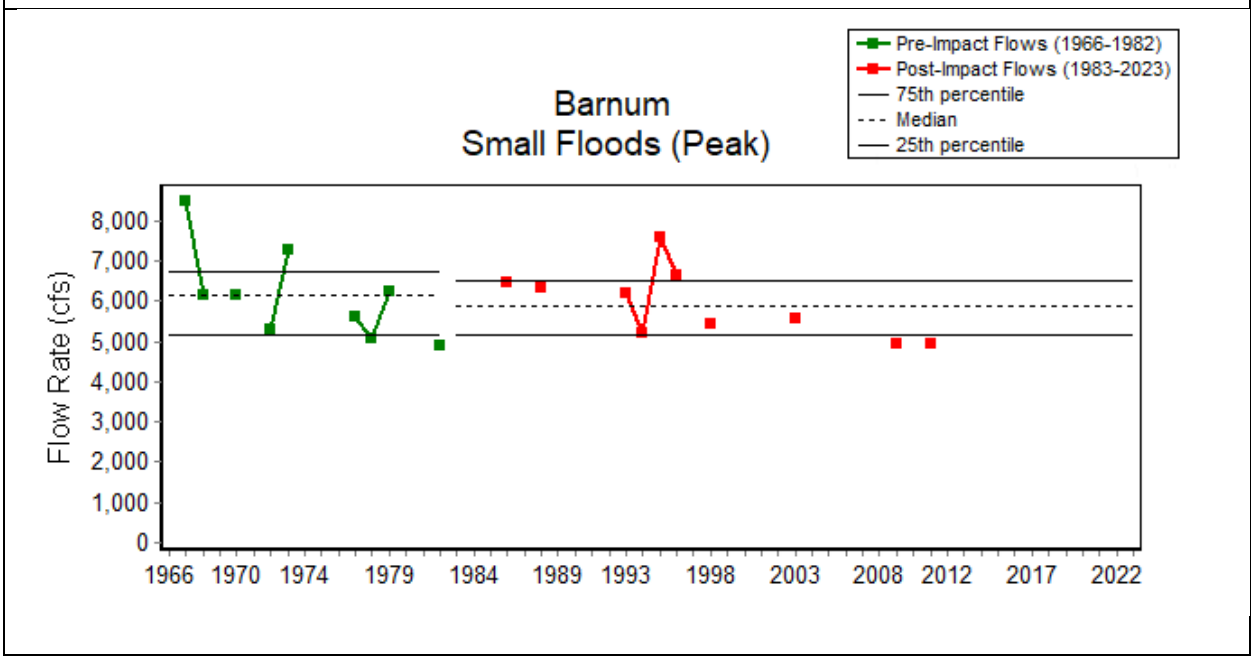
6.3.1.3. Barnum

As expected, due to the Jennings Randolph Dam operation there are no large floods downstream at Barnum after 1984. The magnitude of small floods has generally decreased.

Figure 6.12 - Barnum, Large vs Small Floods



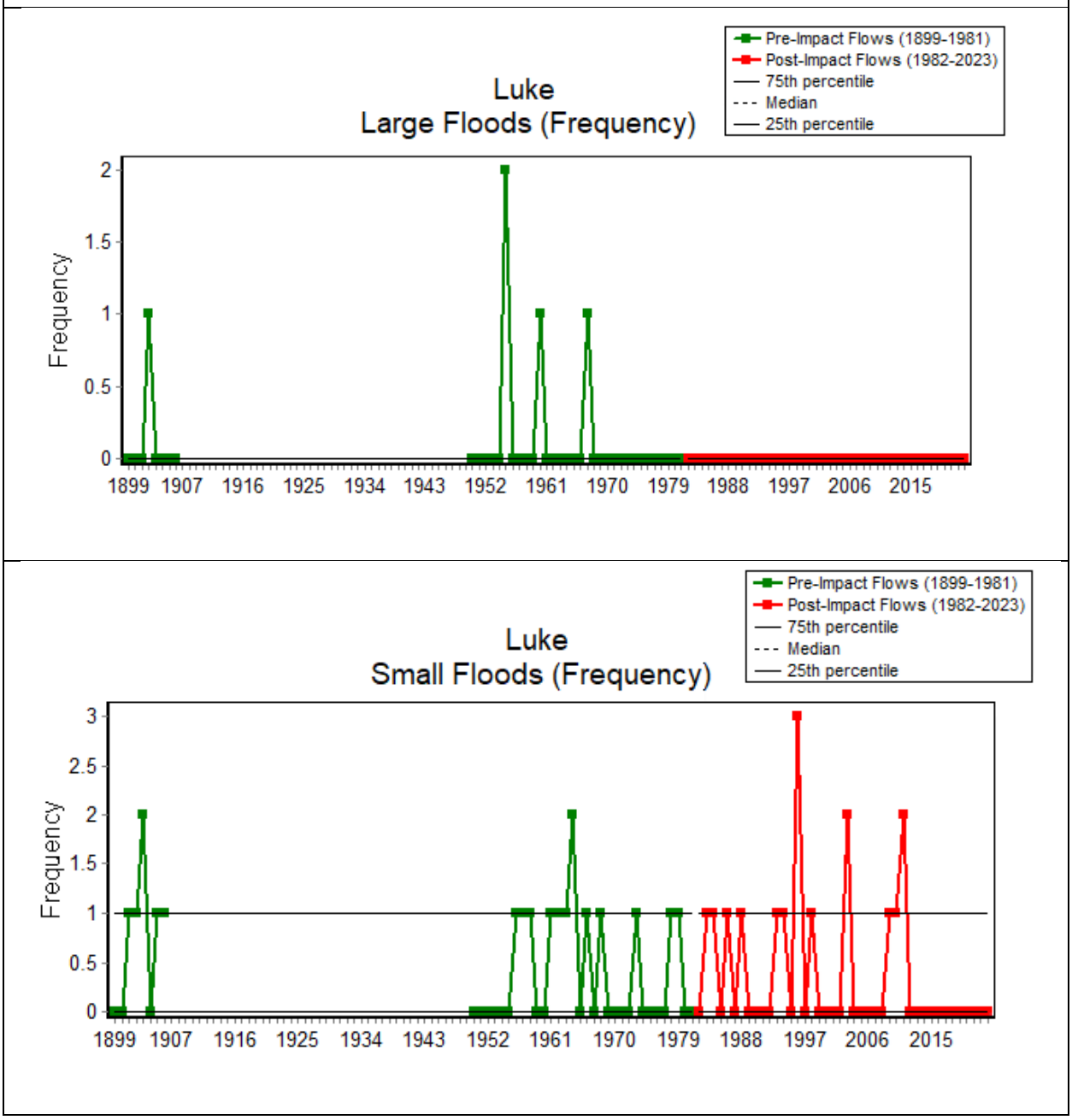

Barnum
Small Floods (Frequency)
Pre-Impact Flows (1966-1982)
Post-Impact Flows (1983-2023)
75th percentile
Median
25th percentile
Frequency
0
1
2
3
4
5
6
1966
1970
1974
1978
1982
1986
1990
1994
1998
2002
2006
2010
2014
2018
2022

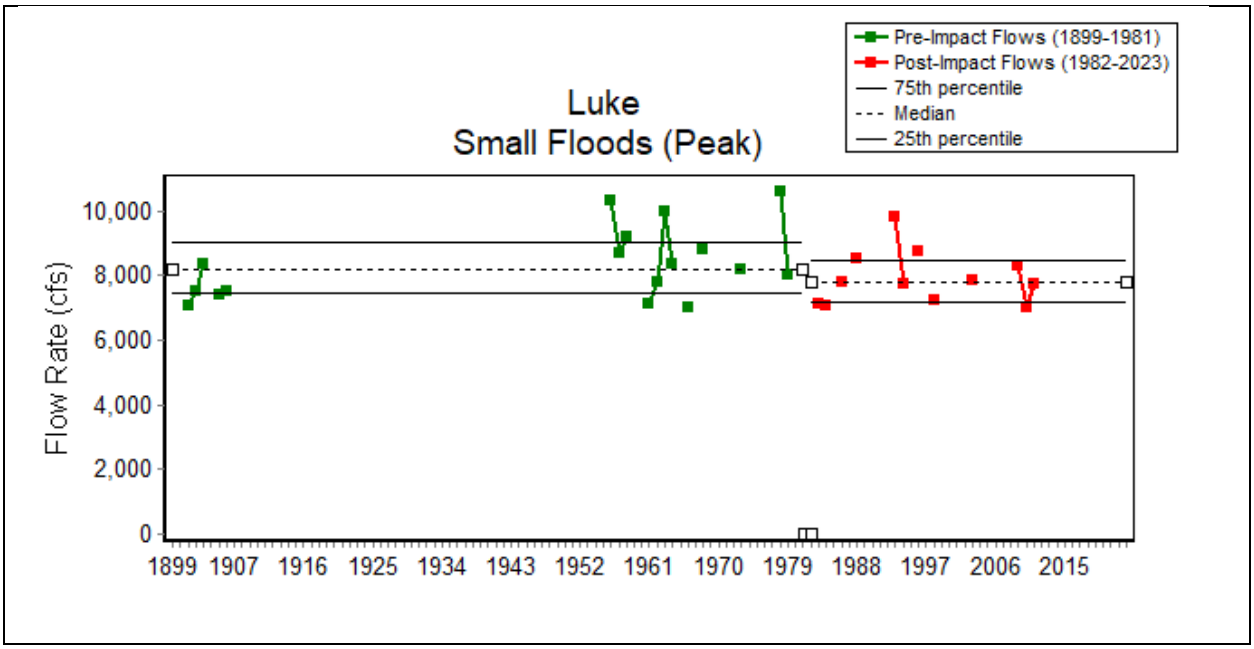

Barnum
Small Floods (Peak)
Pre-Impact Flows (1966-1982)
Post-Impact Flows (1983-2023)
75th percentile
Median
25th percentile
Flow Rate (cfs)
0
1,000
2,000
3,000
4,000
5,000
6,000
7,000
8,000
1966
1970
1974
1979
1984
1989
1993
1998
2003
2008
2012
2017
2022

6.3.1.4. Luke

Similarly, to Barnum, there are no large floods downstream at Luke after 1984, but the frequency of small floods has increased. The magnitude of small floods has decreased.

Figure 6.13 - Luke, Large vs Small Floods



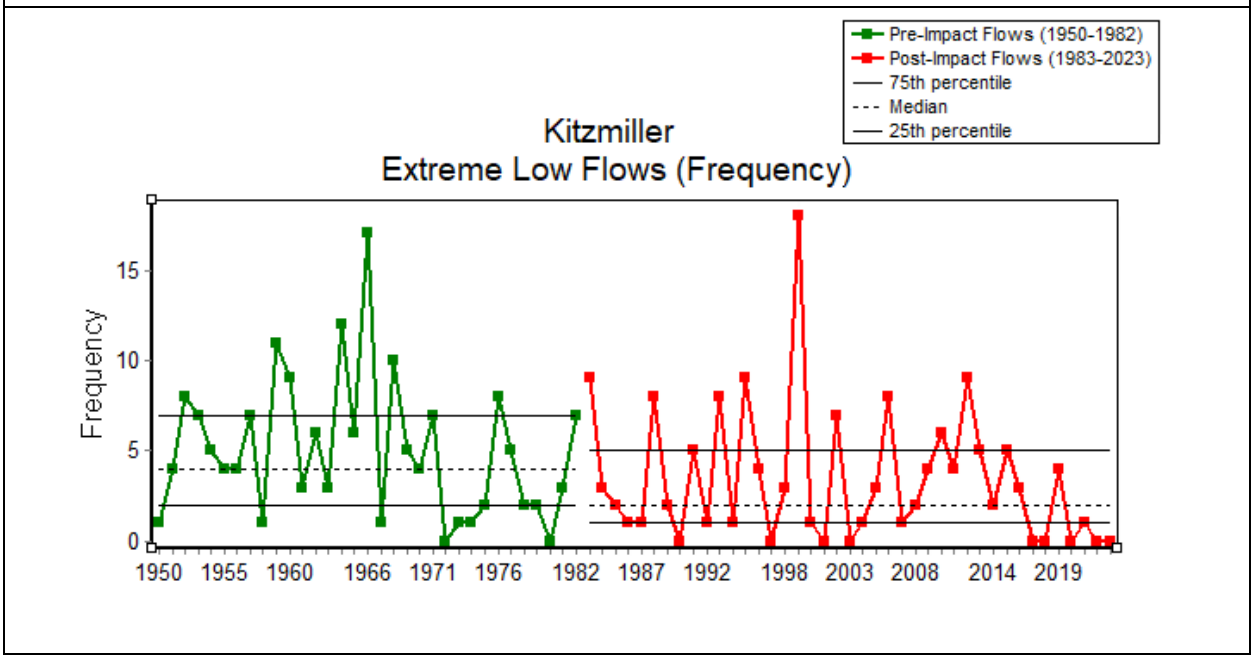


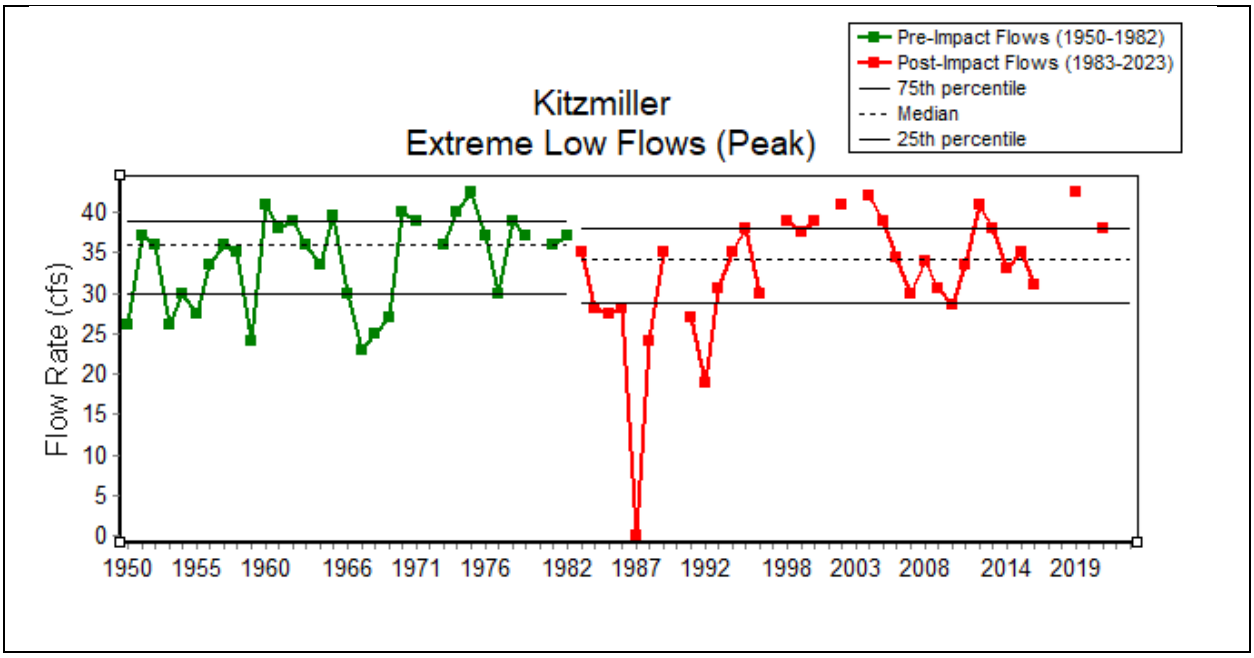
6.3.2. Low Flows

6.3.2.1. Kitzmiller

Kitzmiller extreme low flow magnitudes have stayed approximately the same after dam construction. The frequency has slightly decreased.

Figure 6.14 - Kitzmiller, Extreme Low Flows

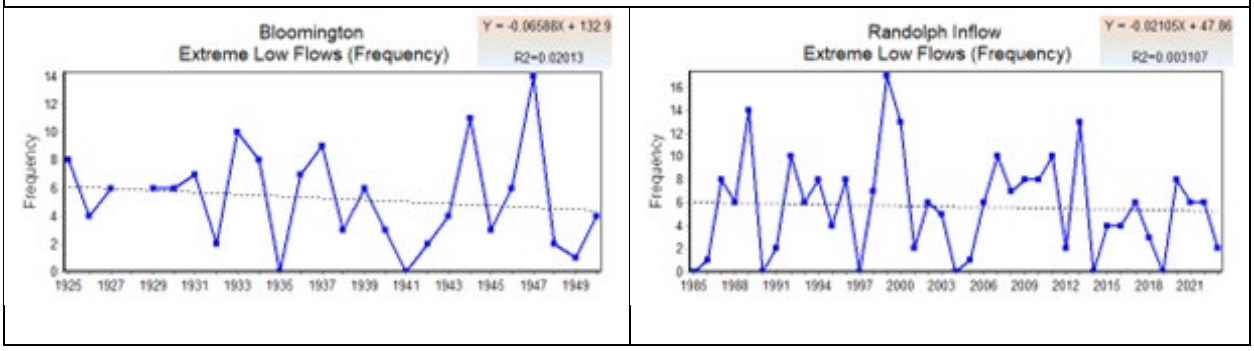




6.3.2.2. Dam Site - Bloomington and Randolph

Extreme low flows magnitude and frequency have stayed relatively the same.

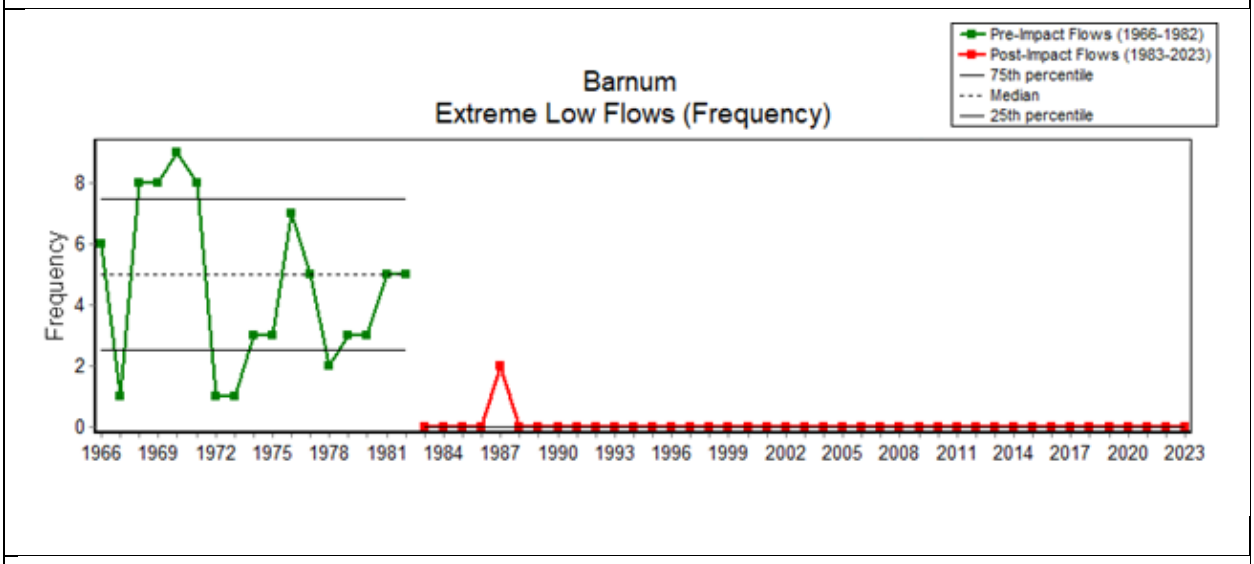
Figure 6.15 - Bloomington and Randolph Inflow, Extreme Low Flows

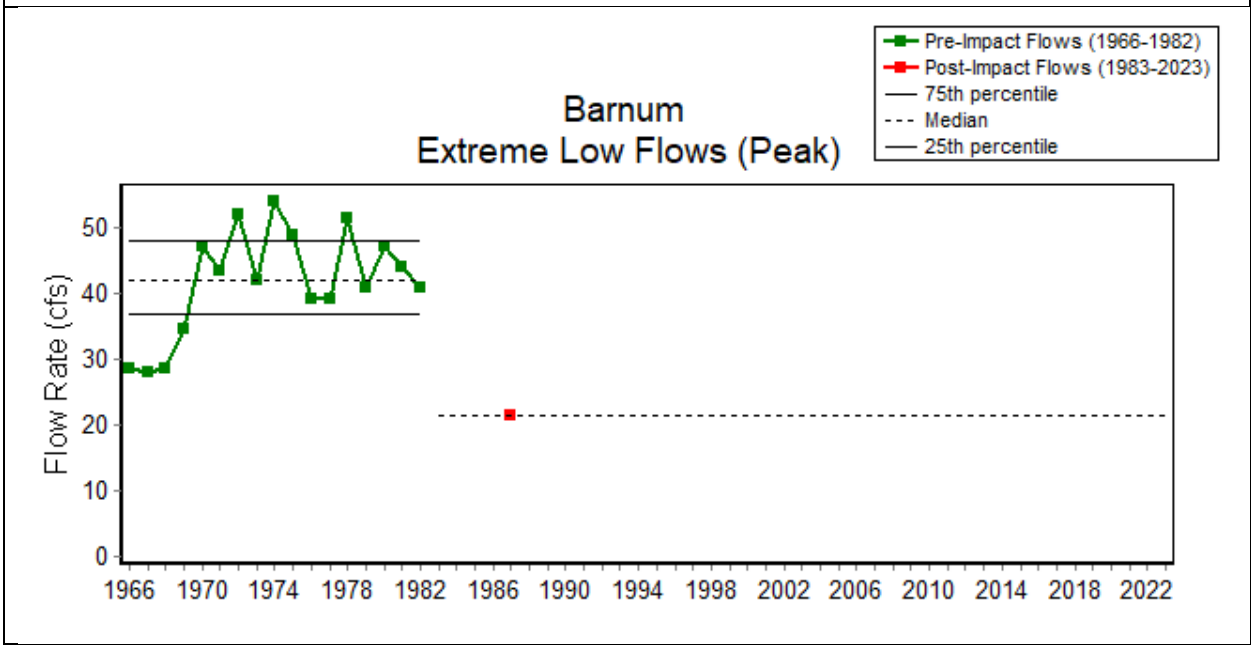


6.3.2.3. Barnum

There is only one instance of extreme low flows at Barnum after 1984. This is primarily a result of the typical 250 cfs release made from JRL during the summer months.

Figure 6.16 - Barnum, Extreme Low Flows

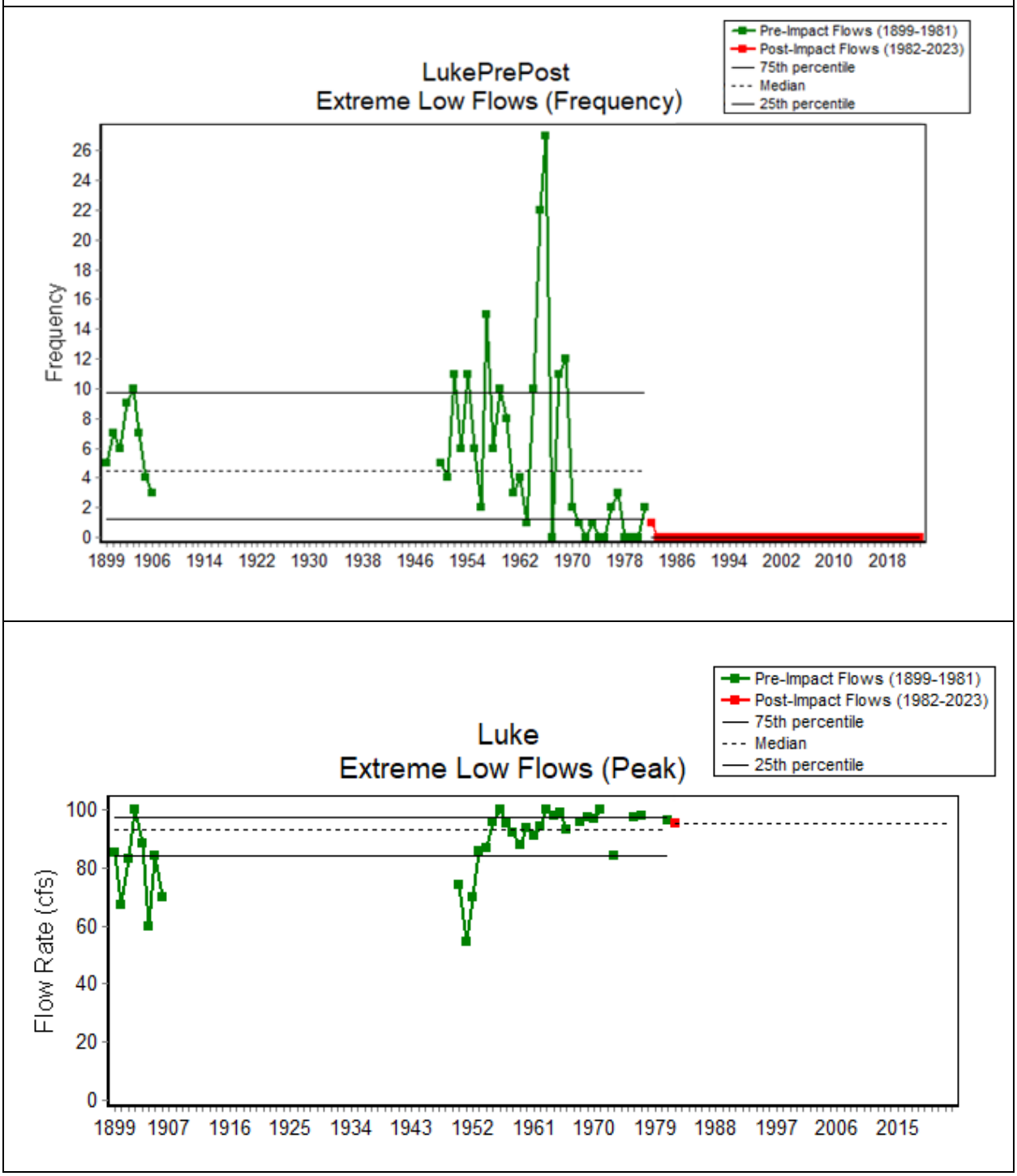




6.3.2.4. Luke

Similarly, to Barnum, there is only one instance of extreme low flows at Barnum after 1984. This is primarily a result of the typical 250 cfs release made from JRL during the summer months.

Figure 6.17 - Luke, Extreme Low Flows



6.3.3. IHA Flow Regimes Results

The results from the IHA analysis can be used as a guideline for environmental flows at the downstream gages of Barnum and Luke. It is important to note that damage at Luke begins at 16,200 cfs and that Water Management typically tries to keep Luke above 300 cfs, with a minimum target of 120 cfs. Flood damage stages are not calculated at Barnum, but the IHA large flood (8,972 cfs) is about 9 ft in stage which corresponds to historic crests at Barnum which range from 9 ft to 13.3 ft.

The following table defines the flow regimes, called environmental flow components in IHA (peak flows, low flows etc.) for each of the relevant gages. See Appendix A1 for the full definitions.

Table 6.2 - Flow Regimes and Environmental Flow Components

	Abbreviated Definition	Flow (cfs)				
		Kitzmiller	Bloomington	Randolph Computed Inflow	Barnum	Luke
EFC low flow threshold:	Normal conditions	250	260	328	279	358
EFC high flow threshold:	Normal precipitation or snow melt	562	580	663.8	627.5	828
EFC extreme low flow threshold:	Drought, dry conditions	44	53	64	54	100
EFC small flood minimum peak flow:	Heavy rain exceeding banks	4,710	4,710	5,259	4,910	7,025
EFC large flood minimum peak flow:	Extreme and rare flooding	7,800	10,800	14,100	8,972	10,690

Chapter 7. Dependencies Between the Ecological Indicators and Specific Environmental Flow Components

7.1. Flow Components

Due to lack of studies on the North Branch, studies from the Middle Potomac and Lower Potomac will be referenced. Below are the effects of different flow regimes.

The components of flow that effect wildlife are flashiness, magnitude (low flow, high flows), variability, and velocity. **Table 7.1 lists the ecological response to changes in the components of flow.**

7.1.1. Flashiness

Due to the steep topography and increased development flash flooding is common in the North Branch and the greater Potomac. Flashiness (streams rising and falling quickly) is correlated with negative effects on stream macroinvertebrate (U.S. Army Corps of Engineers et al., 2014).

7.1.2. Low Flows

Macroinvertebrate are found not to be sensitive to low flows, but fish and mussels are (U.S. Army Corps of Engineers et al., 2014).

7.1.3. Variability

Variability is defined as temporal patterns or variation in flow. In a system regulated by a dam flow may change several times in a day due to changes in releases. In an unregulated system flow may change over the course of days.

7.1.4. Velocity

Velocity is defined as the speed of water in a stream. In any given section of a stream flow may be the same, but velocity is different in pool, riffles, bends, etc. Each species is affected by stream velocity during their life cycle. For example, too high of a velocity due to increased flow to alleviate dry conditions may scour away eggs.

Table 7.1 - Alterations in Flow Components and Common Ecological Response (modified after Poff et al., 1997; Poff & Zimmerman, 2010)			
Flow component	**Alteration**	**Ecological response**	
Magnitude	Flow stabilization (loss of extreme high and/or low flows)	(a)	• Reduced diversity • Loss of sensitive species • Altered assemblages and dominant taxa • Reduced abundance Increase in non-natives
		(r)	• Seedling desiccation • Ineffective seed dispersal • Terrestrialization of flora • Lower species richness • Encroachment of vegetation into channels Increased riparian cover • Altered assemblages
	Greater magnitude of extreme high and/or low flows	(a)	• Life cycle disruption • Reduced species richness • Altered assemblages and relative abundance of taxa • Loss of sensitive species
Frequency	Decreased frequency of peak flows	(a)	• Aseasonal reproduction • Reduced reproduction Decreased abundance or extirpation of native fishes • Decreased richness of endemic and sensitive species • Reduced habitat for young fishes
		(r)	• Shift in community composition • Reductions in species richness • Increase in wood production
Duration	Decreased duration of floodplain inundation	(a)	• Decreased abundance of young fish • Change in juvenile fish assemblage • Loss of floodplain specialists in mollusk assemblage
		(r)	• Reduced growth rate or mortality • Altered assemblages • Terrestrialization or desertification of species composition • Reduced area of riparian plant or forest cover
	Prolonged low flows	(a)	• Concentration of organisms • Downstream loss of floating eggs
		(r)	• Reduction or elimination of plant cover • Diminished plant species diversity

			• Desertification of species composition
	Prolonged inundation	(a)	• Loss of riffle habitat
		(r)	• Change in vegetation functional type Tree mortality
Timing	Shifts in seasonality of peak flows	(a)	• Disruption of spawning cues • Decreased reproduction and recruitment • Change in assemblage structure
	Increased predictability	(a)	• Change in diversity and assemblages structure • Disruption of spawning cues • Decreased reproduction and recruitment
	Loss of seasonal flow peaks	(a)	• Disruption of migration cues • Loss of accessibility to wetlands and backwaters • Modification of food web structure
		(r)	• Reduced riparian plant recruitment Invasion of exotic riparian plant species • Reduced plant growth and increased mortality • Reduction in species richness and plant cover
Rate of change	Rapid changes in river stage	(a)	• Drift (washout) and stranding
	Accelerated flood recession	(r)	• Failure of seedling establishment

7.2. Macroinvertebrate

See Figure 7.1. below for a summary of flow effects on macroinvertebrates.

Figure 7.1 - Flow Alteration Macroinvertebrate Relationship

Table 5. Summary of flow alteration – macroinvertebrate relationships identified.
Possible underlying mechanisms that have been suggested in the literature are also listed.

Alteration in these aspects of the flow regime is associated with:	Degradation in these macroinvertebrate metrics:	Possible mechanisms that could explain the association are:
• Higher maximum flows • Shorter duration of high flows • Shorter duration of low flows • More low flow pulses • More high flow pulses • Faster rates of change in flow (flashier)	All 19 family-level macroinvertebrate metrics and the Chessie BIBI multi-metric index	• Scour of periphyton and organic matter (food) during high flows • Catastrophic accidental drift during floods • Displacement from habitat and stranding when waters recede • Physical alteration of stream bed habitat • Indirect effects of poor runoff water quality (sedimentation, pollutants) • Interruption of development or dispersal cues
• Lower middle and low magnitude flows, includes median flow, August median flow, summer Q85 flow, baseflow index, 3-day and 1-day annual minima, and 7Q10	None of the biometrics	• Swift recovery due to adaptations to low flow (drought resistant or diapausing life stages) • Multi-voltine (short) life cycles • High mobility, able to find refugia and later recolonize

Source: Middle Potomac Report (U.S. Army Corps of Engineers et al., 2014).

Chapter 8. Developing Environmental Flow Hypothesis

8.1. Methodology

Due to limited species based environmental flow information in the literature for the North Branch, the designer flow model has been chosen as the best method for developing environmental flows. The designer flows model is chosen over mimicking natural flows because designer flows: "…favor native over nonnative fishes while rarely, if ever, encroaching on human water needs. The predicted benefits surpass those generated by natural flow mimicry and were retained across periods of heightened drought" (Chen & Olden, 2017). Designer flow is defined as dam releases that are calculated with a multi-objective framework to balance human water needs and the ecology. As mentioned in Section 1.1, natural flow is defined as the flows the river system would have without any impoundments or human action (development, dams, levees, etc.). These values are either observed, calculated or a combination of both.

8.2. Natural Flows

North Branch Potomac River natural flows were calculated by multiplying JRL inflow by drainage area factors from the upstream gage (Kitzmiller) and the downstream locations (Barnum, Luke). For Barnum, the drainage area factor is 1.04 (266 sq mi at Barnum divided by 255 sq mi at Kitzmiller). For Luke, the drainage area factor for only the North Branch Potomac River (not including the Savage River tributary) is 1.17 (404 sq mi at Luke minus 106 sq mi from Savage River divided by 255 sq mi at Kitzmiller). The final calculation for Barnum natural flow is observed Kitzmiller flow multiplied by a factor of 1.04. The final calculation for Luke natural flow is observed Kitzmiller flow multiplied by a factor 1.17.

8.3. Limitations

Benefits from releases from JRL may already be maximized on a monthly and seasonal scale with current environmental strategies (Table 8.1). Since in March the average outflow is already about 300 cfs less than the average inflow, the best months to increase the storage in the lake may be April and May. During the JASON months outflow is greater than inflow due to existing policies such as, temperature mitigation, water quality releases during low flows, AVF releases. Flow recommendations may make the most benefit by focusing on sub-weekly and sub-daily flows. Table 8.1 exhibits the average monthly inflows and outflows at JRL. These flows should be considered during the workshop.

Table 8.1 - Randolph Average Monthly Flows 1997-2022 (cfs)		
Months	**Inflow**	**Outflow**
January	698	662
February	817	702
March	1048	761
April	816	772
May	765	746
June	376	432
July	285	337
August	196	269
September	249	365
October	185	266
November	350	367
December	575	525
*Inflow values are computed from an elevation gage Source: USACE, Baltimore District, Water Management Database		

8.3.1. Randolph Flow limitations

8.3.1.1. Physical release limitations

The tower at Jennings Randolph has two independent wet wells with five pairs of portals at center line elevations 1342, 1375, 1400, 1426, and 1449 feet PCD to provide releases for water quality control. Each wet well has a low flow control gate referred to as the Quality Control (QC) gates. When the discharge from the QC gates exceeds 7,000 cfs, an unstable flow condition begins to develop in the outlet tunnel. This unstable flow condition becomes worse with increasing outflows, and opening the QC gates worsens this condition. The QC gates are normally closed whenever outflow exceeds 7,000 cfs, and discharge through the service gates is normally limited to 10,000 cfs. Discharges greater than 10,000 cfs tend to produce large rocking waves in the stilling basin which overtop the wingwalls and displace adjacent riprap.

8.3.1.2. Supersaturation

As mentioned previously, dissolved oxygen levels in the outflow approach the saturation level when the discharge reaches about 1,000 cfs, and approach 115% when the discharge reaches about 2,500 cfs. When fish are exposed to prolonged periods of water contacting

supersaturated gases (above 110%), gas bubble disease would occur in fish and result in fish kills. Fish are no longer stocked in the stilling basin, but the regulation plan for Jennings Randolph Lake limits outflow to 2,500 cfs or less as much as possible.

8.3.1.3. Luke, Westernport, and Keyser Area

The river-reach from 10 to 20 miles downstream of Jennings Randolph Lake encompasses numerous industrial, commercial, and residential areas. Flood control regulation for these areas is complicated by the fact that several uncontrolled tributaries (principally Georges Creek and New Creek) enter the North Branch and comprise almost 50% of the drainage area by the time the river reaches Keyser. These tributaries tend to run rapidly following intense storms, and releases from JRL must be carefully monitored to avoid flooding in the downstream areas. The maximum non-damaging river stage at Luke is 10.5 feet (about 16,200 cfs). The combined discharge from JRL and Savage River Dam is normally adjusted to keep the stage at Luke below 10.5 feet. During low flow periods, releases from Jennings Randolph Lake and Savage River Dam are usually adjusted to provide a combined flow of at least 120 cfs at Luke for water quality purposes.

8.3.1.4. Cumberland - Ridgeley Areas

A densely populated residential area exists in the floodplain at Cumberland, about 40 miles downstream of Jennings Randolph Lake. Portions of Cumberland and Ridgeley are protected by a local flood protection project. Uncontrolled flow from Wills Creek enters the North Branch Potomac River in this area and contributes enough flow to occasionally restrict project releases. About 30% of the drainage area at Cumberland is controlled by JRL. The NWS flood stage at Cumberland is 17 feet (about 22,300 cfs), and Jennings Randolph Lake is regulated to keep the stage below 17 feet when possible.

8.3.1.5. Mount Storm Dam

Additionally, upstream of Jennings Randolph Dam is a coal power plant, Mount Storm. The facilities include a dam for the water used by the plant to cool. Due to West Virginia state regulations the dam intermittently must make unscheduled, large releases. Due to this, JRL may need to respond to the increased inflow if it causes the lake to rise too high.

8.4. Hypothesis

Section 8.4 to be completed after conclusion of workshop.

Chapter 9. Recommendations

9.1. Initial Recommendations

Due to lack of data and studies in the North Branch the flow variability method to build natural habitat structures is recommended. In addition, increased monitoring in the NBPR is recommended. More specifically a goal of the Baltimore District is monitoring of summer AVF flows more quantitatively to assess their benefits and effects. Below is a summary of recommendation for monitoring.

Recommendations from the Middle Potomac River Watershed Assessment (U.S. Army Corps of Engineers et al., 2014)

Table 9.1 - Initial Recommendations

Flow Variability	• Use IHA flow regime statistics to create flows with the appropriate rate of rise, rate of fall, peak, and duration. • Variable flow regimes move different types of debris to form over time a diverse range of natural habitat structures for various biota
Large river	• Long-term floodplain vegetation monitoring • Mussel and fish studies • Surveys on cormorants, • Macroinvertebrates, amphibians, and reptiles

Flow Data needs	• Fund sustained river monitoring
	• Continued long-term operation of stream gages
	• Flow metric verification
	• Additional observed streamflow data is needed on:
	  ○ Smaller streams,
	  ○ In karst watersheds
	  ○ Undisturbed or reference watershed

Biological Data needs	• Basin-wide datasets of fish, mussels, and vegetation (etc.) • Biological factors unrelated to flow or the stream environment significantly stress and alter fish populations (including fish stocking programs, introductions of non-native species, and disease)
Standardization	• State and local monitoring programs use different methodologies to collect fish.

	• Ensure the dataset assembled for flow analysis contains comparable data

AVF -
Artificial Varied Flow
	• Continuous or daily monitoring before and after AVF release of:
	  ○ DO
	  ○ Didymo
	  ○ Benthic macroinvertebrates
	  ○ Fish

Additionally, assumptions and knowledge gaps are listed in Table 9.2.

Table 9.2 - Assumptions	
Location	• Middle Potomac studies translate the same to North Branch Potomac (U.S. Army Corps of Engineers et al., 2014)
Pre-dam flow regimes	• Due to the lack of information on species flow needs and life cycle, this paper assumes replicating flow regimes pre-dam will benefit native species • The discontinued Bloomington gage It provides the direct pre-dam inflow data from 1924 to 1948.

9.2. Workshop Recommendations

Table 9.3 to be completed after conclusion of workshop.

Table 9.3 - Workshop Recommendations	
Flow Variability	
Flow Data needs	
Biological Data needs	
Standardization	
AVF - Artificial Varied Flow	
Temperature	

References

Baltimore District. (1997). Reservoir Regulation Manual Jennings Randolph Lake. USACE.

Bencala, H. M. P. (2014). Development of a Potomac Basin Water Quality Data Inventory, 2014. ICPRB. https://www.potomacriver.org/wp-content/uploads/2014/12/ICP14-2_Moltz.pdf

Blum, A. G., Kanno, Y., & Letcher, B. H. (2018). Seasonal streamflow extremes are key drivers of Brook Trout young-of-the-year abundance. https://doi.org/10.1002/ecs2.2356

Bogan, A., & Ashton, M. (2016). Manual of the freshwater bivalves of Maryland. Maryland Department of Natural resources. https://dnr.maryland.gov/streams/Publications/Manual-Freshwater%20Mussels%20of%20MD.pdf

Bothwell, M. L., Taylor, B. W., & Kilroy, C. (2014). The Didymo story: The role of low dissolved phosphorus in the formation of Didymosphenia geminata blooms, Diatom Research, 29:3, 229-236, https://doi.org/10.1080/0269249X.2014.889041

Buchanan, C., & Selckmann, G. M. (2019). The Influence of Jennings Randolph Lake and Dam Operations on River Flow and Water Quality in the North Branch Potomac River. https://www.potomacriver.org/wp-content/uploads/2019/12/NBPR_TechRpt_FlowWQ_12_18_2019_b.pdf

Chen, W. & Olden, J.D. (2017). Designing flows to resolve human and environmental water needs in a dam-regulated river: Nature Communications 8, 2158, https://doi.org/10.1038/s41467-017-02226-4.

Cruise, J. F., Laymon C. A., & Al-Hamdan O. Z. (2010). Impact of 20 years of land-cover change on the hydrology of streams in the southeastern United States. Journal of the American Water Resources Association 46(6):1159-1170.

Cummins, J., Buchanan, C., Haywood, C., Moltz, H., Griggs, A., Jones, R. C., ... & Branch, A. E. (2010). Potomac basin large river environmental flow needs. Interstate Commission on the Potomac River Basin. Publication number ICPRB, 10-03.

Cutler, M. (2024). EU_Determination_NB-Potomac-Allegany_Proposed. *Maryland Department of the Environment*. https://mde.maryland.gov/programs/water/TMDL/WaterQualityStandards/Documents/Exisiting_Use_Documents/EU_Determination_NB-Potomac-Allegany_Proposed_8-2024.pdf

Everson, D. A., & H. Boucher, D. (1998). Tree species-richness and topographic complexity along the riparian edge of the Potomac River. *Forest Ecology and Management*, *109*(1–3), 305–314. https://doi.org/10.1016/s0378-1127(98)00264-3

Gallup, J. (2019). Jennings Randolph Lake: Master Plan. https://hdl.handle.net/11681/41561

Gerhart, J. M., & Brakebill, J. W. (1996). Design and implementation of a sampling strategy for a water-quality assessment of the Potomac River Basin (Vol. 96, No. 4034). US Department of the Interior, US Geological Survey. https://doi.org/10.3133/wri964034

Hansen, E., Collins, A., Zegre, S., & Hereford, A. (2010). The benefits of acid mine drainage remediation on the North Branch Potomac River. Morgantown (WV). Downstream Strategies.

Jones, M.T., Roberts H. P., & Willey L. L. (2018). Conservation Plan for the Wood Turtle in the Northeastern United States. Report to the Massachusetts Division of Fisheries & Wildlife and the U.S. Fish & Wildlife Service.

Jones, M., & Willey, L. (2021). A Guide to Habitat Management for Wood Turtles. Northeast Association of Fish and Wildlife Agencies. Retrieved August 16, 2023, from northeastturtles.org

Hitt, N. P., Landsman, A. P., & Raesly, R. L. (2022). Life history strategies of stream fishes linked to predictors of hydrologic stability. Ecology and Evolution, 12(5), e8861.

Kaesler, R. L., & Cairns Jr, J. (1972). Cluster analysis of data from limnological surveys of the upper Potomac River. American Midland Naturalist, 56-67.

Kennan, J. G., Riva-Murray K., & Beaulieu K. M. (2009). Determining Hydrologic Factors That Influence Stream Macroinvertebrate Assemblages in The Northeastern US. Ecohydrology DOI: 10.1002/eco.

Maloney, K. O., Krause, K. P., Buchanan, C., Hay, L. E., McCabe, G. J., Smith, Z. M., ... & Young, J. A. (2020). Disentangling The Potential Effects of Land-Use and Climate Change on Stream Conditions. Global change biology, 26(4), 2251-2269.

Maryland Department of Natural Resources (MD DNR). (2021). Section "North Branch Potomac River from Jennings Randolph Lake to Cumberland." Federal Aid Report.

Maryland Department of Natural Resources (MD DNR). (2010). Maryland Biological Stream Survey Sentinel Site Network: A Multipurpose Monitoring Program. Accessed 9 September 2022. https://dnr.maryland.gov/streams/Publications/2010SentinelSiteReport.pdf

Morgan, R. P., II, Jacobsen, R. E., Weisberg, S. B., McDowell, L. A., & Wilson, H. T. (1991). Effects of flow alteration on benthic macroinvertebrate communities below the Brighton Hydroelectric Dam. *Journal of Freshwater Ecology*, *6*(4), 419–429. https://doi.org/10.1080/02705060.1991.9665321

National Water Summary Wetland Resources: Maryland. https://www.fws.gov/sites/default/files/documents/National-Water-Summary-Wetland-Resources-maryland.pdf

National Water Summary Wetland Resources: West Virginia. https://www.fws.gov/sites/default/files/documents/National-Water-Summary-Wetland-Resources-West-Virgnia.pdf

NOAA. (n.d.). *SC ACIS2*. SC ACIS2; NOAA Regional climate Centers. Retrieved July 16, 2025, from https://scacis.rcc-acis.org/

NPS. (2019, April). Invasive species: Didymo or "Rock snot" - nps.gov. National Park Service. https://www.nps.gov/yose/learn/nature/didymo.htm

Poff N. L., Allan J. D., Bain M. B., Karr J. R., Prestegaard K. L., Richter B. D., Sparks R. E., & Stromberg J. C. (1997). The Natural Flow Regime. Bioscience 47(11):769–784

Poff N. L. & Zimmerman J. K. (2010). Ecological Responses to Altered Flow Regimes: A Literature Review to Inform the Science and Management of Environmental Flows. Freshw Biol 55(1):194–205

Richter, B., Baumgartner, J., Wigington, R., & Braun, D. (1997). How much water does a river need? *Freshwater biology*, *37*(1), 231-249.

USACE, (2019). Jennings Randolph Lake Master Plan. https://usace.contentdm.oclc.org/utils/getfile/collection/p16021coll7/id/15091

U.S. Army Corps of Engineers, The Nature Conservancy, and Interstate Commission on the Potomac River Basin. (2014). Middle Potomac River Watershed Assessment: Potomac River Sustainable Flow and Water Resources Analysis. Final Report. 107 pp. and 11 appendices.

U.S. Geological Survey. (2015). "Genetics Provide New Hope for Freshwater Mussel Species." Accessed 9 September 2022. https://www.usgs.gov/news/national-news-release/genetics-provide-new-hope-endangered-freshwater-mussels

USGS. (n.d.-c). *freshwater golden clam*. Collections; United State Geological Survey. Retrieved July 21, 2025, from https://nas.er.usgs.gov/queries/CollectionInfo.aspx?SpeciesID=92&State=MD&HUCNumber=02070002

Smithsonian Environmental Research Center. (n. d.). Corbicula fluminea in Ches Report. Accessed 9 September 2022. https://invasions.si.edu/nemesis/chesreport/species_summary/Corbicula%20fluminea

Stagge, J. H. (2012). Optimization of Multi-Reservoir Management Rules Subject to Climate and Demand Change in the Potomac River Basin (Doctoral dissertation, Virginia Tech).

Stagge, J. H. & Moglen, G. E. (2014). Evolutionary Algorithm Optimization of a Multireservoir System with Long Lag Times. Journal of Hydrologic Engineering. 19(9)

Stagge, J.H. & Moglen, G. E. (2017). Water Resources Adaptation to Climate and Demand Change in the Potomac River. Journal of Hydrologic Engineering. 22(11)

State wildlife action plan - WVDNR. (2021, May 14). West Virginia Division of Natural Resources. https://wvdnr.gov/state-wildlife-action-plan/

Stauffer, J., Boltz, J., White, L. (2007). Fishes of West Virginia. Academy of Natural Sciences.

SWAP submission. (2015). Maryland Department of Natural Resources. https://dnr.maryland.gov/wildlife/Pages/plants_wildlife/SWAP_Submission.aspx

TNC. (n.d.). *Conservation Gateway - Home*. www.conservationgateway.org. https://www.conservationgateway.org

The Nature Conservancy. (2009). Indicators of Hydrologic Alteration Version 7.1 User's Manual. Smythe Scientific Software: Boulder, CO.

Wolfe, T. A. (1974). A Problematic Analysis of The Tasks of An Environmental Manager for The Upper Potomac River. American University.

Zeiringer, B., Seliger, C., Greimel, F., & Schmutz, S. (2018). River hydrology, flow alteration, and environmental flow. Riverine ecosystem management: Science for governing towards a sustainable future, 67-89. https://doi.org/10.1007/978-3-319-73250-3

Appendices

A1- IHA Flow Regime Definitions

IHA Flow Regime Definition
Low flows – This is the dominant flow condition in most rivers. In natural rivers, after a rainfall event or snowmelt period has passed and associated surface runoff from the catchment has subsided, the river returns to its base- or low-flow level. These low-flow levels are sustained by groundwater discharge into the river. The seasonally varying low-flow levels in a river imposes a fundamental constraint on a river's aquatic communities because it determines the amount of aquatic habitat available for most of the year. This has a strong influence on the diversity and number of organisms that can live in the river. **Extreme low flows** – During drought periods, rivers drop to very low levels that can be stressful for many organisms but may provide necessary conditions for other species. Water chemistry, temperature, and dissolved oxygen availability can become highly stressful to many organisms during extreme low flows, to the point that these conditions can cause considerable mortality. On the other hand, extreme low flows may concentrate aquatic prey for some species or may be necessary to dry out low-lying floodplain areas and enable certain species of plants such as bald cypress to regenerate. **High-flow pulses** – During rainstorms or brief periods of snowmelt, a river will rise above its low-flow level. As defined here, high-flow pulses include any water rises that do not overtop the channel banks. These pulses provide important and necessary disruptions in low flows. Even a small or brief flush of fresh water can provide much-needed relief from higher water temperatures or low oxygen conditions that typify low-flow periods, and deliver a nourishing subsidy of organic material or other food to support the aquatic food web. High-flow pulses also provide fish and other mobile creatures with increased access to up- and downstream areas. **Small floods** – During floods, fish and other mobile organisms are able to move upstream, downstream, and out into floodplains or flooded wetlands to access additional habitats such as secondary channels, backwaters, sloughs, and shallow flooded areas. These usually inaccessible areas can provide substantial food resources. Shallow flooded areas are typically warmer than the main channel and full of nutrients and insects that fuel rapid growth in aquatic organisms. As used here, a "small flood" includes all river rises that overtop the main channel but does not include more extreme, and less frequent, floods. **Large floods** – Extreme floods will typically re-arrange both the biological and physical structure of a river and its floodplain. These large floods can literally flush away many organisms, thereby depleting some populations but in many cases also creating new competitive advantages for some species. Extreme floods may also be important in forming key habitats such as oxbow lakes and floodplain wetlands.

A2 - IHA Median Flow Regimes

Non-Parametric IHA Scorecard- Median Flow Regimes						
Site	**Flow regime**	**Rate of rise cfs/day**	**Peak**	**Rate of fall cfs/day**	**Julian Date**	**Day no. / 365 days a year**
Barnum	Extreme Low	-	42	-	12-Sep	255
	High Flow	268.5	771	-124.2	17-Mar	76
	Small Flood	735	6140	-317.7	4-Mar	63
	Large Flood	1539	10900	-947.9	4-Jul	185
Bloomington	Extreme Low	-	40	-	16-Aug	228
	High Flow	273	816.5	-131.3	16-May	136
	Small Flood	999.6	5530	-440.9	11-Mar	70
	Large Flood	1173	15300	-761.9	7-Jan	7
Kitzmiller	Extreme Low	-	36	-	6-Sep	249
	High Flow	246	750	-120	7-Mar	66
	Small Flood	836.7	5150	-284.5	4-Mar	63
	Large Flood	1935	8310	-562.3	4-Jul	185
Luke	Extreme Low	-	93	-	29-Aug	241
	High Flow	308	1010	-162.5	16-Mar	75
	Small Flood	971.4	8190	-482.3	7-Mar	66
	Large Flood	3340	12200	-495.4	20-Mar	79
Randolph inflow	Extreme Low	-	50.5	-	16-Sep	259
	High Flow	276.1	855	-142.9	9-Feb	40
	Small Flood	1153	6659	-594.2	1-Apr	91
	Large Flood	7012	17310	-2154	18-Jan	18